Virtuous Intentions and Forced Labour: Long-term Impacts of Palm Oil Production in the Belgian Congo

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Abstract: From 1911 onward, the company Lever Brothers, now known as Unilever, established five concession areas in the Belgian Congo to extract palm oil. Lever Brothers had to develop these concession areas by investing in infrastructure, health, and education, but, similar to other colonies, the company also had to resort to violence and forced labour to mobilize enough workers. To examine the potential long-term effects of these concessions on wealth, education, and health, a spatial regression discontinuity design is employed, using the arbitrarily chosen borders of these concessions as geographic cutoffs. The results show that the majority of the estimates lack statistical significance, suggesting that there are no persistent significant effects on socio-economic development outcomes. However, methodological as well as historical context-related caveats, such as robust statistical inference, migration, adjacent concession areas, or uncertainty on whether the exposure to the concession only occurred in its designated areas, prevent definite conclusions from being drawn about long-term impacts.

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List of Abbreviations

DHS Demographic and Health Survey

GDP Gross domestic product

HCB Huileries du Congo Belge

MSE Mean squared error

OLS Ordinary least square

RD Regression Discontinuity

RDD Regression Discontinuity Design

TSI TseTse suitability index

1 Introduction

Although the standard of living in Africa has improved considerably in recent years, it is still the continent with the lowest level of economic and institutional development, hosting over 70% of the least developed countries (UNCTAD, 2022). Moreover, despite economic growth, not all countries and regions benefited equally from it (see Alesina et al., 2021; Fuje and Yao, 2022; UNDP, 2019). Economic research has tried to identify the causes for Africa's relatively low and uneven economic progress and one important explanation is the impact of the European colonial legacy on Africa's development (Michalopoulos and Papaioannou, 2020). Important scientific work has already shown that, for example, the establishment of sustainably profitable or harmful exploitative institutions (see Acemoglu et al., 2001, 2002), the slave trade from Africa to other continents (see Nunn, 2008), or artificially partitioning African ethnic groups into different countries (see Alesina et al., 2011; Michalopoulos and Papaioannou, 2016) had significant impacts on Africa's economic performance.

In this thesis, two aspects of the colonial legacy in Africa are going to be examined more deeply: On the one hand, forced labour as a type of exploitative institutions and, on the other hand, investments in human capital and infrastructure. Both features of European presence can be found in the Belgian Congo since the company Lever Brothers, now known as the multinational company Unilever, settled there to produce palm oil which they required for their soap production. In 1911, Lever Brothers agreed with the Belgian government that their subsidiary Huileries du Congo Belge (HCB) could harvest palm fruits in five concession areas and then process them into palm oil (Port Sunlight Village Trust, 2022; Fieldhouse, 1978, pp.497-502).

The same agreement also included that HCB was obliged to invest in infrastructure, health facilities, and education. William Lever, the head of Lever Brothers, who was known for his philanthropic and paternalistic views regarding his businesses and employees, wanted to launch a business that was beneficial for both the Congolese people as well as for Lever Brothers (see Henriet, 2021, pp.37-38; Fieldhouse, 1978, p.505; Loffman and Henriet, 2019, pp.4-5). However, he and HCB had to realize that the Congolese people were quite unwilling to work for them, mainly due to low wages and bad experiences in the preceding Leopold II era. HCB was not able to pay higher wages, mainly because of emerging international competition. Nevertheless, HCB wanted to continue its business

in the Congo and therefore forced the Congolese people to work for them so that the company could achieve its goals. Fruit cutters were coerced to pay high taxes which they could only pay by working for HCB. If they refused to work, they were whipped, persecuted or their wives were taken captive. Children and women also had to do hard work (see Henriet, 2021, pp.37-38; Fieldhouse, 1978, p.516; Tierney, 2021, p.17).

Considering all this information, it is unclear to what extent the intervention of HCB impacted the Congolese people living in the concession areas in the long run (Tierney, 2021, p.20). On the one hand, the influence could be positive since HCB invested in infrastructure, health, and education (see Dell and Olken, 2020; Huillery, 2009; Jedwab and Moradi, 2016; Jedwab et al., 2022). On the other hand, economic research has shown that colonial influences such as forced labour could lead to comparatively bad outcomes (see Lowes and Montero, 2021; Acemoglu et al., 2005; Dell, 2010). Therefore, this thesis tries to examine the long-term effects HCB could have had on economic, educational and health variables by comparing the respective data of those who live within the concession areas and those who live outside of it.

The five concession areas that were granted to HCB consisted of five circles with a radius of 60 km and were chosen according to their presumed abundance of palm trees (Fieldhouse, 1978, p.500). Therefore, the borders of those circles were drawn in an exogenous way, meaning that these borders did not align with natural or cultural borders. Taking this information into account, a geographic regression discontinuity design is used to compare observations inside and outside the circles within a specific range so that the effect of the HCB intervention in those concession areas can be estimated. The running variable is the distance to the border and the cutoff is the border itself. The dependent variables are obtained from the Demographic and Health Survey (DHS) data for the years 2007 and 2014. The DHS surveys include information on individual wealth (DHS wealth index), education (total years of education and literacy), and health (heigth/age percentile) from different locations all over the country. Additionally, sociological, geographical, and colonial variables (population density, temperature, elevation, colonial missions etc.) are used as covariates and for balance checks.

The main results of this regression discontinuity approach show that the majority of coefficients are positive but lack significance. Only at a manually chosen bandwidth of 30 km, the estimates for years of education and the height/age percentile of women are slightly

significant. However, this significance does not withstand all the robustness checks and Regression Discontinuity (RD) specifications such as different kernels, a quadratic polynomial or a donut-hole analysis that accounts for random displacement of DHS cluster locations. In contrast, most of the coefficients remain insignificant when several robustness checks are applied. Moreover, the same RD approach is also employed for different age cohorts to check whether potential effects have vanished over time. Apart from the estimates for years of education at a bandwidth of 30 km, almost all other coefficients for all age cohorts are insignificant. Having five concession circles, potential heterogeneous effects among these circles are also examined. Due to a low number of DHS cluster locations, this is only possible for the circles called Lusanga and Alberta, for which the majority of coefficients are also insignificant.

Noting that most of the coefficients lack significance, the logical conclusion would be that there has been neither a significant positive nor a negative effect. However, this interpretation may be flawed since there are methodological and context-related caveats. The current default method for RD designs by Calonico et al. (2017), which also provides the optimal bandwidth selection, cannot be applied. The reason for that may be that all the observations (a total of 8015) are divided into 155 DHS cluster locations, which is why the running variable is not very continuous at the cutoff. Hence, the older and less developed optimal bandwidth selection procedure by Imbens and Kalyanaraman (2012) is used, along with several manually chosen bandwidths. To demonstrate potential differences between the RD method by Calonico et al. (2017) and the RD approach employed in this paper, particularly in terms of statistical inference, the wild bootstrap method for RD designs by He and Bartalotti (2020) is utilized, which yields similar robust confidence intervals and p-values as the RD method by Calonico et al. (2017). However, since this wild bootstrap method does not allow the inclusion of covariables and fixed effects, which are used in the main RD approach of this paper, only a tentative comparison is possible. Moreover, several context-related caveats make a causal interpretation difficult. For instance, aspects such as the existence of a relatively high level of migration, other concession areas in the vicinity of the five circles, and uncertainty about the continuation of forced labour or investments in infrastructure, health, and education even outside the palm oil concession areas after the 1940s do not allow for a definite conclusion about the absence of significant long-term impacts.

While most academic work studies the impact of extractive institutions or investments in human capital and infrastructures across countries or regions, there is little economic research to date that geographically examines those two features of colonial impact based on detailed individual cases.[1] There are only four related papers that also use geographic variation in exposure to colonial attributes that were limited to areas similar to concessions: Dell (2010) finds that the mining mita of silver and mercury in Peru had negative long-term effects on household consumption and health. Dell and Olken (2020) show that the Dutch sugar processing system in Java influenced household consumption, education, and infrastructure in a positive way. Méndez and van Patten (2022) discover that exposure to the banana concessions in Costa Rica is linked to higher levels of health, education, and prosperity. Lowes and Montero (2021) take a close look into the rubber extraction concession in the Congo Free State from 1892 until 1906 and they impressively demonstrate that forced labour, indirect rule, and violence in those rubber concessions led to long-lasting negative effects on wealth, education, and health. Their work can also be regarded as an exemplary academic paper that serves as an orientation for this thesis.

Although there were many concessionary agreements in Africa during the colonial era (see Michalopoulos and Papaioannou, 2020), the work by Lowes and Montero (2021) is the first one that examines the effects of such a colonial event in Africa. Hence, the scientific contribution of this thesis is the presentation and empirical investigation of a second set of concession areas, whose results can be compared with previous findings. In particular, examining these concession areas also helps to understand which fundamental factors lead to persistent negative or positive effects on economic and societal outcomes. Therefore, it also contributes to the comprehensive literature on long-term determinants for economic growth in Africa by illustrating whether forced labour or investments in infrastructure and human capital are important channels for persistent impacts on economic wealth, education, and health.

This master thesis is structured as follows. Section 2 comprises the historical context with a special emphasis on HCB's intention to invest in infrastructure and human capital as well as the use of forced labour. Section 3 presents a short theoretical background about forced labour and extractive institutions on the one hand, and about investments

[1] Michalopoulos and Papaioannou (2020) provide a good overview of economic research that has been conducted with regard to the influence of colonial legacy in Africa.

in infrastructure and human capital on the other hand. This chapter also contains a subsection on the related empirical literature. Section 4 describes the data and the estimation strategy, while Section 5 presents the main results. Section 6 will discuss methodological and context-related caveats. The conclusion can be found in Section 7.

2 Historical Background

2.1 History of Huileries du Congo Belge

The company Lever Brothers, now called Unilever and known for food, detergents and hygiene products such as Knorr, Cornetto, Magnum, Domestos, Axe, and Dove (see Unilever, 2023), was already a globally acting soap manufacturer by 1900. The firm's success was partly based on a new soap production process that utilized vegetable oils instead of animal fats, preventing the soap from becoming rancid (Lewis, 2017, p.62) However, key components of vegetable oils such as copra and palm oil needed to be imported from regions such as Africa, Southeast Asia, and the Pacific region. As the global market for these oils became more competitive due to the rising demand for household products and industrial lubricants, William Lever, the head of Lever Brothers, feared that the increased demand for copra and palm oil could jeopardize the company's soap production because of the growing scarcity of these crucial ingredients (Fieldhouse, 1978, p.452)

To address this issue, Lever Brothers wanted to establish its own palm oil production, which could secure a stable supply for the company and influence the global market prices for copra and palm oil (Tierney, 2021, p.11; Byrne, 2016, p.406). Consequently, Lever entered into an agreement with the Belgian government in 1911, granting Lever Brothers' subsidiary, Huileries du Congo Belge (HCB), the rights to harvest palm fruits and establish plantations in five concession areas, each shaped like a circle with a radius of 60 kilometers. The Belgian government had initially approached Lever in 1909, exploring the possibility of Lever Brothers' involvement in its colony. On the other hand, Lever had also previously attempted to identify suitable locations in West Africa for his own palm oil production (Loffman and Henriet, 2019, p.3; Fieldhouse, 1978, p.499).

The five concession areas were called Alberta, Elisabetha, Brabanta, Flandria, and Lusanga, and were formed around the localities of Bumba, Basongo, Barumbu, Igende, and Leverville. Figure 1 shows the location of those five circles in the Congo. According to the

convention of 1911, Lever Brothers was first allowed to choose 75,000 hectares of vacant land in each circle as leasehold land. If certain requirements such as a sufficiently high amount of production output and investments in infrastructure, health, and education were met, the leased land areas could be expanded to 750,000 hectares in total and up to 200,000 hectares in one circle. By 1941, parts of the leased land could have finally been converted into the company's own property if the whole venture had been successful (Henriet, 2021, p.40; Fieldhouse, 1978, pp.503-505).

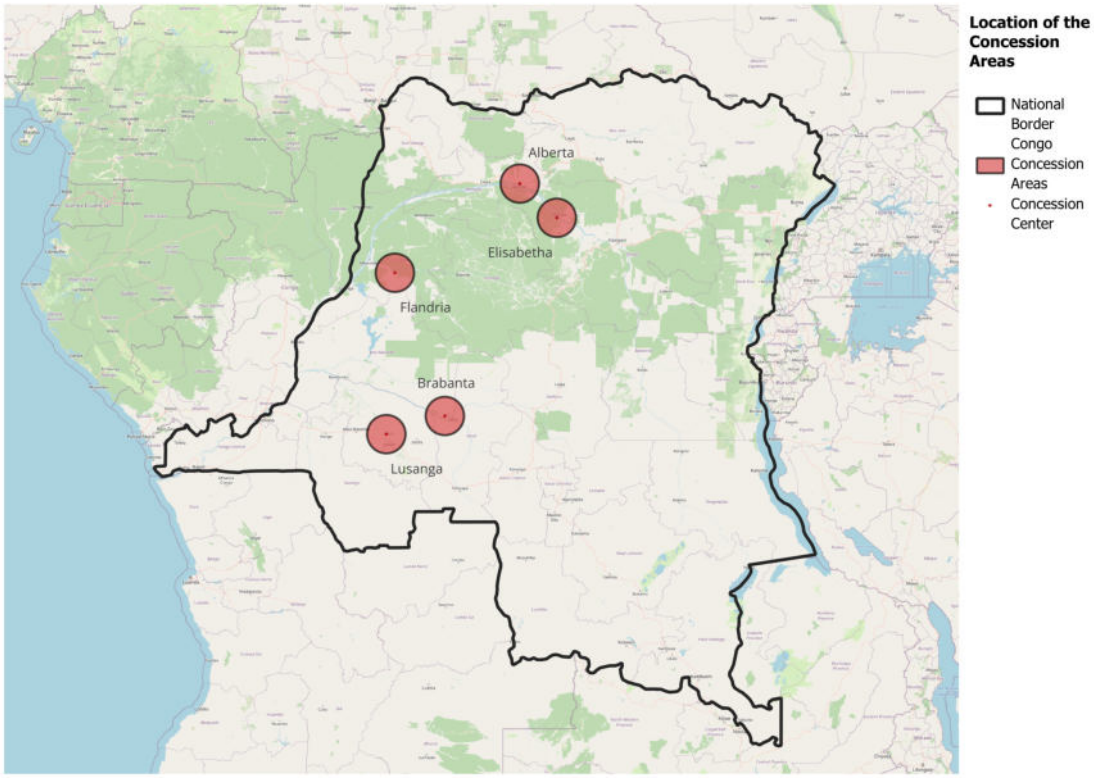


Figure 1: Location of the Concession Areas in Belgian Congo

Notes: This figure shows the location of the palm oil concession in the Belgian Congo according to the convention of 1911. Own figure created with the software QGIS.

In the beginning, the principal idea was to collect palm fruits from natural palm tree groves in those leasehold areas within the concession areas. However, extracting palm oil from these five areas posed several challenges, including a lack of infrastructure, high transportation costs due to the long distance to the coast, and a shortage of native workers. Moreover, uncertainty about the availability of a sufficient number of palm trees and undefined native land rights posed further challenges, making it more difficult to turn Lever Brothers' endeavour in the Congo into a successful business (Fieldhouse, 1978, p.507). Nevertheless, William Lever decided to invest in the Belgian Congo, possibly

because the Belgian government granted him additional monopolistic rights that he could not obtain from other colonial governments. These monopolistic rights ensured that neither other companies nor the natives were allowed to harvest and trade palm fruits and palm oil in the leased areas of HCB. This presented a significant constraint for the local population since palm fruits were also used for medical and cultural purposes (Seibert, 2016, p.140ff; Wilson, 1968, p.167ff).

In the following years until 1930, Lever Brothers invested a substantial amount of money into HCB, ensuring that the requirements of the convention were already met by 1925. Initially, Lever Brothers had to invest one million pounds in the Congo, which constituted one-sixth of HCB's total capital worldwide at that time. This amount was greatly exceeded, resulting in a total invested capital of approximately 3 million pounds by 1930. However, the first profits were not realized until 1918 to 1920, totalling about 70,000 pounds, followed by a loss of 330,000 pounds in 1921. Small profits were achieved from 1924 to 1929. The return on capital may not have been acceptable to Lever Brothers, but William Lever continued investing in HCB, hoping that the entire endeavour would eventually become successful (Fieldhouse, 1978, pp.507-509).

In the early years of the 1930s, HCB experienced substantial losses, primarily due to high production costs and a low world market price for palm oil. More efficient palm oil plantations in Nigeria, the Cameroons, Sumatra, or former British Malaya could offer palm oil at a lower price than HCB. Additionally, welfare costs, transportation expenses, and the inefficient collection of dispersed palm fruits in wide areas contributed to the company's increased production costs. Consequently, a significant change was implemented in the 1930s, gradually initiating the establishment of more efficient palm tree plantations, which would first supplement and then replace the practice of collecting palm fruits (Fieldhouse, 1978, pp.521-523).

From the 1940s onward, the company made relatively large profits until Congo's independence in 1960. As stated in the convention of 1911 and subsequent agreements, the company acquired 350,000 hectares of freehold land in 1956, of which only 60,000 hectares were ultimately used for plantations (Fieldhouse, 1978, p.537). In the years that followed, the company's profitability declined due to various factors including political and economic problems, violent conflicts, and international competitive pressure. As a result, palm oil production gradually became restricted to the domestic market, with only some

plantations remaining in the northern region of the country (Fieldhouse, 1978, p.533ff; Ergo, n.d. p.190ff).

Today, three plantations of the company, which is now called Plantations et Huileries du Congo (PHC), are still in operation: Boteka (formerly part of the Flandria circle), Yaligimba (formerly part of the Alberta circle), and Lokutu (formerly part of the Elisabetha circle). The total concession area, which still belongs to PHC, is about 107,000 hectares, of which only 25, 000 hectares are used for plantations. PHC was finally sold by Unilever in 2009. Several development banks from Western countries invested in the successor companies to enable the plantations to continue operating. However, these investments have faced notable criticism due to ongoing land rights conflicts, environmental pollution, low wages, and human rights violations associated with the current plantations (see RIAO-RDC et al., 2021; Currier, 2022; Currier and Mousseau, 2022).

2.2 Virtuous Intentions

The probably most well-known aspect of Lever Brothers' activities in the Congo from 1911 to 1940 is the coercion used by Huileries du Congo Belge (HCB) to make the local natives collect palm fruit for them. However, Lever had no intention of launching his enterprise in the Congo with the purpose of exploiting the land and its inhabitants. Instead, he sought to establish a business that would be mutually profitable for his company and the local population. Lever was already known in England for his paternalistic and philanthropic approach towards his employees. He had already created a model village where workers at his soap factory enjoyed various benefits, including high-quality accommodations and healthcare. In the same spirit, Lever intended to invest in education, healthcare, and infrastructure in the regions where he could establish his subsidiary. The investments in human capital and infrastructure were also desired by the Belgian government and were documented in the convention of 1911. Thus, each circle should have been equipped with roads, railways, telegraphs, a hospital and educational facilities operated and financed by Lever Brothers (Fieldhouse, 1978, p.505ff; Seibert, 2016, p.148; Tierney, 2021, p.26).

Lever regarded the Congolese as willing workers who, if fairly compensated and provided with proper medical care and housing, could contribute to the success of a model colony. The importance of the Congo project to the head of Lever Brothers, and his desire to turn it into a similarly successful venture as his model village in Port Sunlight, is evident in the substantial financial investments made in the Congo, even though the project incurred

significant losses in its initial years (see Henriet, 2021, p.28ff; Lewis, 2017). Activists and the Belgian authorities were pleased to have an entrepreneur who also cared about the well-being of his employees, because on the one hand, the colony could be developed economically, and on the other hand, the previous years under the reign of terror of the Belgian King Leopold could be left behind. Until 1908, the Belgian king owned the Congo as a private colony, in which private companies obtained concessions for exploiting resources such as rubber under extreme violence. It is estimated that up to 10 million people died, which was about half the population of the Congo at that time (see Henriet, 2021, p.35ff; Hochschild, 2012).

2.3 Forced Labour, Intimidation and Violence

It was obvious that there could have been challenges in recruiting enough workers even before the establishment of HCB, as the areas in which the circles were located, especially the territory of the Lusanga circle, were not densely populated. There was a general labour shortage in the Congo at that time, which also affected other companies. Furthermore, the role of a fruit cutter was considered heavy and dangerous slave work since cutters often had to walk into remote areas with low palm density and then climb tall palm trees. Additionally, Congolese labourers were reluctant to work for Europeans due to the negative experiences during the time of the Congo Free State under the rule of King Leopold II (see Fieldhouse, 1978, p.502ff. Boss, 2014, p.29; Seibert, 2009, p.103).

The colonial charter of 1908 also banned forced labour in the Congo to prevent the atrocities of the Congo Free State from happening again. Nevertheless, HCB resorted to forced recruitment because otherwise the agreed goals of the 1911 convention and also William Lever's goal of establishing a successful subsidiary company could not have been achieved. However, in the same convention of 1911, it was stated to take stimulating measures so that the natives would enter into paid labour relations. Forced labour was also justified on the premise that these Congolese were generally regarded as lazy workers who needed to get accustomed to wage labour. Due to international competitive pressure, HCB couldn't afford to pay higher wages to encourage natives to enrol themselves voluntarily into work for the company (see Henriet et al., 2017, p.41; Fieldhouse, 1978, p.512 ; Henriet, 2021, p.105; Marchal, 2008).

Two methods were commonly employed to recruit native Congolese: The first method involved the imposition of a poll tax that had to be paid in cash and not in kind. To meet

this obligation, the natives had to either sell their property or engage in wage labour. The tax was calculated in such a manner that the native workers were forced to spend nearly their entire earnings on it unless they were willing to work longer than originally agreed upon in the employment contract. Failure to pay the tax could result in being imprisoned or sentenced to forced labour again. The second method involved applying official pressure or even using bribery to compel the chiefs of the respective villages to provide their own people as labour force. In many cases, these chiefs were remunerated in return or received a portion of the wages earned by the workers they sent. For this purpose, HCB recruiters were often accompanied by officials to assist with recruitment. However, the colonial government often viewed this accompaniment critically and, as a result, it was not officially permitted (see Fieldhouse, 1978, p.513; Henriet, 2017, 2021, p.104).

The direct use of violence and intimidation can be described as follows. According to various accounts, workers were beaten, whipped, or arbitrarily arrested to increase their productivity or when they violated the rules. HCB also used its own mercenaries and sentries who supervised workers and often acted for their own benefit by demanding bribes and sexually abusing employees' wives. Military posts were established to retrieve escaped workers. Men's wives were captured when tax quotas could not be paid. Labourers were often malnourished and housed in poor accommodations. Medical care was inadequate, although HCB, with its philanthropic obligations, should have provided it. The wages were often far too low, and when workers were paid according to palm fruit harvested, they were often unable to meet the required quotas because they were set far too high. Labour contracts were automatically renewed, which was accepted without resistance because they feared the Europeans. There are reports indicating high death rates at some work sites because of the extremely hard work, diseases and poor sanitary conditions. Women also contributed to HCB's business by carrying the heavy palm fruits although they were not officially employed by HCB. The expectation that women would engage in strenuous carrying tasks was grounded in the belief that such labour was a customary element of African culture. Children did heavy work, but they were paid less for it (see Marchal, 2008; Henriet, 2021, p.95ff. Fieldhouse, 1978, p.509ff. Seibert, 2016, p.152ff. Henriet et al., 2017; Henriet, 2017; Seibert, 2009).[2]

[2] All these sources are based on reports that can be found in several archives. Since it was not possible to visit the archives as part of this thesis, only these secondary sources are cited.

3 Theoretical Considerations and Literature Review

3.1 Extractive Institutions

In economics, a good way to explain the use of strategies like forced labour and the enforcement of monopolies is the concept of extractive institutions. In general, institutions primarily refer to a set of laws, rules, and social norms that help to establish a regulatory framework for a free market economy and prevent humans from acting in a detrimental way for society. Property rights can be regarded as a concrete example of institutions because they ensure that individuals can buy, retain and derive revenues from their property without fearing that it could be easily taken away from them. This security in turn also encourages the owners to invest even more in their property. On the contrary, extractive institutions have often been used to exploit resource-rich areas for the advantage of the elites. Forced labour or monopolistic rights as measures used by HCB in Belgian Congo are good examples since almost only HCB benefited from the extraction of palm oil. The natives did not earn a just wage nor were they able to participate in the trade in palm oil products, which is why they could not derive any advantages from the palm oil business. Instead, they were forced to work for minimal wages that did not increase their wealth. Furthermore, mighty elites tend to strengthen and maintain extractive institutions so that they can benefit from them in the long term. Acemoglu et al. (2001, 2002) have shown that there is a strong linkage between the level of early inclusive or extractive institutions and contemporary wealth among different countries (see Todaro and Smith, 2020, p.62f; Roland, 2016, p.108 and p.176ff; Acemoglu and Robinson, 2017; Leschke, 2015). As described later in Section 5.4, the violation of native land rights or egoistic chiefs not acting in favour of their people may explain potential negative persistent impacts in the palm oil concession areas because, for example, fewer investments in public goods were undertaken, hindering sustainable improvements in socio-economic outcomes. A more detailed explanation of extractive institutions can be found in Section C.1.

3.2 Effect of Health, Education and Infrastructure on Development

Since Lever Brothers had to make investments in education, health, and infrastructure, the hypothesis is that these investments led to economic growth and an improvement in the health and educational situation in the concession areas. Many textbooks stress the importance of infrastructure, health and education for socio-economic development since, for instance, a better infrastructure reduces transport and production costs as

well as facilitates access to new markets. Better education does not only lead to better-paid and more productive jobs, it also enables workers to adopt new technologies and promotes innovations. Better healthcare increases the productivity of workers since they are less affected by diseases and thus, for example, miss fewer working days. While better education and medical care are desirable on their own, these examples illustrate the various ways in which infrastructure, education, and healthcare contribute to overall development (see Perkins et al., 2013, p.257ff; Janvry and Sadoulet, 2016, p.594ff; Todaro and Smith, 2020, p.390ff; Roland, 2016, p.422ff; Meier and Rauch, 2005, p.217). This positive linkage is also supported by several studies (see Ridhwan et al., 2022; Benos and Zotou, 2014; Valero, 2021; Foster et al., 2023). A more detailed description about the impact of health, education and infrastructure on development outcomes can be found in Section C.2.

3.3 Related Literature

As mentioned in the introduction, there are four papers which also examine the impact of colonial presence in areas similar to concession areas, such as those established by Lever Brothers in the Congo. The most important scientific work is the one published by Lowes and Montero (2021) since they have investigated whether the Abir and Anversoise concession areas, which existed from 1892 to 1906 in northern Congo under King Leopold II's oppressive rule, had persistent effects on development outcomes. In these two concession areas, rubber was harvested by coercing native people to work for private rubber companies. The working conditions and atrocities endured by the natives seemed even worse than those committed by HCB just ten years later. Lowes and Montero (2021) also applied a spatial regression discontinuity design and used the same DHS data from 2007 and 2014 as well as the same dependent variables as this thesis. According to their findings, all the estimates display lower values for economic wealth, education, and health, indicating that the exposure to these rubber concession areas had severe, long-lasting effects on the well-being of the natives living in those regions. Due to the similarity to the topic of this thesis, their work serves as an orientation regarding the RD specifications, data and variable modification.

The second paper by Dell (2010) explores the long-lasting effects of the mining mita in Peru, a labour recruitment system that lasted from 1573 to 1812 and also resorted to forced labour for silver mining. The results of this RD approach reveal a 25% reduction

in household consumption and a 6-percentage-point increase in child stunting in subjected districts. In the third paper, Dell and Olken (2020) investigate the long-lasting impacts of the Dutch sugar cultivation system in Java, Indonesia. Between the 1830s and 1870s, the Dutch colonial government compelled Javanese villagers to cultivate sugar, which was then processed in nearby colonial factories. Unlike the other three papers, the authors don't use an RD design as the only main estimation strategy. Instead, they compare areas where a sugar processing factory was established with potential counterfactual areas in which the Dutch could also have set up such factories to identify the effect of this sugar cultivation system. The empirical analysis of this methodology reveals that areas close to the sugar factories experienced improved infrastructure, better educational opportunities, and greater wealth. The latest paper published by Méndez and van Patten (2022) examines the enduring effects of the banana concessions granted to the United Fruit Company from 1899 to 1984 in Costa Rica. The authors demonstrate that exposure to the banana concessions is linked to overall better well-being in terms of health, education, and economic wealth. All these four papers also explore mechanisms that provide explanations for the detected persistent effects. However, these mechanisms are discussed in Section 5.4, as they may also represent potential channels regarding the impact of HCB's palm oil concessions on the development outcomes examined in this thesis.

Apart from the four papers introduced in the preceding paragraphs, there is also other related scientific work which does not explore the effects of colonialism in Africa by only focusing on delimited areas similar to concessions: Starting with the investments in human capital by colonizers in Africa, Christian missions were primarily associated with the provision of education and health services during the colonial era (see Frankema, 2012; Cagé and Rueda, 2020). Therefore, several studies use this relation and find that proximity to missions can be linked to positive effects on outcomes such as education (Nunn, 2014; Wantchekon et al., 2015; Alesina et al., 2021), health (Cagé and Rueda, 2020; Dimico, 2014), and political participation (Cagé and Rueda, 2016; Henn et al., 2023). Jedwab et al. (2022) provide a comprehensive overview of all studies related to the impacts of missions during the colonial era. However, Jedwab et al. (2022) also point out that all these studies probably suffer from an endogeneity problem since the locations were established in geographically more advantageous locations, which is why all the reported effects could be weaker. Without relying on colonial missions for the empirical analysis, Huillery (2009) explores the influence of colonial investments by the colonial state in health and education from 1910 to 1928 in French West Africa. The applied OLS specifications

show that a higher number of employed teachers in the colonial era is linked to higher school attainment of children in 1995 and that a higher number of employed doctors can be related to a decrease in the number of children suffering from stunted growth in 1995.

Turning to infrastructure, the amount of constructed roads and railroads in the African colonies was quite limited since they were mainly built to extract resources and therefore did not connect major cities. However, several studies establish a positive connection between transport infrastructure and development (Michalopoulos and Papaioannou, 2020). Cross-country studies found that colonial transportation has persistent effects on market access, urbanisation and school attainment (Jedwab and Moradi, 2016; Jedwab and Storeygard, 2022). Case studies examining the impact of colonial infrastructure in countries like Kenya, Nigeria and South Africa similarly report positive influences on education, urbanization and economic development (Jedwab et al., 2017; Okoye et al., 2017; Herranz-Loncán and Fourie, 2018; Bertazzini, 2022).

Lastly, more economic papers address forced labour and oppression in the colonial era in Africa, but they do not empirically quantify its effects on development outcomes as done in this thesis or by Lowes and Montero (2021): For instance, some studies estimate the benefit that colonial powers gained by resorting to unpaid forced labour (van Waijenburg, 2018; Archibong and Obikili, 2019; Tadei, 2020) Other researchers examine how attitudes toward more democratic or authoritarian views as well as the appreciation of freedom can be linked with indirect rule or oppression (Lechler and McNamee, 2018; Garcia Ponce and Wantchekon, 2011; Acemoglu et al., 2014). In the context of forced labour in the Belgian Congo, Juif and Frankema (2018) describe how the mining sector initially resorted to forced labour but later experienced an extraordinary rise in wages. To attract natives to remain longer in the mining industry, the mining sector offered a high degree of social services.

4 Empirical Approach

4.1 Data

The main data for investigating potential long-term effects of the palm oil concession areas on socio-economic outcomes stem from the Demographic Health Survey program, which conducted two surveys in 2007 and 2014 in the entire Congo. In addition, these data are assigned to inner territory or the outer territory of each concession by using

maps showing the borders of HCB's five palm oil concessions. Figure 2 displays the DHS survey clusters that are located inside and outside the concession areas within a maximum range of 220 km to the respective circle centre. The distance from the DHS clusters to the respective concession boundary as the running variable for this RD setup was measured in the software QGIS. The complete sample consists of 8015 observations, which are divided into 155 clusters. For a bandwidth of 60 km, the sample contains 3548 observations, divided into 69 clusters. The main dependent variables for wealth are the wealth index, a categorical variable ranging from 1 to 5, where 1 corresponds to the poorest quintile and 5 to the richest quintile, and the logarithm of the wealth factor score, an index calculated by using a principle component analysis for the possession of assets for each individual. For the education category, the two outcome variables are the total number of years of education for each individual and literacy, an indicator of whether somebody can read or not. For health, the height/age percentile, a ratio of each individual's height and age, for which the respective percentile is calculated, is taken as the dependent variable for women and children. Summary Statistics for the dependent variables within a bandwidth of 60 km and 30 km can be found in Table D1.

Moreover, several sociological, geographical and colonial variables are used for conducting the balance checks in Section 4.3. One part of these variables was already attached to the DHS data (e.g., level of urbanization, population density in 2015 or spatial economic output). Since Lowes and Montero (2021) also examined concessions that existed about 10 years earlier in the entire Congo, almost all other variables were taken over from their scientific work, after checking their suitability for the empirical analysis of this paper. The geographic characteristics, adopted from Lowes and Montero (2021), are presented as means of 20 km by 20 km grid cells, while the geographic characteristics attached to the DHS data are mostly based on buffer zones with a radius of 2 km for urban areas and 10 km for rural areas. Unfortunately, not all DHS clusters can be linked to colonial variables, as these are assigned to specific tribes and data is only available for a few tribes. However, the balance checks are carried out with the available data and interpreted accordingly. Figure B4 shows each tribe's area that overlaps with the palm oil concessions. The variable for missionary stations in 1924 was newly created since its original values differed from those used by Lowes and Montero (2021). The locations of the missions can be found in Figure B1 in the Appendix. Moreover, the variable inside AA was also newly created to indicate which of the DHS clusters are located in the area of the former rubber concession examined by Lowes and Montero (2021). More detailed information about the

DHS variables and the variables used for the balance check, including the original source of each variable, can be found in Appendix A.

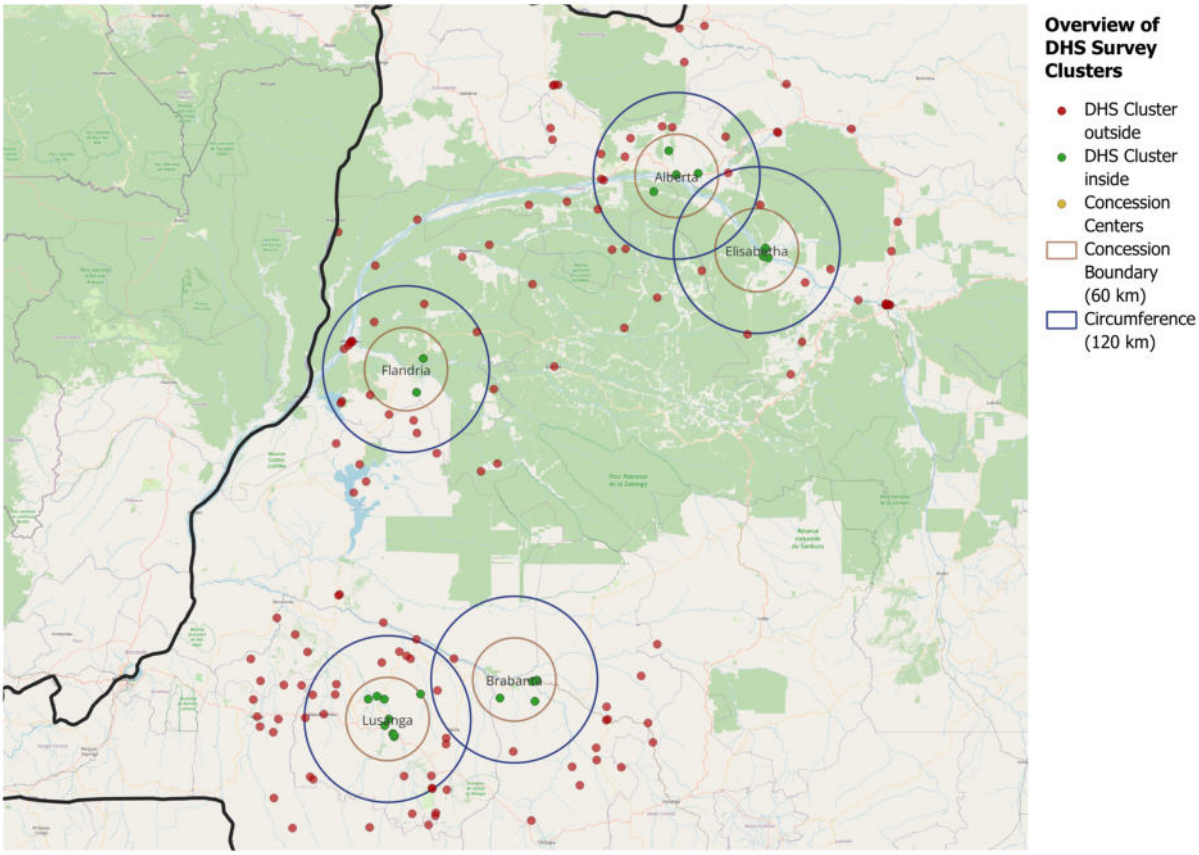


Figure 2: Overview of DHS Clusters inside and outside the Concession Areas

Notes: This figure shows DHS cluster locations around and within the palm oil concessions. The circles outlined in brown represent the concession boundaries, while the circles outlined in blue indicate auxiliary circles/circumferences with a radius of 120 km. Own figure created with the software QGIS.

4.2 Regression Discontinuity Design as Estimation Strategy

To estimate the causal effect of the exposure to the palm oil concession on development outcomes, the following equation for regression discontinuity designs is used:

$$\begin{aligned} y_{ivc} = \alpha + \gamma_1 ConcessionArea_{iv} + \gamma_2 DistanceBorder_v \\ + \gamma_3 ConcessionArea_v * DistanceBorder_v + \beta X_i + \Phi_c + \varepsilon_{ivc} \end{aligned} \tag{1}$$

In this equation, y_{ivc} denotes the outcome variable for each individual i in cluster or village v and concession c, $ConcessionArea_{iv}$ is an indicator equal to 1 if village v is located inside the concession areas and 0 otherwise, X_i is a vector of covariates including gender, age, age squared, and survey year, and Φ_c represents the nearest concession fixed effects.

The expression $\gamma_2 DistanceBorder_v + \gamma_3 ConcessionArea_v * DistanceBorder_v$ is the typical RD polynomial with its interaction term and the running variable $DistanceBorder_v$, which measures the distance from each DHS cluster to the next border point. This RD polynomial controls for smooth function of the geographic location of DHS cluster v and allows a separate regression estimation on both sides of the cutoff. Consequently, a non-parametric local liner RD estimation is applied, using the package "rddapp" by Jin et al. (2023) in R. Unfortunately, the current default methodology for RD estimations introduced by Calonico et al. (2017) is not applicable to this setting because there are probably too few DHS clusters and thus too little variation in the variable $DistanceBorder_v$. This is also the reason why the older optimal bandwidth choice procedure by Imbens and Kalyanaraman (2012) is employed and not the more advanced one by Calonico et al. (2017). The choice of the exact RD methodology, along with the selection of the bandwidth and its implications for statistical inference, will be discussed in more detail in Section 6.1.

In addition to the optimal bandwidth choice, results for the bandwidth of 60 km, which covers the full concession areas, and for 30 km are presented. As the default option, a triangular kernel is used in this RD estimation, and standard errors are clustered at the DHS cluster level. Apart from the utilized estimation command in the respective statistical software and the corresponding bandwidth choice, all the specifications, including the choice of covariables in the regression itself, are the same as in Lowes and Montero (2021). Various robustness checks involving different bandwidths, kernels, RD polynomials, or excluding clustered errors and fixed effects can be found in Appendix E. The coefficient of interest is γ_1 since it measures the effect of being located inside the concession on contemporary wealth, education and health indicators. The assumption behind this estimation strategy is that the exposure to the concession areas is the only aspect that changes abruptly at the concession border and that all other characteristics are approximately equal so that the people living inside the concession areas can be compared with those living outside of it.

A threat to this assumption is that the centers and boundaries of the circles may have been chosen in such a way that certain characteristics predominate within the circles. These characteristics could potentially explain effects on current development indicators, rather than investments made in the circles or the use of forced labour. The intentional selection of circle centers following a specific pattern cannot be confirmed by the available historical sources. The localization of the circles was only predetermined by inaccurate information

from government officials. Expeditions to explore the circles were inadequately conducted in only three circles. Two of the circles could not be inspected at the time of the contract conclusion in 1911 because the corresponding area had not yet come under the control of the Belgian administration. The imprudent selection of sites for the circles also resulted in some circles lacking a significant number of natural palm trees (Fieldhouse, 1978, p.500). The assumption that the boundaries could coincide with any natural or political boundaries is unlikely due to the artificial circular shape of the concession areas. One criterion by which the circles were probably selected is that all centers are located along a river. However, this should not affect whether certain characteristics suddenly change at the circular boundary of the concessions areas. Table 1 and Table D2 therefore present balance checks with summary statistics and RD estimates for variables that should not change abruptly at the boundary. The detailed results of this balance check are presented in the following Section 4.3, indicating that balance is found for a range of variables.

Another precondition that must hold is that there was no selective sorting of villages or their inhabitants to the other side of the border. Regarding the location of villages, selective sorting is unlikely, as this would imply that natives could have influenced the border's shape and chosen whether they want to belong to the concession areas or not. Historical records indicate some plans by HCB to move villages inside the circles, but these plans were not implemented (see Henriet, 2021, p.80ff; Marchal, 2008, p.23ff). Nevertheless, there are some caveats that pose serious threats to interpreting the results of this RD strategy as causal. Migration, adjacent concession areas, and uncertainty on whether forced labour or virtuous investments were limited only to the concession areas play a major role when drawing conclusions from this empirical analysis. These drawbacks are discussed in more detail in Section 6.

4.3 Balance Check

Since it is an important assumption for an RD analysis that all other observable characteristics except the treatment itself are approximately equal within and outside the circles, Table 1 shows RD coefficients for important sociological, geographic and colonial factors as balance check. The respective standard errors of those RD regressions are clustered at the DHS cluster level for sociological variables (gender, age), at the territory level for geographic variables or at ethnic group level for colonial variables. Territories are administrative units in each province of the Democratic Republic of the Congo. The regressions

also include concession fixed effects and use a triangular kernel.

Table 1: Balance Check for RD Coefficients

	Balance Check RD Estimates 60km			Balance Check RD Estimates 30km		
	RD Coefficient	Standard Error	P-Value	RD Coefficient	Standard Error	P-Value
DHS Covariables						
Gender	-0.032	0.041	0.428	0.005	0.052	0.920
Age	1.410	1.099	0.199	1.387	1.236	0.262
Age squared	113.972	66.688	0.087	132.304	78.757	0.093
Dummy Survey Year	-0.167	0.338	0.621	-0.287	0.334	0.390
Population Density 2015	-609.974	347.832	0.079	86.715	69.258	0.211
Level of Urbanization	-0.042	0.042	0.315	0.158	0.165	0.339
Spatial Economic Output	-2.561	8.056	0.751	12.446	12.770	0.330
Travel Times 2015	11.292	48.278	0.815	2.202	127.733	0.986
Mean Temperature 2015	0.026	0.144	0.856	-0.154	0.178	0.388
Enhanced Vegetation Index	-3.470	4.197	0.408	-4.400	3.676	0.231
Further Geographic Covariables						
Elevation	-153.871	118.252	0.193	-24.011	106.175	0.821
Ruggedness	-37.631	22.740	0.098	-26.545	28.662	0.354
Land Suitability	-0.023	0.048	0.627	-0.003	0.040	0.935
Average Rainfall	3.360	6.489	0.605	10.204	12.382	0.410
Standard Deviation Rainfall	0.312	0.391	0.424	0.836	0.597	0.162
Malaria Suitability	0.053	0.603	0.930	-0.552	0.759	0.467
Tsetse Fly Suitability	0.033	0.016	0.032	0.054	0.026	0.038
Road Density	12.696	35.550	0.721	83.122	54.009	0.124
Distance to Kinshasa	-13.395	25.425	0.598	-13.927	40.530	0.731
Distance to Coast	76.204	27.718	0.006	135.315	33.121	0.000
Colonial Covariables						
Inside AA	-0.134	0.216	0.534	0.049	0.223	0.824
Missionary Stations 1924	0.004	0.189	0.982	0.035	0.283	0.902
Colonial Population Density	0.613	0.296	0.039	0.654	0.032	0.000
Number Enslaved in 1000	18.549	14.797	0.210	9.854	9.870	0.318
Centralization	-0.022	0.291	0.941	0.040	0.142	0.776
Polygynous	-0.343	0.239	0.150	-0.197	0.190	0.299

Notes: Column 1 shows the RD coefficient of the corresponding variable for a bandwidth of 60 km. The associated standard errors and p-values are reported in columns 2 and 3. The same pattern is applied to a bandwidth of 30 km in columns 4 - 6. Standard errors are clustered at the DHS cluster level for sociological variables, at the territory level for geographic variables or at ethnic group level for colonial variables. The same RD estimation procedure as for the main results is used, incorporating fixed effects and a triangular kernel. More information on the variable definitions and data sources can be found in Appendix A.

As Table 1 indicates, the majority of those coefficients are insignificant, which means that the requirement of further balanced observable characteristics holds. However, three variables do show a significant RD estimate at a 5% level, but their relevance as an important precondition is questionable: The first variable, distance to coast is geographically predetermined in such a way that it may not be entirely clear to what extent this could affect this RD estimation on a more local level. As Table D2 in the Appendix shows, the difference in means for distance to the coast is only about six kilometers, and also all other covariables, such as the number of enslaved natives or the spatial economic output

for each DHS cluster, that may be related to the distance to the coast through, for example, potential trade routes, are insignificant.

The second variable, the tsetse fly suitability, is more important in the context of this paper because several reports mentioned by Marchal (2008) describe that many natives in HCB's concession areas suffered from sleeping sickness caused by pathogens that were transmitted by the tsetse fly. The variable tsetse fly suitability is again taken over from Lowes and Montero (2021) but its calculation is based on the TseTse suitability index (TSI) created by Alsan (2015). Historical climate data such as temperature and humidity are used to construct a measure for potential tsetse fly steady-state populations for different areas in Africa. Alsan (2015) then runs several regressions with the TSI as the explanatory variable to investigate whether the tsetse fly could also be a determinant for Africa's development. As a result, strong negative correlations between the TSI and human population density, advanced agricultural methods, and political centralization were found. However, the author also states that these results do not allow causal conclusions since, on the one hand, only a correlation is measured and, on the other hand, the TSI still may capture some tropical effects as it can only predict a potential tsetse fly population. Therefore, considering the balance of tsetse fly suitability as an indispensable precondition for this RD analysis may be too demanding. Moreover, the balance table includes some geographic and climate variables such as temperature, rainfall or the enhanced vegetation index that show insignificant RD estimates. Running the same RD regression with tsetse fly suitability as the dependent variable but using a rectangular kernel or excluding fixed effects results in insignificant coefficients. Since unbalanced variables cannot be easily incorporated into an RD regression as control variables to restore causal inference (see Cattaneo et al., 2022), this relevant aspect may require further investigation. For instance, using more precise data than the mean value of a 20 km by 20 km grid cell, as constructed by Lowes and Montero (2021), may shed a different light on this RD precondition.

The last unbalanced variable is the colonial population density. Again, this measure is not reliable, as only a few observations, mainly for the Flandria circle, are available within the 30 km bandwidth. These observations stem from the Ethnographic Atlas by Murdock et al. (1999), and thus, they are assigned to a few ethnic groups, as depicted in Figure B4. For the 30 km bandwidth, data for the ethnic groups Lele, Ekonda, Ngala, and Nkundo are available, covering 13 out of 34 clusters. When extending the bandwidth to 60 km, there is additional data for the Lokele and Pende ethnic groups, covering 26 out of 69

clusters. However, historical sources indicate that indigenous people were living in the concession areas, although they were very few (Fieldhouse, 1978, p.513).

One colonial variable not presented in the table is hereditary local headman selection, an indicator of whether the leader is chosen through family inheritance. For all ethnic groups within the 60 km and 30 km bandwidth, this indicator consistently holds the value 1. Due to this lack of variation, estimating an RD coefficient for this variable does not make sense. However, this still means that it is balanced on both sides.

Moreover, it is worth mentioning that historical records also highlight the role of Christian missions in the concession areas. On one hand, these missions advocated for locals, expressing concerns about the harsh working conditions and high death rates among the natives. On the other hand, missions played a crucial role in providing health and education (see Henriet, 2021, p.22; Tierney, 2021, p.28; Marchal, 2008, p.35 and p.56). Therefore, having a balanced variable that measures whether missions were close to the DHS clusters is important since it ensures that possible positive effects regarding health and education cannot be solely explained through the presence of missions inside the concession areas.

Finally, the variable inside AA indicates whether a DHS cluster is located inside the rubber concessions Abir and Anversoise that were examined by Lowes and Montero (2021). As mentioned in Section 3.3, they found out that the exposure to these two concession areas had persistent negative effects on all the outcome variables that are also examined in this paper. Since the RD coefficients for inside AA are insignificant, the precondition of balanced covariates is not violated by observations that were located inside Abir and Anversoise 10 to 20 years earlier.

5 Regression Discontinuity Results

5.1 Main Results

The effects of the exposure to the palm oil concessions on contemporary wealth, education and health outcomes, as estimated by Equation 1, are presented in Table 2. As can be seen, most of the coefficients are not significant, indicating that the concession areas with their characteristics do not affect the investigated dependent variables. Moreover, it is difficult to conclude whether the overall impact is positive or negative. For the wealth outcomes,

Table 2: Main Results

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.4979	0.1212	0.6765	−0.2374	−0.0603	0.1531
	(0.4792)	(0.3805)	(0.5300)	(0.2577)	(0.1869)	(0.2820)
Num.Obs.	2541	3548	1708	1254	3548	1708
Bandwidth	39.54	60.00	30.00	26.85	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.545	2.540	2.343	10.845	11.045	10.958
SD Dep. Var.	1.274	1.238	1.204	0.513	0.602	0.569
Total Number Clusters	49	69	34	25	69	34
Clusters within Con. Area	16	23	10	8	23	10

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.7369	1.2214	2.8910*	−0.0462	0.0820	0.2096
	(1.2030)	(1.2599)	(1.4625)	(0.1067)	(0.1142)	(0.1327)
Num.Obs.	3819	3544	1704	4312	3532	1701
Bandwidth	70.97	60.00	30.00	85.88	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.831	6.908	6.183	0.712	0.732	0.666
SD Dep. Var.	3.984	3.981	4.112	0.453	0.443	0.472
Total Number Clusters	75	69	34	85	69	34
Clusters within Con. Area	23	23	10	23	23	10

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.1036	0.0523	0.1801**	0.0503	0.0620	0.0414
	(0.0638)	(0.0719)	(0.0664)	(0.0797)	(0.0886)	(0.0607)
Num.Obs.	866	1190	578	1821	1042	527
Bandwidth	39.41	60.00	30.00	128.40	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.229	0.236	0.231	0.282	0.298	0.289
SD Dep. Var.	0.258	0.255	0.257	0.348	0.352	0.351
Total Number Clusters	49	69	34	116	69	34
Clusters within Con. Area	16	23	10	23	23	10

Notes: Standard errors are clustered at the DHS cluster level and reported in parentheses. The RD regressions include concession fixed effects as well as age, age squared, gender and survey years as covariables. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

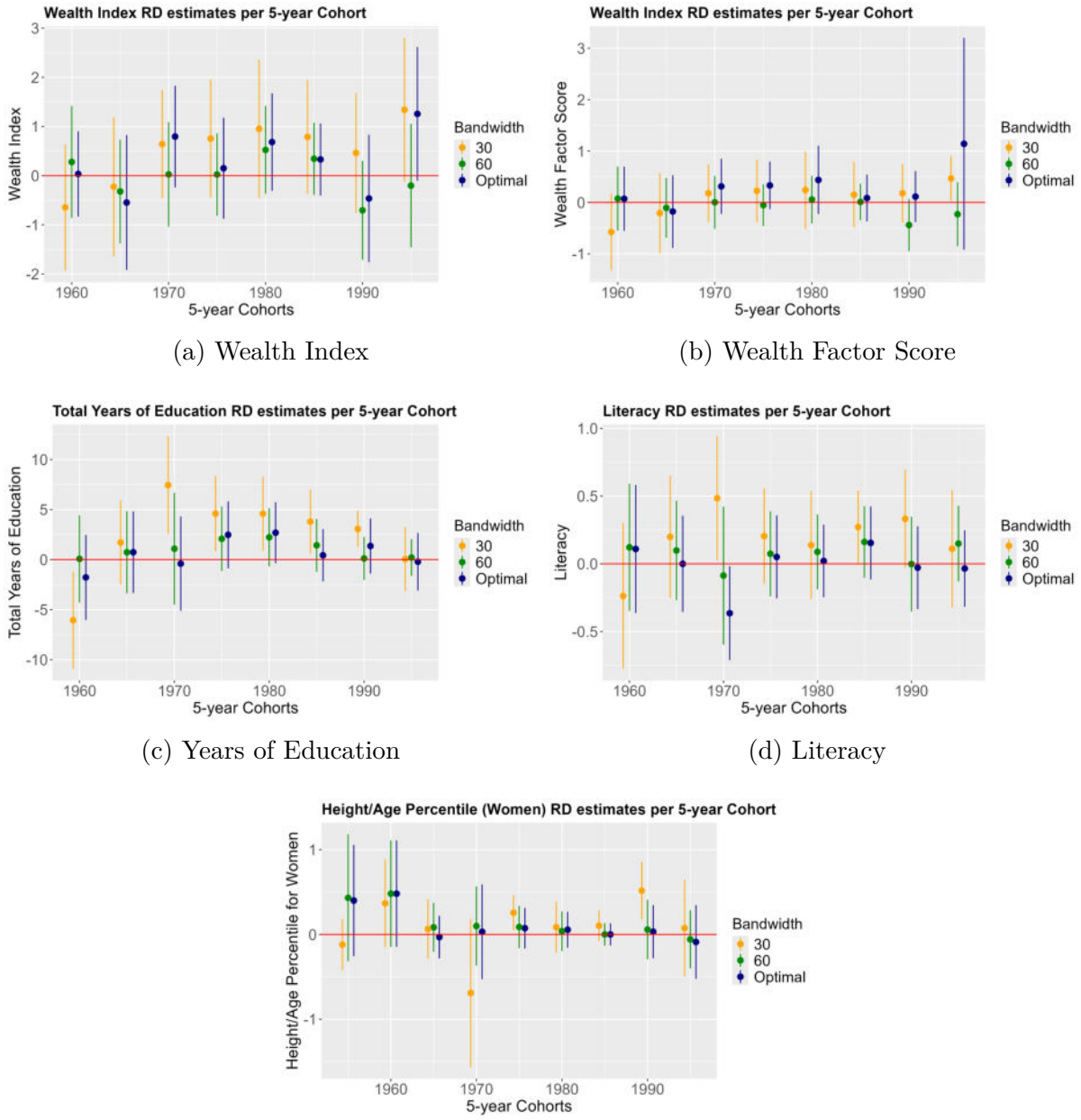
which are depicted in panel A, positive coefficients are shown for the wealth index, while the coefficients for the logarithm of the wealth factor score for the optimal and 60 km bandwidth are negative. However, it should be noted that the computation of the wealth index is based on the wealth factor score. The reasons for the different signs could be that the quantiles of the wealth index were divided in such a way that observations within a quantile just barely fall below or exceed the quantile threshold. The choice of the kernel could also play a role since, as shown in Table E.1, the use of a rectangular kernel results

in a positive value for the 60 km bandwidth. Regarding the education category, most effects are positive, except for literature at the optimal bandwidth of 85.88 km, where a negative value is observed. The 85.88 km and every other reported bandwidth larger than 60 km only refer to a distance to DHS clusters outside the circles. The coefficients for the women's and children's height/age percentile are all positive.

For the variables years of education and women's height/age percentile, the coefficients for the 30 km bandwidth are significant at the 5% level and the 1% level. Furthermore, the same dependent variables are also significant at a 5% level for the bandwidth of 35 km, as shown in Table D3. This suggests that there is a pattern of significant positive estimates for those two outcome variables at bandwidths of 30 km and 35 km. Moreover, apart from the children's height/age percentile, all the other estimates show the highest value at a bandwidth of 30 km. At least for the significant coefficients, this would mean that people within concession areas have, on average, almost three more years of education, and women within circles have an 18 percentage points higher height/age percentile than women outside the circles. However, in comparison to the other coefficients, the magnitude of these coefficients is relatively high. Table D3 presents the RD estimates for more bandwidths within the range of 20 km to 60 km and it also exhibits the same general pattern just described in the previous lines. The typical RD plots, as depicted in Figure G1 and Figure G2, illustrate that the discontinuities for a bandwidth of 60 km are not as obvious as those for a bandwidth of 30 km. Table D5 also shows that the significant results at 30 km are not explained by one circle being overrepresented compared to a bandwidth of 60 km.

In Appendix E, various adjustments to the estimation Equation 1 are presented to check robustness of the observed findings. Most estimates for the majority of the bandwidths remain insignificant when using a rectangular kernel, an Epanechnikov kernel, a quadratic polynomial, or when covariates and fixed effects are omitted. Some coefficients occasionally become significant for certain bandwidths, but sometimes these are the optimal bandwidths that are relatively small and therefore have too few clusters to provide reliable results. Regarding the coefficients that are significant at a bandwidth of approximately 30 km, the pattern exhibited in the main results is not clearly discernible in the robustness checks. For example, with a rectangular kernel, only the coefficient for the variable years of education at a bandwidth of 30 km is significant, while when using a quadratic polynomial, this is only the case for the height/age percentile of women.

5.2 RD Estimates for Age Cohorts



(a) Wealth Index

(b) Wealth Factor Score

(c) Years of Education

(d) Literacy

(e) Height/Age Percentile (Women)

Figure 3: RD Estimations Coefficients for each 5-year Age Cohort.

Notes: The sub-figures in the figure above show the RD coefficient and the corresponding 95% confidence interval for every 5-year age cohort of the respective dependent variable. Standard errors are clustered at the DHS cluster level. The RD regressions include concession fixed effects as well as age, age squared, gender and survey years as covariables. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.

Figure 3 displays the coefficients of 5-year age cohorts for the optimal bandwidth as well as for the bandwidths of 60 km and 30 km. The underlying assumption is that older individuals are more likely to be affected by the exposure to the concession areas, which could result in significant estimates for older age groups. The oldest group shown in the graphs is born between 1955 and 1959 (1960 and 1964 for women's height/age

percentile), as the observations for even older individuals are limited. The age cohort coefficients for children are not calculated since the DHS data only include children aged up to 5 years. Figure 3 indicates that almost all estimates for all age groups remain insignificant. However, within a bandwidth of 30 km, significant values are observed for the variable years of education for some age cohorts, which is consistent with the main results. In terms of the effects per age group, it can therefore be ruled out that there has been an impact that has dissipated over the years.

5.3 Heterogeneity among the Circles

Another way of analysing the potential long-term impacts of HCB's concession areas is to examine the effects of each circle. Unfortunately, this is only feasible for Lusanga and Alberta due to the small number of clusters. Specifically for Lusanga, historical sources suggest that the level of forced recruitment and violence was notably high because the area was sparsely populated and locals were reluctant to work for HCB due to past negative experiences with other European companies. However, the density of palm trees in Lusanga is reported to have been the highest compared to the other circles (see Fieldhouse, 1978, p.513ff; Henriet, 2021, p.95ff). Therefore, Lusanga was probably also the most important concession area in economic terms, as most of the palm fruits were harvested and processed there and thus most of the people were employed in this circle (Marchal, 2008, p.87 and p.173). With this information, the empirical strategy outlined in Section 4.2 can be applied to the Lusanga circle in order to investigate whether there was a negative effect due to intensive forced recruitment or a positive effect due to its economic importance.

In Table 3, it can be seen that the results are mostly insignificant. For the wealth category, the wealth index variable is the only one with a positive and significant estimate at a bandwidth of 60 km. For the wealth factor score, the estimated range of 22.65 km is probably too small to yield a useful coefficient. Unlike the main results for all five territories in Table 2, the pattern of significant values at a bandwidth of 30 km cannot be observed again. Notably, the height/age percentile variable for females is significant and negative for the optimal bandwidth of 44.23 km and the bandwidth of 60 km, whereas it is positive for the main results and only significant at a bandwidth of 30 km. For Lusanga, it can be concluded that, compared to the main results, the estimation results do not tend to indicate significant uniform effects. However, there are again occasional significant coefficients but they do differ from those in the main results.

Table 3: RD Results for Lusanga

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	1.2722	0.8611*	3.4544+	5.2591***	0.2710	1.4569
	(3.0766)	(0.4376)	(1.9025)	(0.0345)	(0.2285)	(0.9350)
Num.Obs.	344	1328	434	181	1328	434
Bandwidth	27.90	60.00	30.00	22.65	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.340	2.750	2.415	10.809	11.137	10.918
SD Dep. Var.	1.090	1.261	1.104	0.375	0.653	0.542
Total Number Clusters	7	25	9	4	25	9
Clusters within Con. Area	3	9	3	2	9	3

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	5.3637	2.2898	3.1329	0.1579	0.1829	0.4770
	(3.8954)	(2.5601)	(10.6514)	(0.2066)	(0.2155)	(0.8734)
Num.Obs.	1006	1327	433	1379	1323	434
Bandwidth	44.23	60.00	30.00	61.15	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	8.200	7.914	7.032	0.786	0.785	0.703
SD Dep. Var.	3.896	3.917	4.137	0.410	0.411	0.458
Total Number Clusters	19	25	9	26	25	9
Clusters within Con. Area	7	9	3	9	9	3

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	−0.1116*	−0.1019**	−0.1318	−0.0678	−0.0529	−0.2707
	(0.0555)	(0.0392)	(0.1033)	(0.0435)	(0.0617)	(0.1953)
Num.Obs.	354	459	148	773	353	115
Bandwidth	44.04	60.00	30.00	154.55	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.196	0.198	0.142	0.255	0.282	0.234
SD Dep. Var.	0.248	0.241	0.199	0.338	0.349	0.332
Total Number Clusters	19	25	34	50	25	9
Clusters within Con. Area	7	9	10	9	9	3

Notes: Standard errors are clustered at the DHS cluster level and reported in parentheses. The regressions include age, age squared, gender and survey years as covariables. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

The only other circle for which reasonable results can be obtained is Alberta. However, little information can be found in the historical sources and archives about the general working conditions in this circle (Marchal, 2008, p.91). In terms of economic output, it was just under half the size of Lusanga (Marchal, 2008, p.87). Table D4 presents the RD results for Alberta, and similar to the previous findings, most coefficients are not significant. The only significant coefficients are linked to a bandwidth of 30 km or an even smaller optimal bandwidth. This raises concerns that the number of clusters and

observations might be too small to yield reliable estimates. Hence, the estimation results from Alberta also tend to support the hypothesis that the palm oil concessions had no long-term impact on socio-economic development in its areas.

5.4 Potential Mechanisms

Unfortunately, it is not possible to investigate potential mechanisms since there is no adequate data available in the context of HCB's concession areas. Lowes and Montero (2021) went to the Congo to conduct a study that helped to identify such mechanisms. Although the results are insignificant, potential mechanisms could shed light on whether the effects of virtuous intentions or atrocities can be detected through specific channels. Examining such mechanisms could also clarify if those two opposite features of HCB's concession areas outweigh each other so that the main results are rendered insignificant. Additionally, it would be interesting if potential channels exist that are similar to those found by Lowes and Montero (2021) and other scientific works introduced in Section 3.3. Based on findings of other scientific works and also based on historical records, this subsection will explore potential channels that could have explained certain impacts.

First of all, the mechanisms investigated by Lowes and Montero (2021) may also reveal potential negative effects of HCB's concession areas because many characteristics of the rubber concessions and the cruel rule of King Leopold II also existed in the circles (see Henriet et al., 2017). Chiefs collaborated with HCB by accepting bribes in exchange for providing workers, which is why it is likely that they did not entirely act in favour of their people and thus provided less public goods such as maintenance of roads and schools or conflict arbitration (see Fieldhouse, 1978, p.513; Henriet, 2021, p.104). Moreover, as suggested by Lowes and Montero (2021), more pro-social behaviour could also be found among the people in the former concession, as it may substitute extractive or weak institutions.

Another potential mechanism may entail the violation of native land rights in the concession areas. According to the Belgian Congo legislation, native land rights should have been respected. This meant that, within the designated circles, HCB should only lease "vacant land" that was not used by the natives. However, natives were forced to give up land with palm trees that they cultivated before (p.48 Marchal, 2008). Contrary to the assumption of HCB that palm trees would simply grow in nature, palm trees needed to be taken care of, which is why the majority of the existing palm trees were already utilized by

the natives. In the first years of HCB's operations, the Belgian administration lacked the capacity to demarcate native land, which is why it was difficult for HCB to choose these "vacant lands". This problem should be addressed by the so-called "tripartite agreements", which allowed the Belgian administration and HCB to declare land within the circles as temporarily undefined land until the demarcations were carried out. Consequently, HCB was granted permission to harvest palm fruits on these undefined lands, enabling the company to operate in a territory that was bigger than the originally designated land blocks supposed to be leased by HCB. Furthermore, the tripartite agreements also implied that HCB obtained a monopoly on collecting and trading palm fruits in an area larger than the original 750,000 hectares. Natives were forced to sell palm fruits that were not cultivated on leased land blocks to the company for a fixed price, which was usually below market price. Despite protests against this monopoly, HCB vigorously tried to assert its monopolistic rights since they would avoid increased competition and ensure higher profits for the company. For example, the demarcation of native land was neglected by the Belgian administration and HCB, as it enabled HCB to continue to benefit from its monopoly (see Henriet, 2021, p.150ff; Fieldhouse, 1978, p.518ff; Marchal, 2008, p. 47ff, p. 100ff and p. 196ff). As already described in Section 3.1, property rights play a crucial role in economic development, as they provide incentives to invest in one's own property and also benefit from the resulting revenues. In addition to that, Dell (2010) shows that secure land rights outside the mining concession area are likely to have led to a higher level of provision of public goods such as infrastructure and education. Therefore, secure property rights for land may constitute a mechanism that explains higher development outcomes outside the mining concessions in Peru. These two findings show that depriving the natives of their land rights within the palm oil concession areas may represent a potential channel that could have led to contemporary poorer socio-economic living conditions.

Apart from the channels that lead to negative outcomes, Dell and Olken (2020) illustrate that setting up such big ventures in the colonial era may also attract other related businesses so that economic and industrial centers evolved. The palm oil extracted in HCB's concession areas was mainly exported, which is why it is unlikely that related businesses that processed palm oil further emerged in the first half of the 20th century. However, since subsistence agriculture has likely been the only notable economic sector before the arrival of HCB, newly constructed palm oil mills and factories for processing palm fruits, as well as hospitals, schools, and camps for hundreds of workers could have contributed to the emergence of economic centers. These agglomeration effects are also in line with

the main findings of the literature on the impact of colonial investments in infrastructure presented in Section 3.3. Even after railroads lost their importance, economic centres persisted.

The channels for potentially better health and educational outcomes may be mainly attributed to the establishment of schools and hospitals. This is also consistent with the mechanism found by Méndez and van Patten (2022), who demonstrate that long-term improvements were driven by investments in health and education to attract workers. The literature about the positive effects of Christian missions on education and health in Africa presented in Section 3.3 is also in line with this linkage since European missions played crucial role in spreading Western education and medicine in Africa (see Cagé and Rueda, 2020; Meier zu Selhausen, 2019).

6 Methodological and Context-Related Caveats

6.1 Bandwidth Choice and Statistical Inference

Apart from some historical aspects that make a casual interpretation of the main results presented in Section 5.1 difficult, there is also a methodological caveat that needs to be discussed: As already mentioned in Section 4.2, the default RD method by Calonico et al. (2017) cannot be applied because the observations are probably spread over a too small number of clusters, leading to a small variation in the running variable. When using the same RD specifications as Lowes and Montero (2021), who also employ the default RD method by Calonico et al. (2017), R produces the following error message: "The leading minor of order 1 is not positive". This error likely suggests that certain requirements for finding an optimal bandwidth that also takes into account clustered errors, covariables and fixed effects, are not fulfilled due to insufficient variation in the forcing variable. When shifting the border from 60 km to placebo borders of 30 km and 120 km, the optimal bandwidth selection procedure from Calonico et al. (2017) works if clustered errors are not applied. This may further indicate that having only one village within the closest 17 km to the border in the concession areas also prevents this MSE-minimizing procedure from performing properly.[3] The scientific work by Lowes and Montero (2021) shows that having only 85 clusters within a bandwidth of 100 km and 32 clusters within an optimal

[3]The replication package includes the attempts of using the command by Calonico et al. (2017) in R.

bandwidth of approximately 29 km is sufficient for the MSE-minimizing procedure by Calonico et al. (2017) to work properly.

To understand the importance of Calonico et al. (2017)'s RD method in the context of this paper, it is essential to comprehend the underlying idea behind this method. The main idea of the MSE-minimizing selection process is to choose a bandwidth that minimizes the bias for the RD point estimate. Two types of bias lead to inaccurate estimates: One type of bias is caused by using a bandwidth that is too wide and thus may have an overly large sample size. This results in a misspecification error, as the regression function fails to adequately approximate the true course of the observations close to the cutoff. The other bias stems from choosing a small bandwidth, resulting in a high variance due to the low number of observations. The mean squared error (MSE) bandwidth selection method addresses this issue by determining a specific bandwidth that provides the best trade-off between those two biases. Hence, the same MSE-minimizing procedure selects a bandwidth that yields the optimal point estimate for a potential effect measured by an RD design (see Cattaneo et al., 2019; Calonico et al., 2014).

However, apart from incorporating covariates and clustered errors, the main difference between the bandwidth selection method by Calonico et al. (2017) and the one by Imbens and Kalyanaraman (2012), which is used in the empirical estimation strategy of this paper, is the inclusion of robust bias-corrected confidence intervals and p-values. If the bandwidth is small enough, the misspecification error should converge to zero, and conversely, if there are more observations, which is most likely the case when the bandwidth is wider, the bias stemming from a large variance should also converge to zero. However, in practise, these asymptotic properties do not hold when a MSE-minimizing point estimation is chosen because, for example, the bandwidth is probably not small enough to neglect the misspecification error. Hence, a bias is very likely to remain, which should be taken into consideration when drawing statistical conclusions because this bias influences the confidence intervals and p-values. A possible solution to this problem is provided by Calonico et al. (2017) through their robust-bias-correction estimation. To create suitable confidence intervals, they correct the point estimate by the estimated bias and also enlarge the confidence intervals by adding the newly created variation stemming from the corrected point estimate. In other words, the point estimate will be recentered by adding or subtracting the respective bias and the corresponding confidence intervals will be created around this bias-corrected point estimate. These new confidence intervals

are also larger, as they also incorporate the new variation created by this bias correction (Cattaneo et al., 2023; Cattaneo and Titiunik, 2022).

It is important to highlight the lack of robust bias-corrected statistical inference in the context of the RD setting examined in this paper because the robust bias-corrected confidence intervals tend to be larger than the standard OLS confidence intervals used in the estimation strategy presented in Section 4.2. Consequently, relying on the normal t-distribution for statistical inference in an RD setting is likely to result in an overrejection of the null hypothesis. Since some of the results shown in Section 5.1 are only slightly significant, this would mean that there is a relatively high probability that these results turn out to be insignificant if the more appropriate RD procedure by Calonico et al. (2017) is applied. However, if an RD coefficient would be less significant also depends on the extent to which the point estimate would be corrected by its bias.

To show the difference between the conventional statistical inference and the robust bias-corrected statistical inference in the context of HCB's palm concession areas, a wild bootstrap method for RD settings is applied. The basic idea of a bootstrap method is to redraw observations from the original sample and then run the statistical model over the resampled data. Repeating this procedure several times yields many estimates, for which the standard error and the confidence intervals can be calculated by using a cumulative distribution function. These statistical values are usually at least as accurate as their conventional asymptotic counterparts (Horowitz, 2019). This bootstrap procedure was adapted to RD settings by He and Bartalotti (2020), and they showed that similar robust bias-corrected point estimates and confidence intervals as those generated by the method by Calonico et al. (2017) can be obtained. However, the same wild bootstrap method cannot yet include covariates and fixed effects, which is why it cannot be used to fully replicate the estimation strategy presented in Section 4.2. He and Bartalotti (2020) mention that it should be intuitively easy to account for covariables. However, since Cattaneo et al. (2022) have shown that adding further covariables to an RD estimation already requires some attention, adequately incorporating covariates in this wild bootstrap method would expand the scope of this thesis. Hence, the wild bootstrap confidence intervals and p-values for the RD setting examined in this paper are calculated without including covariables and fixed effects. Furthermore, this wild bootstrap method cannot select the optimal bandwidth on its own, which is why it relies on other optimal bandwidth selection procedures or arbitrarily chosen bandwidths. Therefore, the bandwidth selection

procedure by Imbens and Kalyanaraman (2012) is applied, which yields the same optimal bandwidths utilized for the main results presented in Table 5.1. To be able to apply this wild bootstrap procedure to the RD design of this paper, the R command was further modified.[4]

Table 4 represents the results of this RD bootstrap procedure. The first four rows of each panel display the treatment effect and the corresponding conventional standard errors, confidence intervals and p-values that are obtained by applying the estimation strategy presented in Section 4.2 without incorporating covariables and fixed effects. The subsequent three rows show the robust bias-corrected point estimate, confidence intervals and p-values for the respective RD specification and outcome variable. Comparing these results with the previous ones, it becomes clear that the majority of coefficients are again insignificant for both approaches. This is especially the case for both health variables. However, excluding fixed effects in the main estimation strategy also renders the height/age percentile variable at a bandwidth of 30 km insignificant, which has not been the case when the original empirical strategy of this paper is applied. Moreover, Table 4 also indicates that some coefficients are significant for one approach but not for the other. For instance, this can be seen for the log(wealth factor score) variable at a bandwidth of 30 km as well as for the literacy variable at a bandwidth of 85.88 km and 30 km. The estimate for years of education at a bandwidth of 30 km is also significant when the main estimation approach of this paper is used, but it is not according to the wild bootstraps results. This suggests that the pattern of significant results at a bandwidth of 30 km, as shown in Table 2, may not be detectable when robust bias-corrected confidence intervals are utilized. Interestingly, all other coefficients for years of education are positive and significant for both approaches, indicating that it might be worth investigating this further when, for example, the wild bootstrap method for RD designs is extended by the inclusion of fixed effects.

In summary, the wild bootstrap results are not directly comparable with the main results presented in Section 5.1 because fixed effects and covariables could not be included. However, they show how certain coefficients can differ if a more appropriate method for statistical inference is used, suggesting that the RD method by Calonico et al. (2017)

[4]The corresponding programming codes created by Bartalotti et al. (2017) and He and Bartalotti (2020) were adapted in a way so that a sharp RD design with clustered errors could be estimated.

Table 4: RD Main Results based on Wild Bootstrap Data

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.5798	0.2946	0.7289+	−0.0908	−0.0240	0.0304
Standard Error	0.4911	0.4126	0.3918	0.0737	0.1486	0.1052
Conventional CI	[−0.383, 1.542]	[−0.514, 1.103]	[−0.039, 1.497]	[−0.235, 0.054]	[−0.315, 0.267]	[−0.176, 0.237]
Conventional P-Value	(0.238)	(0.475)	(0.063)	(0.217)	(0.872)	(0.772)
Robust Estimate	0.4843	0.7107	−0.2324	−0.2536	0.0116	−0.4301***
Robust CI	[−0.407, 1.56]	[−0.371, 1.874]	[−1.355, 0.82]	[−0.58, 0.132]	[−0.435, 0.401]	[−0.764, −0.093]
Robust P-Value	0.3183	0.2222	0.5325	0.2422	0.9209	0
Num.Obs.	2541	3548	1708	1254	3548	1708
Bandwidth	39.54	60.00	30.00	26.85	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.545	2.540	2.343	10.845	11.045	10.958
SD Dep. Var.	1.274	1.238	1.204	0.513	0.602	0.569
Total Number Clusters	49	69	34	25	69	34
Clusters within Con. Area	16	23	10	8	23	10

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	2.2135*	2.7884*	3.8814***	0.0567	0.1861+	0.2463**
Standard Error	1.0779	1.1503	0.7875	0.0991	0.1124	0.0898
Conventional CI	[0.101, 4.326]	[0.534, 5.043]	[2.339, 5.425]	[−0.136, 0.251]	[−0.034, 0.406]	[0.070, 0.422]
Conventional P-Value	(0.0400)	(0.0153)	(0.0000)	(0.5670)	(0.0979)	(0.0061)
Robust Estimate	4.2362***	4.2721***	1.1739	0.2939*	0.2559+	−0.4301
Robust CI	[1.599, 6.562]	[1.421, 7.292]	[−0.216, 2.756]	[0.014, 0.561]	[−0.005, 0.562]	[−0.764, −0.093]
Robust P-Value	0.0000	0.0000	0.1081	0.036	0.0601	0.7047
Num.Obs.	3819	3544	1704	4312	3532	1701
Bandwidth	70.97	60.00	30.00	85.88	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.831	6.908	6.183	0.712	0.732	0.666
SD Dep. Var.	3.984	3.981	4.112	0.453	0.443	0.472
Total Number Clusters	75	69	34	85	69	34
Clusters within Con. Area	23	23	10	23	23	10

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	−0.0107	0.0029	0.0035	0.0200	0.0427	0.0411
Standard Error	0.0454	0.0652	0.0421	0.0623	0.0597	0.0499
Conventional CI	[−0.0998, 0.078]	[−0.123, 0.131]	[−0.079, 0.086]	[−0.102, 0.142]	[−0.074, 0.160]	[−0.057, 0.139]
Conventional P-Value	(0.814)	(0.965)	(0.934)	(0.748)	(0.475)	(0.411)
Robust Estimate	0.0624	−0.0158	0.0121	0.0478	0.0421	−0.0182
Robust CI	[−0.043, 0.194]	[−0.116, 0.074]	[−0.122, 0.161]	[−0.042, 0.136]	[−0.047, 0.147]	[−0.16, 0.141]
Robust P-Value	0.2923	0.7287	0.9129	0.3003	0.4545	0.6547
Num.Obs.	866	1190	578	1821	1042	527
Bandwidth	39.41	60.00	30.00	128.40	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.229	0.236	0.231	0.282	0.298	0.289
SD Dep. Var.	0.258	0.255	0.257	0.348	0.352	0.351
Total Number Clusters	49	69	34	116	69	34
Clusters within Con. Area	16	23	10	23	23	10

Notes: This table presents the main results of the wild bootstrap method adapted to the RD estimation strategy of this paper. The first four rows of each panel show the main treatment effect of the concession areas, along with conventional standard errors, confidence intervals, and p-values for the respective RD specification and outcome variable. The subsequent three rows display the robust bias-corrected point estimate, confidence interval and p-values that are obtained from the wild bootstrap method. Standard errors are clustered at the DHS cluster level. The RD results are calculated without concession fixed effects and covariables. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Section A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

would yield different results. Nevertheless, the tendency of insignificant effects is also evident for this approach. Moreover, since this wild bootstrap method still relies on other MSE-minimizing procedures, using the more advanced method by Calonico et al. (2017) would have yielded even more reliable results.

6.2 Migration

One aspect that could explain the insignificant results is that historical sources clearly report on labor migration (Henriet, 2015; Marchal, 2008). To address the shortage of labour force, locals from villages and tribes outside the original concession areas were recruited. For instance, people came from locations that were 50 to 150 km away from the Lusanga circle to work there. The recruited workers remained in the concession areas for a period of a few months to three years until they had collected a certain number of palm fruits and thus fulfilled their employment contract. After their contracts ended, the fruit cutters returned to their home villages and were replaced by new recruits. This constant flow of people also led to the emergence of unofficial villages, known as "villages doublures", where people lived for a few months and even took their family members with them. For HCB, these unofficial settlements represented a notable advantage, as they did not have to provide social benefits such as education and health services there. There were also attempts to establish model villages where recruits could stay permanently, but the sources do not report a successful implementation of these villages. Furthermore, efforts were made to retain people in their employment by automatically extending their work contracts, but this was reluctantly accepted by the locals. Therefore, it can be assumed that at least some of the recruited labor force left the concession areas (see Henriet, 2021, p.75ff; Henriet et al., 2017).

The historical sources report various figures regarding the recruitment of people from outside the circles. For instance, Marchal (2008) describes, based on a report from that time, that around 10,000 people worked for HCB in the Lusanga circle in 1927, of which approximately 4,000 came from outside the concession area. With a total population of 20,000 inhabitants in the Lusanga territory, almost all local able-bodied men had to work for HCB, and a proportion of 20% of Lusanga's population had to immigrate to meet the labour demand (Marchal, 2008, pp.94-95). Other historical sources for Lusanga in 1934 indicate that, within one year, 33,000 recruits resided in these unofficial "villages doublures". As fruit cutters were often accompanied by their family members, who also

supported them in their work, it can be assumed that there were even more migrants in the Lusanga circle. A census in 1935 reported the number of recruited fruit cutters was about 32,000. However, it is not clear from these numbers whether they only refer to people from outside the concessions or if they also include local workers who already lived in the circle and just moved to another palm grove (Henriet, 2015, p.76). Although these figures differ, they indicate that the scale of migration was high.

The information on the extent of migration is an aspect that makes a causal interpretation of the results difficult, as migrant workers who suffered from forced labour and violence, but who may also have benefited from the availability of medicine and education, may now live outside the original circles. Thus, the possible long-term effects of the concession areas would no longer be clearly measurable, as people living now outside of them were also affected. The assumption that exposure to the concession areas is the only characteristic that does not change abruptly at the border would be violated. If the palm oil concession areas have indeed had a long-term impact on socio-economic development, migration could explain why the results are mostly insignificant, because the group of returnees may have mixed with the normal population outside the circles.

6.3 Further Historical Caveats

Apart from migration, there are several other historical aspects that need to be considered in this RD setting. As mentioned in the introduction, HCB was only allowed to use a maximum of 750,000 hectares within the circles, which corresponded to about 13 percent of the total concession area. A few years later, this area was further reduced. It is therefore at least questionable to what extent HCB's virtuous intentions reached every part of these circles. Forced labor may have affected the entire territory of the concession areas since natives were also recruited from outside the circles, and HCB's monopoly on harvesting and trading with palm fruits covered large parts of the circles. However, investments in human capital and infrastructure may have been limited to the areas in which HCB was operating. Historical reports, which can be found in the book by Marchal (2008), indicate that at least the provision of health care was not sufficient to treat all the sick people who, for example, suffered from sleeping sickness, syphilis, or worms (Marchal, 2008, p.33 and p.204).

Another aspect that influences the validity of this empirical analysis is the lack of clarity about what happened after the 1940s regarding the continuation of forced labour and the

investments in infrastructure and human capital. Historical sources indicate that forced labor and violence in HCB's concession areas were primarily associated with the years between 1911 and the 1930s. The last certain cases of recorded forced labour were the establishment of plantations and during the Second World War, when palm oil and other resources were needed to support the Allies (see Marchal, 2008, p.211; Henriet, 2021, p.164). However, there are indications that forced labour was continued at least in the Lusanga circle until the 1960s, but there is limited information in the historical sources about forced recruitment and the labour situation after 1931 (Marchal, 2008, p.217). Therefore, one possible solution to the labour problem could have been the slow switch to more efficient plantations, which enabled HCB to pay higher wages and thus employ a permanent and voluntary labour force (Fieldhouse, 1978, p.517). Nevertheless, HCB's oppressive monopolistic rights were renewed in 1956 for ten more years (Fieldhouse, 1978, p.537). On the other hand, HCB kept running its schools and health facilities until the 1990s (Ergo, n.d. p.193). The persistence of these two features of the palm oil concessions would not matter as long as they continued to occur within the confines of the original circles. However, as part of the extension of the tripartite contracts in 1938, HCB was allowed to choose areas for harvesting palm fruits and establishing plantations outside the former circles. For example, one important plantation was situated 160 km north of Brabanta in the territory of Oshwe (Fieldhouse, 1978, p.530). Again, this would violate the assumption that exposure to the concessions abruptly changes at its borders since people from inside and outside the original concession areas could have suffered or benefited from being exposed to HCB's palm fruit business operations. Hence, drawing causal conclusions by using the estimation strategy presented in Section 4.2 would be flawed.

A third issue that requires consideration is the existence of other concessions that were used to harvest palm fruits or extract various resources in the same decades as HCB (see Nicolaï, 2013; Juif and Frankema, 2018). There is evidence that concession areas of at least one other company were located next to Lusanga. As Figure B7 shows, palm oil concessions were also situated to the northwest in the immediate vicinity of Lusanga. These concessions belonged to the company Compagnie du Kasai, which, according to historical sources, treated its workers even worse than HCB (Marchal, 2008, p.116ff). This would not pose a threat to the statistical analysis if the impact of HCB's concession areas was positive and forced labour equally occurred outside and inside the circles. A causal effect would still be measurable as long as the level of forced labour and violence

were balanced on both sides. However, there were at least some health facilities in these adjacent concessions, which is why it can be assumed that investments in health were also made there. Hence, people outside HCB's circles may have experienced a similar treatment as those who resided inside. Therefore, using the neighbourhood of HCB's circles as counterfactual territories may not be appropriate, as an RD estimation would primarily measure the different effects among the concession areas. This could also explain the insignificant coefficients presented in the main results since there would not be any substantial differences among the people living inside and outside the circles if the treatment for both sides had been similar. Lowes and Montero (2021) also examined the long-term effects of all the concession areas that existed at the same time as the Abir and Anversoise concessions. Although only some coefficients are significant, they find that all these concessions tend to have a negative effect on the same dependent variables used in the estimation strategy of this thesis. In that regard, the persistent effects of the Abir and Anversoise concessions found by Lowes and Montero (2021) could also influence the results of this paper since their territory overlapped with the area of Alberta and Elisabetha. However, they revealed strong negative significant effects, while the estimates of the main results presented in Section 5.1 tend to be positive. Within a bandwidth of 60 km, as can be seen in Figure B5, there are seven DHS clusters located inside both the rubber and the palm oil concession areas. Table F1 shows that the main results hardly change when Equation 1 includes a dummy that accounts for the Abir and Anversoise concessions. Thus, the findings of this paper are not driven by the negative impacts of the Abir and Anversoise concessions.

6.4 Exact Locations of the Circles

Unfortunately, the historical sources are not entirely consistent about the centers of each circle. For example, the place Barumbu, which is, according to the convention of 1911, the center of Elisabetha, may be difficult to find on ordinary maps, since it seems that it belongs to the city Basoko. More accurate maps of the Thsopo province show the exact location of Barumbu (Médecins Sans Frontières, 2022). In general, the centers for the main results were chosen on the basis of sources that all refer to the convention of 1911 and are also consistent with each other. Hence, all the sources that contain deviations from the places agreed upon in the convention of 1911 seem to be randomly erroneous. However, Table F5 shows results that are based on alternative concession centers that also seem plausible when looking at the corresponding maps. The following potential

concession centers were chosen: In the Flandria circle, there is a place called Boteka that is surrounded by palm tree plantations, in the Brabanta circle, there is a place also called Brabanta and in the Alberta circle, there is Ebonda, formerly known as Alberta. In the Elisabetha circle, Basoko is the alternative, while Leverville remains the center since there are no indications of other potential centers in the Lusanga circle. The alternative concession areas can be seen in Figure B6. The results show that most of the coefficients are insignificant. In comparison to the main results presented in Table 2, the dependent variable literacy is significant at a 10% level for a bandwidth of 60 km and at a 1% level for a bandwidth of 30 km. The coefficient for years of education is once again significant for a bandwidth of 30 km, while this is not the case for the height/age percentile of women for the same bandwidth.

Furthermore, there are some sources such as Kisangani (2023) and Nicolaï (2013) that report a radius of 50 km for each circle. However, this also seems incorrect since the majority of the literature, whose content is based on the archives, mention a radius of 60 km. Nonetheless, Table F3 contains the coefficients that are calculated with a radius of 50 km. Similar to the previous findings, the majority of the estimates lack statistical significance. Unlike the main results, the values for years of education and literacy are negative and significant for a bandwidth of approximately 30 km. This suggests that clusters close to the 60 km border are now just outside the treatment area of HCB's investments in education. Hence, under the assumption that the radius of 60 km is right, this would support the hypothesis that there are positive effects regarding education inside the concession areas. However, there is only one DHS cluster that would swap sides, which is why it is unlikely that these results are in line with the hypothesis of positive impacts for education. The same exercise was also conducted for a hypothetical radius of 70 km. The results are presented in Table F4 and are also insignificant for almost all RD specifications.

Finally, due to confidentiality reasons, the clusters of the DHS program are displaced by up to 10 km in rural areas and up to 2 km in urban areas. Normally, this is a classical measurement error that would bias the coefficients towards zero. However, this issue may raise some concerns since, on the one hand, the main results tend to represent insignificant results and, on the other hand, one might be worried that the estimates are strongly influenced by those DHS clusters that are incorrectly assigned to the wrong side of the concession border. Following Lowes and Montero (2021), a "donut-hole" approach is

applied that excludes all the clusters that are closer than 10 km to the concession border. Table F2 shows that, apart from a narrow optimal bandwidth for the logarithm of the wealth factor score, no coefficient is significant, which means that the main results are not driven by the DHS displacement.

6.5 Discussion

Considering all the results and the robustness checks, the most obvious conclusion is that the exposure to the palm oil concessions did not have a measurable long-term impact on socio-economic outcomes since the majority of coefficients are insignificant. Nevertheless, there are a few coefficients that are significant for some bandwidths in almost every RD specification. With only 69 clusters for the 60 km bandwidth and 155 clusters for the entire sample, the main RD regression of this thesis can also be compared with an OLS regression with the same relatively small number of observations (see Bartalotti and Brummet, 2017; Cattaneo et al., 2024). This is important in the sense that running a regression with a relatively small number of observations makes it more difficult to detect significant results since the statistical power would be relatively low and the likelihood of having a type II error relatively high if the null hypothesis is truly false. Hence, running the RD regressions for the 30 km bandwidth with a total number of 34 clusters makes it less likely to find significant effects. Therefore, the significant coefficients for years of education and the height/age percentile of women at a bandwidth of 30 km in the main results could indicate that there is a real positive significant impact.

On the other hand, having a significance level of 5% implies that there are chances to detect significant effects by coincidence although the null hypothesis is true. This may explain the rather random occurrence of some significant effects in the sub-results such as those for the Lusanga circle as well as those that are based on the wild bootstrap method. All these significant estimates mostly differ from those detected for years of education and the height/age percentile of women at a bandwidth of 30 km in the main results presented in Table 2. Hence, there is no consistent pattern that always shows the same significant coefficients.

Moreover, these significant estimates in the main results lose their significance when an RD specification with a rectangular kernel or with a quadratic polynomial is applied. This strengthens the notion that their significance could be random. Only for years of education, the coefficient at a bandwidth of 30 km in the main results may not be random since

several age cohorts for the same narrow bandwidth in Figure 3 show significant estimates. However, this significance could also be driven by certain DHS clusters, especially when the number of clusters is low within a bandwidth of 30 km.

Although having a narrow bandwidth is a basic assumption of an RD setting, the number of clusters within a bandwidth of 30 km may already be too small to adequately reflect the data and the true underlying relationship between the exposure to the concession areas and examined outcomes. As a result, the estimates may be biased due to a mis-specification error, especially when there are only 10 clusters on one side, which are not evenly distributed along the distance to the concession center. This bias may also explain why the magnitude of most of the estimates for a bandwidth of 30 km presented in Table 5.1 and visualized in Figure G1 and Figure G2 is relatively high in comparison to wider bandwidths. Appendix H shows different OLS regressions in which a dummy variable is included that equals 1 when the DHS cluster is located inside the concession areas. None of these OLS estimates is significant at a level of at least 5%.

Furthermore, if there is a significant coefficient for a certain RD specification, one needs to keep in mind, that these results are estimated without using the robust bias-corrected confidence intervals and p-values incorporated in the RD method by Calonico et al. (2017). Since this method usually calculates larger confidence intervals, it would be even more likely that those significant estimates are not significant when this more advanced method is applied. While writing this thesis, the 8th round of the DHS survey is being conducted in the Congo. Once this data is available, it could be added to the existing DHS data and the RD approach of this paper could be repeated with more clusters. Having more cluster locations and thus more variation in the running variable makes it more likely that the optimal bandwidth choice and the robust bias-corrected RD method by Calonico et al. (2017) works.

Normally, an RD strategy detects causal effects if all required preconditions are met. However, since the majority of the coefficients are not significant, there is no causal relationship in the sense that it can be ensured that there is only one variable that influences the outcome. Nevertheless, an RD approach yielding insignificant results can ascertain that there is no significant impact. In the context of HCB's palm oil concessions, this would mean that the exposure to these concessions is likely to have neither positive nor negative effects. An RD estimation can ensure that insignificant effects are truly not

significant by excluding the possibility that the results are diluted by other confounding variables so that lack of significance just occurs by coincidence. However, as shown in Section 6, historical context-related caveats such as migration or the uncertainty about whether the exposure to the palm oil concession was only limited to its original territories may violate the preconditions necessary to ensure that there are no significant effects. It is unclear whether the border of the concession areas can be really regarded as an abrupt change in the exposure to the HCB's concessions. In the historical sources used in the thesis, it is never mentioned that these borders played a crucial role or were strictly adhered to. Therefore, this RD approach might not be the best method to detect effects that ensure insignificant impacts in a causal sense. One method that might be more suitable is the use of many counterfactual concession areas with similar characteristics as done by Dell and Olken (2020). As described in Section 2, the selection of the locations for HCB's concession areas was only based on vague assumptions about places that contain palm trees and were accessible by boat on rivers.

All in all, the most plausible conclusion from all the results presented in this paper is that there is at least a tendency towards insignificant persistent effects regarding the exposure to palm oil concessions. However, due to several caveats, this conclusion cannot be regarded as completely certain.

7 Conclusion

In this thesis, the potential long-term effects of palm oil concessions established by Lever Brothers in the Belgian Congo in the 1910s are examined. As many other colonial concessions, Huileries du Congo Belge (HCB), a subsidiary of Lever Brothers had to resort to forced labour and violence to recruit workers for their palm oil business. However, the company was also obliged by the Belgian administration to develop these areas by investing in infrastructure, health, and education. Since the concessions had the shape of a circle, their borders were randomly drawn and did not align with other geographic or cultural borders. This exogenous feature is used by a spatial regression discontinuity design as geographic cutoff to estimate persistent effects of the exposure to palm oil concession on contemporary wealth, educational, and health outcomes. The main results show that the majority of estimates for different bandwidths are positive but lack significance, indicating there is no significant impact. This finding also holds for coefficients estimated for age cohorts as well as for individual concessions. There might be a chance to

detect positive significant results for years of education and female health at a bandwidth of 30 km, but due to methodological caveats regarding robust statistical inference of the applied RD method as well as inconsistent occurrence of these significant estimates for sub-results and robustness checks, their significance is not very convincing. Furthermore, context-related caveats such as migration, other concession areas in the close vicinity, or uncertainty if the exposure to the palm oil concession was limited to its designated area make it difficult to draw definite conclusions. Therefore, the applied RD approach may not be the best method for this setting and doubts remain that the overall effect is clearly insignificant.

References

Acemoglu, Daron, Simon Johnson, and James A. Robinson (2001): The Colonial Origins of Comparative Development: An Empirical Investigation, *American Economic Review*, 91(5): 1369–1401.

Acemoglu, Daron, Simon Johnson, and James A. Robinson (2002): Reversal of Fortune: Geography and Institutions in the Making of the Modern World Income Distribution, *The Quarterly Journal of Economics*, 117(4): 1231–1294.

Acemoglu, Daron, Simon Johnson, and James A. Robinson (2005): Institutions as a Fundamental Cause of Long-Run Growth, in Aghion, Philippe, and Steven N. Durlauf, ed., *Handbook of Economic Growth*, Elsevier, p. 385–472.

Acemoglu, Daron, Tristan Reed, and James A. Robinson (2014): Chiefs: Economic Development and Elite Control of Civil Society in Sierra Leone, *Journal of Political Economy*, 122(2): 319–368.

Acemoglu, Daron, and James A. Robinson (2013): *Why nations fail: The origins of power, prosperity, and poverty*, Paperback edition, Profile Books, London.

Acemoglu, Daron, and James A. Robinson (2017): The Economic Impact of Colonialism, in Michalopoulos, Stelios, and Elias Papaioannou, ed., *The Long Economic and Political Shadow of History*, CEPR Press, London, p. 81–89.

Agenor, Pierre-Richard, and Blanca Moreno-Dodson (2006): Public Infrastructure And Growth: New Channels And Policy Implications, *World Bank Policy Research Working Paper*, No. 4064, World Bank, Washington, DC.

Alesina, Alberto, Sebastian Hohmann, Stelios Michalopoulos, and Elias Papaioannou (2021): Intergenerational Mobility in Africa, *Econometrica*, 89(1): 1–35.

Alesina, Alberto, Janina Matuszeski, and William Easterly (2011): ARTIFICIAL STATES, *Journal of the European Economic Association*, 9(2): 246–277.

Alsan, Marcella (2015): The Effect of the TseTse Fly on African Development, *American Economic Review*, 105(1): 382–410.

Archibong, Belinda, and Nonso Obikili (2019): Convict Labor and the Costs of Colonial Infrastructure: Evidence from Prisons in British Nigeria, 1920-1938, available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3395458, Accessed on 20.02.2024.

Bartalotti, Otavio, Gray Calhoun, and Yang He (2017): Bootstrap Confidence Intervals for Sharp Regression Discontinuity Designs, in Cattaneo, Matias D., and Juan Carlos Escanciano, ed., *Regression discontinuity designs*, Emerald Publishing, Bingley, p. 421–453.

Bartalotti, Otávio, and Quentin Brummet (2017): Regression Discontinuity Designs with Clustered Data, in Cattaneo, Matias D., and Juan Carlos Escanciano, ed., *Regression discontinuity designs*, Emerald Publishing, Bingley, p. 383–420.

Benos, Nikos, and Stefania Zotou (2014): Education and Economic Growth: A Meta-Regression Analysis, *World Development*, 64: 669–689.

Bertazzini, Mattia C. (2022): The long-term impact of Italian colonial roads in the Horn of Africa, 1935–2015, *Journal of Economic Geography*, 22(1): 181–214.

Boss, Eelco (2014): The Scramble for Palms, A Comparative Historical Investigation: Two Companies Sourcing Two Commodities in Two Colonies, available at https://studenttheses.uu.nl/handle/20.500.12932/17501, Accessed on 20.02.2024.

Byrne, Liam (2016): *Dragons*, Head of Zeus, London.

Cagé, Julia, and Valeria Rueda (2016): The Long-Term Effects of the Printing Press in Sub-Saharan Africa. *American Economic Journal: Applied Economics*, 8(3): 69–99.

Cagé, Julia, and Valeria Rueda (2020): Sex and the mission: the conflicting effects of early Christian missions on HIV in sub-Saharan Africa, *Journal of Demographic Economics*, 86(3): 213–257.

Calderón, César, Enrique Moral-Benito, and Luis Servén (2015): Is Infrastructure Capital Productive? A Dynamic Heterogeneous Approach. *Journal of Applied Econometrics*, 30(2): 177–198.

Calonico, Sebastian, Matias D. Cattaneo, Max H. Farrell, and Rocío Titiunik (2017): Rdrobust: Software for Regression-discontinuity Designs, *The Stata Journal*, 17(2): 372–404.

Calonico, Sebastian, Matias D. Cattaneo, and Rocio Titiunik (2014): Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs, *Econometrica*, 82(6): 2295–2326.

Cattaneo, Matias D., Nicolas Idrobo, and Rocio Titiunik (2024): *A Practical Introduction to Regression Discontinuity Designs: Extensions*, Cambridge University Press.

Cattaneo, Matias D., Nicolás Idrobo, and Rocío Titiunik (2019): *A Practical Introduction to Regression Discontinuity Designs*, Cambridge University Press.

Cattaneo, Matias D., Luke Keele, and Rocío Titiunik (2023): A guide to regression discontinuity designs in medical applications, *Statistics in Medicine*, 42(24): 4484–4513.

Cattaneo, Matias D., Luke Keele, and Rocio Titiunik (2022): Covariate Adjustment in Regression Discontinuity Designs, available at http://arxiv.org/pdf/2110.08410.pdf, Accessed on 20.02.2024.

Cattaneo, Matias D., and Rocío Titiunik (2022): Regression Discontinuity Designs, *Annual Review of Economics*, 14(1): 821–851.

Center for International Earth Science Information Network - Columbia University (2016): Gridded Population of the World, Version 4 (GPWv4): Population Density Adjusted to Match 2015 Revision of UN WPP Country Totals, available at https://data.jrc.ec.europa.eu/dataset/jrc-ghsl-ghs_built_ldsmt_globe_r2015b, Accessed on 20.02.2024.

Currier, Andy (2022): Colonial Legacy of Exploitation Thrives Today on the PHC Oil Palm Plantations, available at https://www.oaklandinstitute.org/blog/colonial-legacy-exploitation-phc-oil-palm-plantations, Accessed on 16.10.2023.

Currier, Andy, and Frédéric Mousseau (2022): Meet the Investors Behind the PHC Oil Palm Plantations in DRC, available at https://www.oaklandinstitute.org/meet-investors-behind-phc-oil-palm-plantations-drc, Accessed on 17.10.2023.

Dell, Melissa (2010): The Persistent Effects of Peru's Mining Mita, *Econometrica*, 78(6): 1863–1903.

Dell, Melissa, and Benjamin Olken (2020): The Development Effects of the Extractive Colonial Economy: The Dutch Cultivation System in Java, *Review of Economic Studies*, 87(1): 164–203.

Didan, Kamel, and Armando Barreto (2016): NASA MEaSUREs Vegetation Index and Phenology (VIP) Phenology EVI2 Yearly Global 0.05Deg CMG, available at https://lpdaac.usgs.gov/products/vipphen_evi2v004/, Accessed on 11.02.2024.

Dimico, Arcangelo (2014): Poverty trap and educational shock: Evidence from missionary fields, *QUCEH Working Paper Series*, No. 14-07, Queen's University Centre for Economic History, Belfast.

Dissou, Yazid, and Selma Didic (2013): Infrastructure and Growth, in Dissou, Yazid, Jean-Yves Duclos, Luca Tiberti, and John Cockburn, ed., *Infrastructure and Economic Growth in Asia*, Springer Open, Cham et al., p. 5–45.

Ergo, André-Bernard (n.d.): Histoire de l'Elaéiculture au Congo Belge, available at https://www.congoforum.be/Upldocs/palmhisco-5.pdf, Accessed on 15.10.2023.

Erlei, Mathias (2016): *Neue Institutionenökonomik*, 3rd edition, Schäffer-Poeschel Verlag für Wirtschaft Steuern Recht GmbH, Freiburg.

Fieldhouse, David K. (1978): *Unilever Overseas: The anatomy of a multinational, 1895 - 1965*, Croom Helm, London.

Foster, Vivien, Nisan Gorgulu, Stéphane Straub, and Maria Vagliasindi (2023): The Impact of Infrastructure on Development Outcomes: A Qualitative Review of Four Decades of Literature, *World Bank Policy Research Working Papers*, No. 10343, World Bank.

Frankema, Ewout (2012): The origins of formal education in sub-Saharan Africa: was British rule more benign?, *European Review of Economic History*, 16(4): 335–355.

Fuje, Habtamu, and Jiaxiong Yao (2022): Africa's Rapid Economic Growth Hasn't Fully Closed Income Gaps, available at https://www.imf.org/en/Blogs/Articles/2022/09/20/africas-rapid-economic-growth-hasnt-fully-closed-income-gaps, Accessed on 10.09.2023.

Garcia Ponce, Omar, and Leonard Wantchekon (2011): Echoes of Colonial Repression: The Long-Term Effects of the 1947 Revolt upon Political Attitudes in Madagascar, *APSA 2011 Annual Meeting Paper*, Princenton University, available at http://www.princeton.edu/~lwantche/Echoes_of_Colonial_Repression, Accessed on 20.02.2024.

Günther, Isabel, Kenneth Harttgen, and Katharina Michaelowa (2021): *Einführung in die Entwicklungsökonomik*, UVK Verlag, Konstanz.

Harris, Ian, Philip Jones, Timothy Osborn, and David Lister (2014): Updated high–resolution grids of monthly climatic observations – the CRU TS3 .10 Dataset, *International Journal of Climatology*, 34(3): 623–642.

He, Yang, and Otávio Bartalotti (2020): Wild bootstrap for fuzzy regression discontinuity designs: obtaining robust bias-corrected confidence intervals, *The Econometrics Journal*, 23(2): 211–231.

Henn, Soeren J., Horacio Larreguy, and Carlos Schmidt-Padilla (2023): Missionary Activity, Education, and Long-run Political Development: Evidence from Africa, available at https://soerenhenn.com/files/Henn_Missionaries.pdf, Accessed on 20.02.2023.

Henriet, Benoît (2015): "Elusive natives": escaping colonial control in the Leverville oil palm concession, Belgian Congo, 1923–1941, *Canadian Journal of African Studies*, 49(2): 339–361.

Henriet, Benoît (2017): Ordering the Wetlands. Policing and Legitimate Violence in the Leverville Concession (Belgian Congo, 1911-1920), in Blanchard, Emmanuel, Marieke Bloembergen, and Amandine Lauro, ed., *Policing in Colonial Empires*, Peter Lang Verlag, Brussels, p. 41–61.

Henriet, Benoît (2021): *Colonial Impotence: Virtue and Violence in a Congolese Concession (1911-1940)*, De Gruyter Oldenbourg, München and Wien.

Henriet, Benoît, Bérengère Piret, and Nathalie Tousignant (2017): "Travail et Progrès". Labour Legislation in Belgian Congo (1906-1960), *Rechtskultur. European Journal of Legal History*, 5: 36–47.

Herranz-Loncán, Alfonso, and Johan Fourie (2018): "For the public benefit"? Railways in the British Cape Colony, *European Review of Economic History*, 22(1): 73–100.

Hijmans, Robert J., Susan E. Cameron, Juan L. Parra, Peter G. Jones, and Andy Jarvis (2005): Very high resolution interpolated climate surfaces for global land areas, *International Journal of Climatology*, 25(15): 1965–1978.

Hochschild, Adam (2012): *King Leopold's ghost: A story of greed, terror, and heroism in Colonial Africa*, Paperback edition, Pan Books, London.

Holmgren, Johan, and Axel Merkel (2017): Much ado about nothing? – A meta-analysis of the relationship between infrastructure and economic growth, *Research in Transportation Economics*, 63: 13–26.

Horowitz, Joel L. (2019): Bootstrap Methods in Econometrics, *Annual Review of Economics*, 11(1): 193–224.

Huillery, Élise (2009): History Matters: The Long-Term Impact of Colonial Public Investments in French West Africa, *American Economic Journal: Applied Economics*, 1: 176–215.

Imbens, G., and K. Kalyanaraman (2012): Optimal Bandwidth Choice for the Regression Discontinuity Estimator, *The Review of Economic Studies*, 79(3): 933–959.

Janvry, Alain de, and Elisabeth Sadoulet (2016): *Development economics: Theory and practice*, Routledge, London and New York NY.

Jedwab, Remi, Edward Kerby, and Alexander Moradi (2017): History, Path Dependence and Development: Evidence from Colonial Railways, Settlers and Cities In Kenya, *The Economic Journal*, 127(603): 1467–1494.

Jedwab, Remi, Felix Meier zu Selhausen, and Alexander Moradi (2022): The economics of missionary expansion: evidence from Africa and implications for development, *Journal of Economic Growth*, 27(2): 149–192.

Jedwab, Remi, and Alexander Moradi (2016): The Permanent Effects of Transportation Revolutions in Poor Countries: Evidence from Africa, *The Review of Economics and Statistics*, 98(2): 268–284.

Jedwab, Remi, and Adam Storeygard (2022): The Average and Heterogeneous Effects of Transportation Investments: Evidence from Sub-Saharan Africa 1960-2010, *Journal of the European Economic Association*, 20(1): 1–38.

Jin, Ze et al. (2023): Regression Discontinuity Design Application (R package rddapp version 1.3.2), available at https://cran.r-project.org/web/packages/rddapp/index.html, Accessed on 15.01.2023.

Juif, Dácil, and Ewout Frankema (2018): From Coercion to Compensation: Institutional Responses to Labour Scarcity in the Central African Copperbelt, *Journal of Institutional Economics*, 14(2): 313–343.

Kessides, Christine (1996): A Review of Infrastructure's Impact on Economic Development, in Batten, David F., and Charlie Karlsson, ed., *Infrastructure and the Complexity of Economic Development*, Springer, Berlin and Heidelberg, p. 213–230.

Kisangani, Emizet F. (2023): *The Belgian Congo as Developmental State: Revisiting colonialism*, Routledge, Abingdon, Oxon and New York, NY.

Kiszewski, Anthony et al. (2004): A global index representing the stability of malaria transmission, *The American Journal of Tropical Medicine and Hygiene*, 70(5): 486–498.

Lechler, Marie, and Lachlan McNamee (2018): Indirect Colonial Rule Undermines Support for Democracy: Evidence From a Natural Experiment in Namibia, *Comparative Political Studies*, 51(14): 1858–1898.

Leschke, Martin (2015): *Ökonomik der Entwicklung: Eine Einführung aus institutionenökonomischer Sicht*, 2nd edition, Verlag für Nationalökonomie Management und Politikberatung (NMP), Bayreuth.

Lewis, Brian (2017): *So Clean: Lord Leverhulme, Soap and Civilisation*, Manchester University Press, Oxford.

Loffman, Reuben, and Benoît Henriet (2019): 'We Are Left with Barely Anything': Colonial Rule, Dependency, and the Lever Brothers in the Belgian Congo, 1911–1960, *The Journal of Imperial and Commonwealth History*, 48(1): 71–100.

Lowes, Sara, and Eduardo Montero (2021): Concessions, Violence, and Indirect Rule: Evidence from the Congo Free State, *The Quarterly Journal of Economics*, 136(4): 2047–2091.

Marchal, Jules (2008): *Lord Leverhulme's ghosts: Colonial exploitation in the Congo*, Verso, London and New York.

Martino Pesaresi et al. (2015): GHS-BUILT R2015B - GHS built-up grid, derived from Landsat, multitemporal (1975, 1990, 2000, 2014) - OBSOLETE RELEASE, available at https://data.jrc.ec.europa.eu/dataset/jrc-ghsl-ghs_built_ldsmt_globe_r2015b, Accessed on 20.02.2024.

Mayala, Benjamin, Thomas D. Fish, David Eitelberg, and Trinadh Dontamsetti (2018): The DHS Program Geospatial Covariate Datasets Manual (Second Edition). available at https://spatialdata.dhsprogram.com/covariates/, Accessed on 10.02.2024.

Médecins Sans Frontières (2022): République Démocratique du Congo - Province de la Tshopo, available at https://reliefweb.int/map/democratic-republic-congo/republique-democratique-du-congo-province-de-la-tshopo-carte-de-reference-16052022, Accessed on 12.02.2024.

Meier, Gerald M., and James E. Rauch (2005): *Leading Issues in Economic Development*, 8th edition, Oxford University Press, New York.

Meier zu Selhausen, Felix (2019): Missions, Education and Conversion in Colonial Africa, in Mitch, David, ed., *Globalization and the Rise of Mass Education*, Springer International Publishing AG, Cham, p. 25–59.

Méndez, Esteban, and Diana van Patten (2022): Multinationals, Monopsony, and Local Development: Evidence From the United Fruit Company, *Econometrica*, 90(6): 2685–2721.

Michalopoulos, Stelios, and Elias Papaioannou (2016): The Long-Run Effects of the Scramble for Africa, *American Economic Review*, 106(7): 1802–1848.

Michalopoulos, Stelios, and Elias Papaioannou (2020): Historical Legacies and African Development, *Journal of Economic Literature*, 58(1): 53–128.

Miguel, Edward, and Michael Kremer (2004): Worms: Identifying Impacts on Education and Health in the Presence of Treatment Externalities, *Econometrica*, 72(1): 159–217.

Murdock, Georg Peter et al. (1999): D-PLACE dataset derived from Murdock et al. 1999 'Ethnographic Atlas', available at https://d-place.org/contributions/EA, Accessed on 20.02.2024.

Nicolaï, Henri (2013): Le Congo et l'huile de palme. Un siècle. Un cycle ?, *Belgeo*, (4).

Nordhaus, Wiliam, and Xi Chen (2008): Geographically based Economic data, available at https://gecon.yale.edu/, Accessed on 20.02.2024.

Nunn, Nathan (2008): The Long Term Effects of Africa's Slave Trades, *Quarterly Journal of Economics*, 123(1): 139–176.

Nunn, Nathan (2010): Religious Conversion in Colonial Africa, *American Economic Review*, 100(2): 147–152.

Nunn, Nathan (2014): Gender and Missionary Inluence in Colonial Africa, in Akyeampong, Emmanuel, Nathan Nunn, Robert H. Bates, and James Robinson, ed., *Africa's Development in Historical Perspective*, Cambridge University Press, New York, p. 489–509.

Nunn, Nathan, and Diego Puga (2012): Ruggedness: The Blessing of Bad Geography in Africa, *The Review of Economics and Statistics*, 94(1): 20–36.

Nunn, Nathan, and Leonard Wantchekon (2011): The Slave Trade and the Origins of Mistrust in Africa, *American Economic Review*, 101(7): 3221–3252.

Okoye, Dozie, Roland Pongou, and Tite Yokossi (2017): On the Dispensability of New Transportation Technologies: Evidence from the Heterogeneous Impact of Railroads in Nigeria, available at https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2926912, Accessed on 20.02.2024.

Perkins, Dwight H., Steven C. Radelet, David L. Lindauer, and Steven Block (2013): *Economics of Development*, 7th edition, Norton, New York.

Port Sunlight Village Trust (2022): Racism, the Belgian Congo, and William Lever, available at https://www.portsunlightvillage.com/wp-content/uploads/2022/06/PSVT_Booklet_Racism_the_Belgian_Congo_and_William_Lever_-_V_FINAL_14.06.22.pdf, Accessed on 20.02.2024.

Ramankutty, Navin, Jonathan A. Foley, John Norman, and Kevin McSweeney (2002): The global distribution of cultivable lands: current patterns and sensitivity to possible climate change, *Global Ecology and Biogeography*, 11(5): 377–392.

Ray, Debraj (1998): *Development Economics*, Princeton University Press, Princeton, N.J.

RIAO-RDC et al. (2021): Development Finance as Agro-Colonialism: European Development Bank funding of Feronia-PHC oil palm plantations in the Democratic Republic of Congo, available at https://www.thecornerhouse.org.uk/sites/thecornerhouse.org.uk/files/Development%20Finance%20as%20Agro_Colonialism_Feronia%20PHC.pdf, Accessed on 16.10.2023.

Ridhwan, Masagus M., Peter Nijkamp, Affandi Ismail, and Luthfi M Irsyad (2022): The effect of health on economic growth: a meta-regression analysis, *Empirical Economics*, 63(6): 3211–3251.

Roland, Gérard (2016): *Development Economics*, Routledge Taylor & Francis Group, London and New York, NY.

Roome, William (1924): Ethnographic Survey of Africa: Showing the Tribes and Languages; Also the Stations of Missionary Societies, available at https://archives.ungeneva.org/ethnographic-survey-of-africa-showing-the-tribes-and-languages-also-the-stations-of-missionary-societies-2, Accessed on 12.02.2024.

Sawada, Yasuyuki (2015): The Impacts of Infrastructure in Development: A Selective Survey, *ADBI Working Paper Series*, No. 511, Asian Development Bank Institute.

Seibert, Julia (2009): Travail Libre ou Travali Forc? Die Arbeiterfrage im belgischen Kongo 1908-1930, *Journal of Modern European History*, 7(1): 93–108.

Seibert, Julia (2016): *In die globale Wirtschaft gezwungen: Arbeit und kolonialer Kapitalismus im Kongo (1885-1960)*, Campus Verlag, Frankfurt and New York.

Tadei, Federico (2020): Measuring extractive institutions: colonial trade and price gaps in French Africa, *European Review of Economic History*, 24(1): 1–23.

Tierney, Josephine (2021): Report of Scoping Survey of the Lever Brothers' Plantations in the Solomon Islands and the Congo, 1900-1930, available at https://royalafricansociety.org/wp-content/uploads/2022/05/University-of-Liverpool-Scoping-Report-Lever-Bros-Plantations-in-the-Congo-and-Solomon-Islands-1900-to-1930.pdf, Accessed on 15.08.2023.

Timilsina, Govinda, Gal Hochman, and Ze Song (2020): Infrastructure, Economic Growth, and Poverty: A Review, *World Bank Policy Research Working Paper*, No. 9258, World Bank.

Timilsina, Govinda, David I. Stern, and Debasish K. Das (2023): Physical infrastructure and economic growth, *Applied Economics*, 56(18): 2142–2157.

Todaro, Michael P., and Stephen C. Smith (2020): *Economic Development*, 13th edition, Pearson, Harlow, England et al.

UNCTAD (2022): UN list of least developed countries, available at https://unctad.org/topic/least-developed-countries/list, Accessed on 20.02.2024.

UNDP (2019): As Africa gains development ground, new inequalities emerge, says new UNDP report | United Nations Development Programme, available at https://www.undp.org/africa/press-releases/africa-gains-development-ground-new-inequalities-emerge-says-new-undp-report, Accessed on 20.02.2023.

Unilever (2023): Brands, available at https://www.unilever.com/brands/, Accessed on 20.02.2024.

Valero, Anna (2021): Education and economic growth, *CEP Discussion Papers*, No. 1764, Centre for Economic Performance.

van Waijenburg, Marlous (2018): Financing the African Colonial State: The Revenue Imperative and Forced Labor, *The Journal of Economic History*, 78(1): 40–80.

Wantchekon, Leonard, Marko Klašnja, and Natalija Novta (2015): Education and Human Capital Externalities: Evidence from Colonial Benin, *The Quarterly Journal of Economics*, 130(2): 703–757.

Weiss, D. J. et al. (2018): A global map of travel time to cities to assess inequalities in accessibility in 2015, *Nature*, 553(7688): 333–336.

Wilson, Charles (1968): *The History of Unilever: A study in economic growth and social change; in two volumes*, Cassell, London.

Appendices

Appendix A Variable Definitions and Data Sources

A.1 Dependent Variables from the DHS Survey Data

The dependent variables used in this paper are taken from the DHS Surveys which were conducted in 2007 and from 2013 to 2014 all over the Congo. The DHS data includes information about asset ownership, education, and health for individuals for each cluster location. For the survey in 2007, 9995 women and 4757 men were asked. For the survey from 2013 to 2014, 18827 women and 8656 men were asked. The women are between 15 and 49 years old, the men are between 15 and 59 years old, and the children are up to 5 years old. Due to confidentiality reasons, the geographic location of the DHS clusters is displaced by up to 10 km in rural areas and up to 2 km in urban areas. Normally, this is a classical measurement error that would bias the coefficients towards zero. More detailed information about the definitions of the variables and the survey data can be found at http://dhsprogram.com/data/Data-Variables-and-Definitions.cfm. The variables are modified in the same way as in Lowes and Montero (2021).

- **Wealth Factor Score:** The wealth factor score is calculated by using a principal component analysis for asset ownership for each respondent in the DHS survey. As in Lowes and Montero (2021), the logarithm of the wealth factor is used following the formula: *Log(wealth factor score) = log(wealth index factor score + min (wealth index factor score) + 1)*. In Lowes and Montero (2021), this variable is mostly called wealth score.

- **Wealth Index:** The wealth index is a categorical variable that ranges from 1 to 5. 1 represents the poorest quintile and 5 represents the richest quintile in the complete sample for 2007 or in the complete sample for 2014. Its calculation is based on the wealth index factor score introduced in the previous bullet point.

- **Years of Education:** This variable represents the total number of years of education for each individual in years.

- **Literacy:** This variable is an indicator variable which equals 0 when the person that is asked cannot read at all and 1 when the person can read at least parts of a sentence or a whole sentence.

- **Height/Age Percentile:** The DHS survey data includes information about the height (in centimeters) and weight (in kilograms) of women and children. The height/age percentile is calculated as the quotient of height and weight, for which then the percentile relative to the women or children sub-sample is determined. This variable is normalized so that its values range between 0 and 10,000, but for this paper, the values are divided by 10,000.

A.2 DHS Covariables for Balance Check

These variables were already attached to DHS dataset and apart from gender, age, age squared and dummy survey year, their values are assigned to the respective DHS cluster. The exact description by Mayala et al. (2018) for each variable can be found at https://spatialdata.dhsprogram.com/covariates/.

- **Gender, Age, Age squared and Dummy Survey Year:** The information of these variables is already included in the DHS dataset for every individual and indicates each respondent's gender, age, the squared value of the age, and whether the respondent was surveyed either in the survey round of 2007 or in the survey round of 2013/2014.

- **Population Density 2015:** This variable represents the average population density within a buffer zone of 2 km for urban areas and 10 km for rural areas around the DHS cluster location in 2015. Its original calculation is based on 1 km grid cells. In the description for the DHS covariates, this variable is called UN Population Density Year. More information is provided by Center for International Earth Science Information Network - Columbia University (2016) and can be found at https://data.jrc.ec.europa.eu/dataset/jrc-ghsl-ghs_built_ldsmt_globe_r2015b.

- **Level of Urbanization:** This index measures the degree of urbanization within a buffer zone of 2 km for urban areas and 10 km for rural areas around the DHS cluster location in 2014. The scale of this index ranges from 0, indicating extreme rurality, to 1, indicating extreme urbanization. The original calculation of this variable is based on 1 km grid cells. In the description for the DHS covariates, this variable is called BUILT Population Year. More information is provided by Martino Pesaresi et al. (2015) at can be found at https://data.jrc.ec.europa.eu/dataset/jrc-ghsl-ghs_built_ldsmt_globe_r2015b.

- **Spatial Economic Output:** This variable indicates the mean geographic economic output in the form of the purchasing power parity in 2005 US Dollar within a buffer zone of 2 km for urban areas and 10 km for rural areas around the DHS cluster location. Its original calculation is based on 5 km grid cells, and, unlike conventional economic indicators, demographic and geophysical aspects are also taken into account for its calculation. In the description for the DHS covariates, this variable is called Gross Cell Production. More information is provided by Nordhaus and Chen (2008) and can be found at https://gecon.yale.edu/.

- **Travel Times 2015:** This variable shows the average travel time in minutes to reach the next urban center from DHS survey locations in 2015. The data can be obtained from the Malaria Atlas Project at https://data.malariaatlas.org/maps?layers=Accessibility:201501_Global_Travel_Time_to_Cities,Malaria:202206_Global_Pf_Parasite_Rate. More information is also provided by Weiss et al. (2018).

- **Mean Temperature 2015:** This variable represents the average annual temperature within a buffer zone of 2 km for urban areas and 10 km for rural areas around the DHS cluster location in 2015. Its original calculation is based on 0.5 decimal degrees (approximately 55 km) grid cells. The data can be obtained from the following link https://catalogue.ceda.ac.uk/uuid/58a8802721c94c66ae45c3baa4d814d0. More information is also provided by Harris et al. (2014).

- **Enhanced Vegetation Index:** This index measures the level of vegetation within a buffer zone of 2 km for urban areas and 10 km for rural areas around the DHS cluster location in 2015. It ranges from 0, indicating the lowest level of vegetation, to 10,000, indicating the highest level of vegetation. For this paper, the value was divided by 100. The vegetation broadly quantifies the vegetation greenness by using surface reflectance data. Its original calculation is based on 0.05 decimal degrees (approximately 5 km) grid cells. More information is provided by Didan and Barreto (2016) and can be found at https://lpdaac.usgs.gov/products/vipphen_evi2v004/.

A.3 Further Geographic Covariables for Balance Check

Apart from the land suitability variable, all variables are taken over from the paper by Lowes and Montero (2021), which is why the definitions of the following variables can also be found in the appendix of their paper. All these variables were checked if they can be used for the empirical estimation strategy of this thesis. In addition to the concession areas Abir and Anversoise, Lowes and Montero (2021) also estimated the effects of all concession areas that existed at the same time in the Congo Free State under the rule of King Leopold II. This is the reason why this data is also available for more DHS clusters than those which are inside or close to the Abir and Anversoise concessions. The majority of these variables are presented as average values of 20 km by 20 km grid cells.

- **Elevation:** This variable shows the mean elevation in meters for 20 km by 20 km grid cells. The original values are based on 30 arc-second resolution (approximately 1 km grid cell size at the equator). Since the value for the cluster 185 of the 2007 DHS survey was missing, it is replaced by the value 470 m obtained from Google Earth. The data stems from the Global Climate Database by Hijmans et al. (2005) and can be found at https://www.worldclim.org/.

- **Ruggedness:** This variable indicates the average deviation in elevation from different points within a grid cell. More specifically, the sum of squared differences regarding the elevation between nine neighbouring points within a grid cell of the size of 30 arc-seconds is calculated and then the square root is taken from this sum. The unit of measurement used by Lowes and Montero (2021) and in this paper takes the mean ruggedness value of each 30 arc-seconds cell (approximately 1 km grid cell size at the equator) to construct the average terrain ruggedness for 20 km by 20 km grid cells. This index was created by Nunn and Puga (2012).

- **Land Suitability:** This index represents the probability that a certain grid cell will be cultivated, which is why its values range from 0 to 1. The size of each grid cell is 0.5 degrees (approximately 55 km). The variable called soil suitability used in Lowes and Montero (2021) is probably a component of this index, but unfortunately their dataset only includes values for DHS clusters close to the Abir and Aversoise concessions. The data can be obtained from https://sage.nelson.wisc.edu/global-land-use/grid.php and more information about this index is provided by Ramankutty et al. (2002).

- **Average Rainfall:** This variable represents the average rainfall in millimeters per year within 20 km by 20 km grid cells. The original values of this variable are measured as monthly means which are then taken to construct the average rainfall per year. This variable also stems from the Global Climate Database by Hijmans et al. (2005) and can be found at https://www.worldclim.org/.

- **Standard Deviation Rainfall:** The Standard Deviation for Rainfall is calculated by using the average rainfall per month introduced in the preceding bullet point, resulting in a measure for the variation of rainfall per year.

- **Malaria Suitability:** This variable provides estimated prevalence rates for severe forms of malaria and takes into account the dominant mosquito species present in a specific region. The mean values of this index for a grid cell size of 20 km by 20 km are taken over from Lowes and Montero (2021). This index was created by Kiszewski et al. (2004).

- **Tsetse Fly Suitability:** This index indicates a standardized potential steady-state tsetse fly population. Its calculation is based on historical geographic climate data such as temperature and humidity. The mean values of this index for a grid cell size of 20 km by 20 km are taken over from Lowes and Montero (2021). The original index is called TseTse suitability index (TSI) and stems from Alsan (2015).

- **Road Density**: This index indicates the road network density in 1968 by measuring the total length (in meters) of roads within a 20 km by 20 km grid cell and then dividing it by the surface area (kilometers squared) of the grid cell. This variable was constructed by Lowes and Montero (2021) by using the following map which is available at: https://maps.lib.utexas.edu/maps/ams/africa/africa_index.html.

- **Distance to Kinshasa:** This measure is calculated by Lowes and Montero (2021) and represents the euclidean distance in kilometers from the center of the respective 20 km by 20 km grid cell to the capital city (Kinshasa).

- **Distance to Coast:** This measure is calculated by Lowes and Montero (2021) and represents the euclidean distance in kilometers from the center of the respective 20 km by 20 km grid cell to the coast.

A.4 Colonial Variables for Balance Check

The variables colonial population density, centralization, polygynous and hereditary local headman selection are taken from the Ethnographic Atlas by Murdock et al. (1999). Unfortunately, data is only available for some ethnic tribes, which is why these colonial characteristics are not assigned to every DHS cluster.

- **Inside AA**: Since the Abir and Aversoise concessions are overlapping with the Elisabetha and Alberta circle, there are some DHS clusters that are located within or close to these 4 concession areas. To account for the effect detected by Lowes and Montero (2021), inside AA is an indicator that equals 1 if the DHS cluster used for palm oil concessions is also located inside or close to the Abir and Aversoise concessions. It is 0 otherwise.

- **Number Enslaved in 1000:** This data provides values for the number of enslaved natives (in 1000s) for each ethnic tribe during the Atlantic slave trade. The data stems from Nunn and Wantchekon (2011) and ethnic tribes in Africa are based on the maps provided by Murdock et al. (1999).

- **Missionary Stations 1924:** This index equals 1 if a protestant or catholic mission in 1924 is located closer than 20 km to a DHS cluster location and 0 otherwise. The locations of those missions can be obtained using a GIS shapefile provided by Nunn (2010) and available at https://nathannunn.arts.ubc.ca/data/. This shapefile is based on the map by Roome (1924) available at https://archives.ungeneva.org/ethnographic-survey-of-africa-showing-the-tribes-and-languages-also-the-stations-of-missionary-societies-2/download.

- **Colonial Population Density:** This variable shows approximate values for population density for each tribe in the colonial era. This data is estimated by Murdock et al. (1999) and can be downloaded from https://d-place.org/contributions/EA. The values of the population density are indicated as the average size of the local tribe, categorized into bins including: 0-50, 50-99, 100-199, 200-399, 400-1,000, 1,000-5,000, 5,000-50,000, and over 50,000. As in Lowes and Montero (2021), the measurement approach by Alsan (2015) is applied, which takes the logarithm of inhabitants per square kilometer for each ethnic group.

- **Centralization:** Following Alsan (2015) and Murdock et al. (1999), centralization is an indicator that equals 1 if there is more than one level of jurisdictional or political authority above the local level. The corresponding abbreviation in the Ethnographic Atlas by **Murcoempty citation** is (EA033).

- **Polygnous:** This indicator variable equals 1 if polygyny is present, either in the form of an atypical co-wives pattern or in the form of a typical co-wives pattern, and 0 otherwise. This information stems from the Ethnographic Atlas by Murdock et al. (1999) and the corresponding abbreviation is (EA008).

- **Hereditary Local Headman Selection:** This indicator variable equals 1 if the succession of the local headmen is determined by either matrilineal or patrilineal inheritance, and 0 otherwise. Since all the tribe areas that overlap with the palm oil concessions show the value 1, this variable is not included in the tables for the balance check. This information stems from the Ethnographic Atlas by Murdock et al. (1999) and the corresponding abbreviation is (EA072).

Appendix B Additional Figures

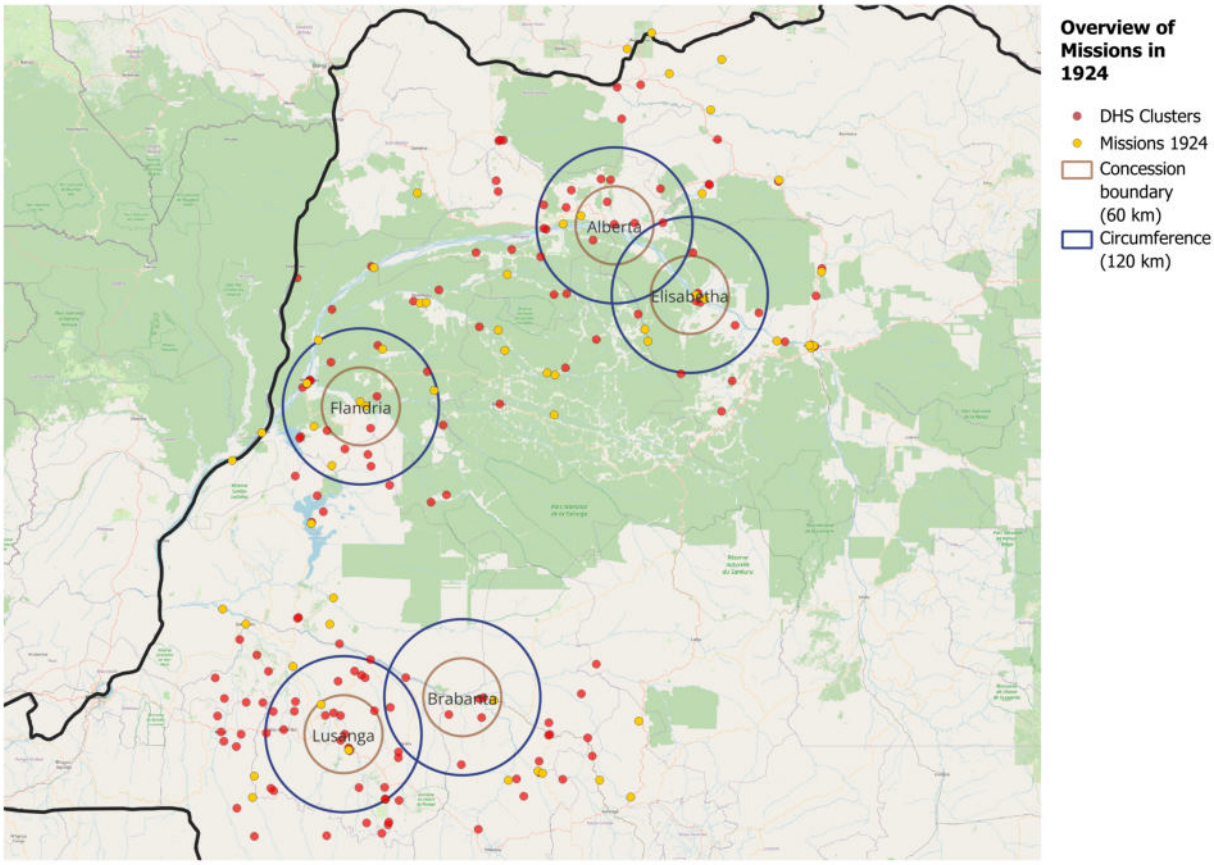


Figure B1: Overview of DHS Clusters and Missions in 1924.

Notes: This figure shows DHS cluster locations around and within the palm oil concessions. The circles outlined in brown represent the concession boundaries, while the circles outlined in blue indicate auxiliary circles/circumferences with a radius of 120 km. The yellow points represent the location of Christian missions in 1924. Own figure created with the software QGIS.

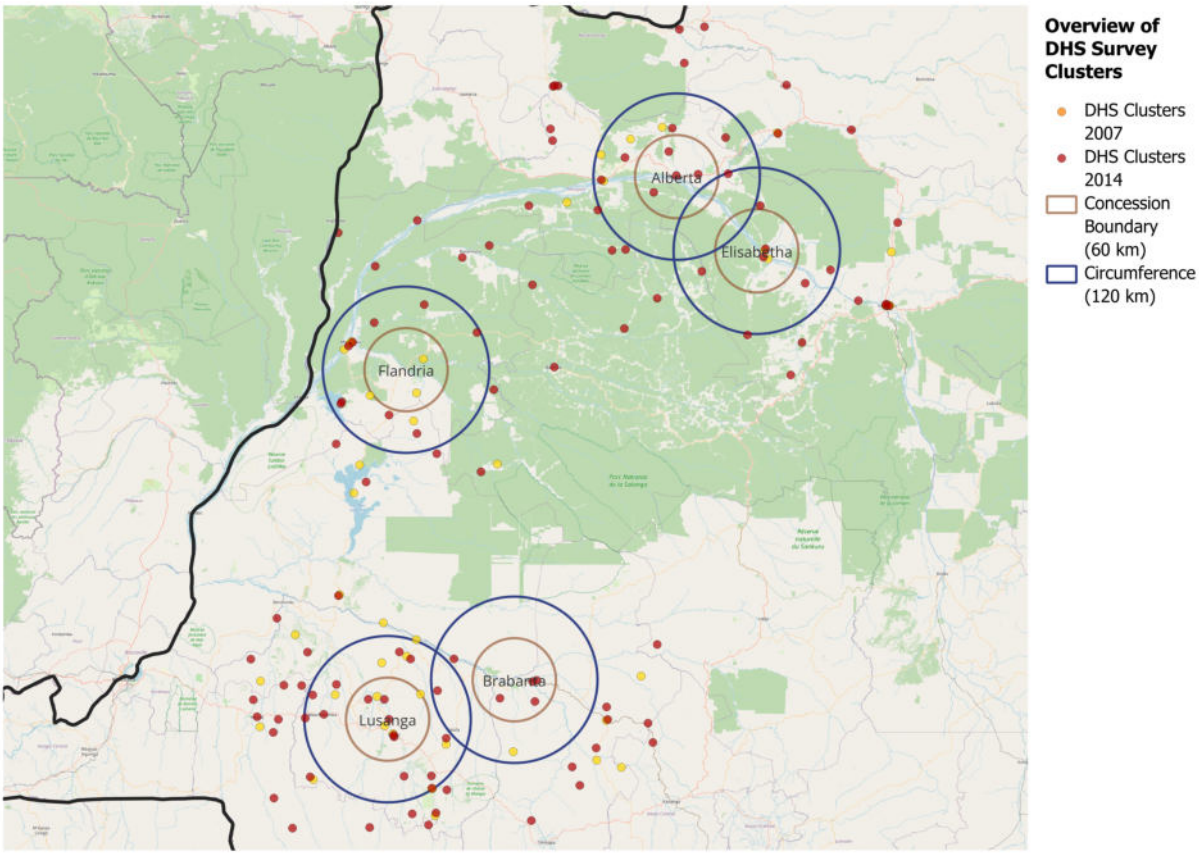


Figure B2: Overview of DHS Survey Locations 2007 and 2014.

Notes: This figure shows the DHS cluster locations of the survey rounds in 2007 and 2013/2014 separately. The circles outlined in brown represent the concession boundaries, while the circles outlined in blue indicate auxiliary circles/circumferences with a radius of 120 km. Own figure created with the software QGIS.

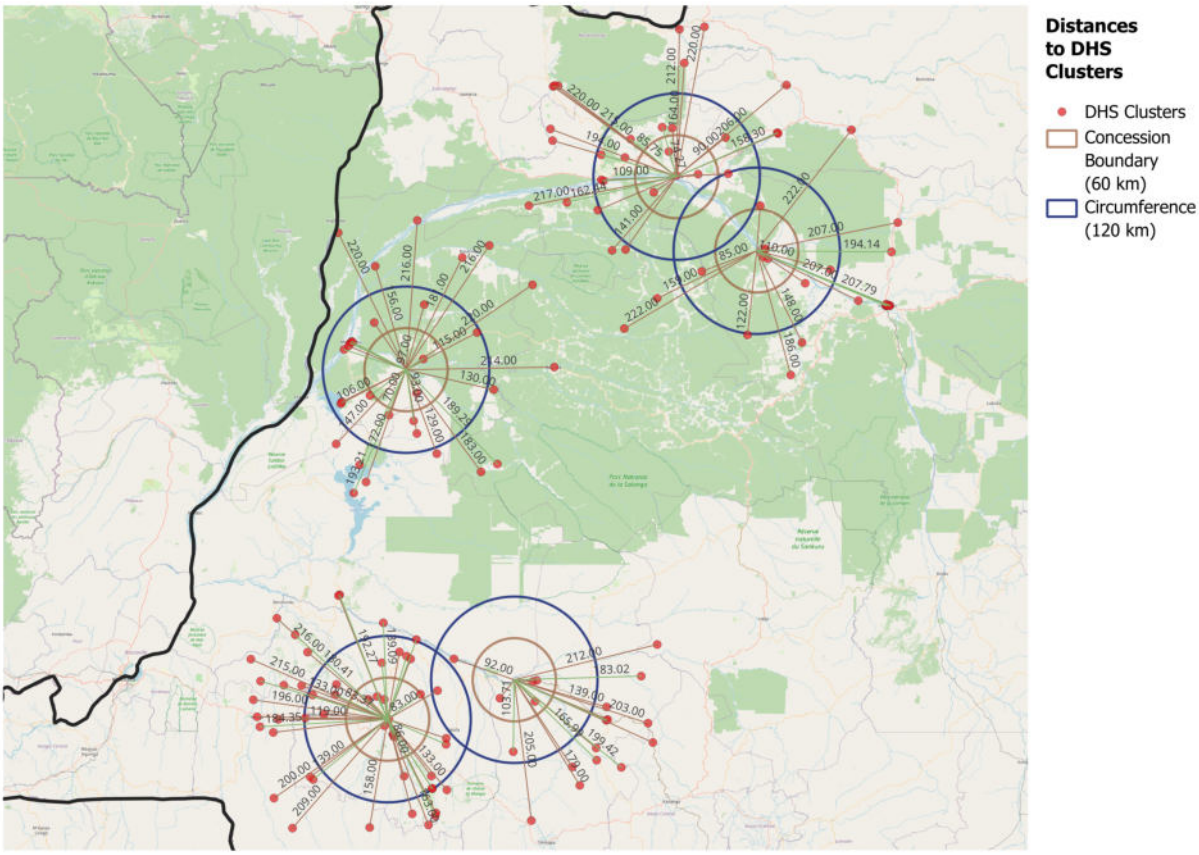


Figure B3: Distances between DHS Clusters and Concession Centers.

Notes: This figure shows the distances between the DHS clusters and the nearest concession center. The circles outlined in brown represent the concession boundaries, while the circles outlined in blue indicate auxiliary circles/circumferences with a radius of 120 km. Own figure created with the software QGIS.

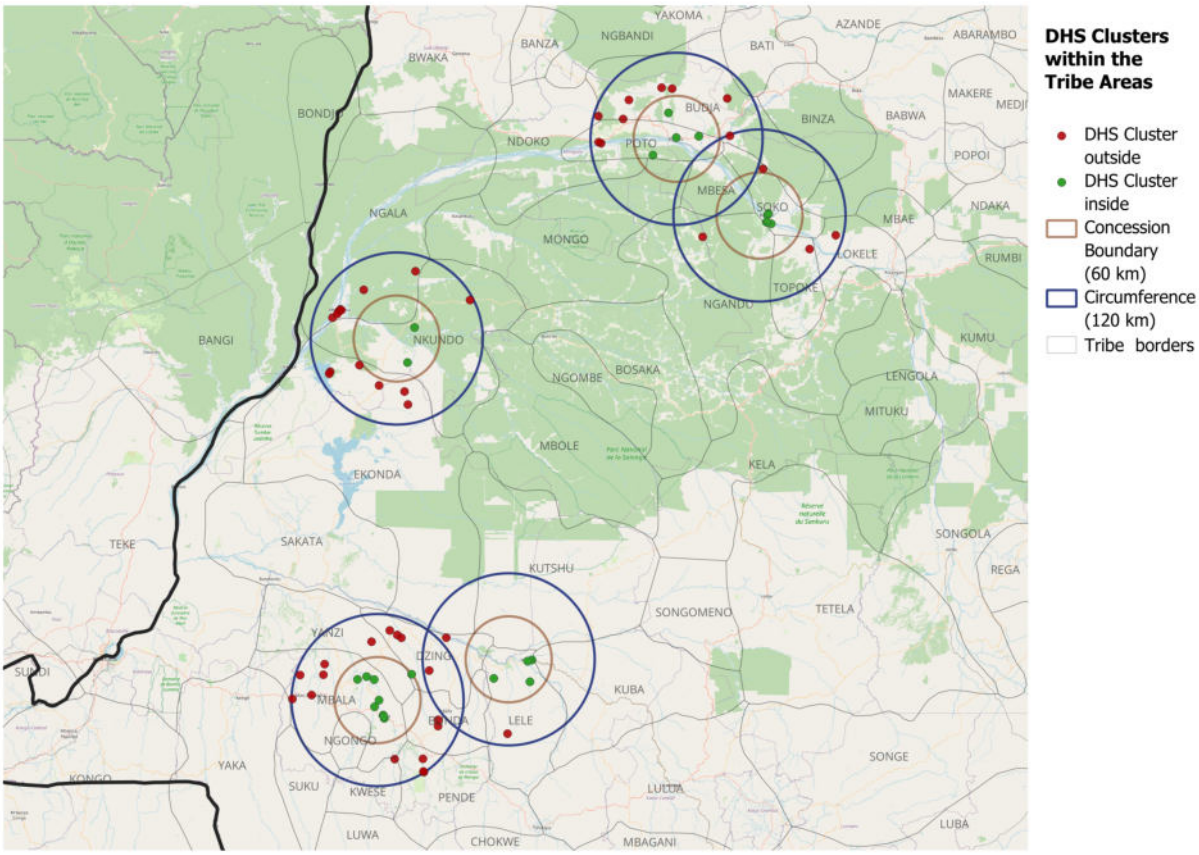


Figure B4: Overview of Ethnic Tribes overlapping with the Concession Areas.

Notes: This figure displays the locations of DHS clusters within the territories of the respective ethnic groups. The circles outlined in brown represent the concession boundaries, while the circles outlined in blue indicate auxiliary circles/circumferences with a radius of 120 km. Own figure created with the software QGIS.

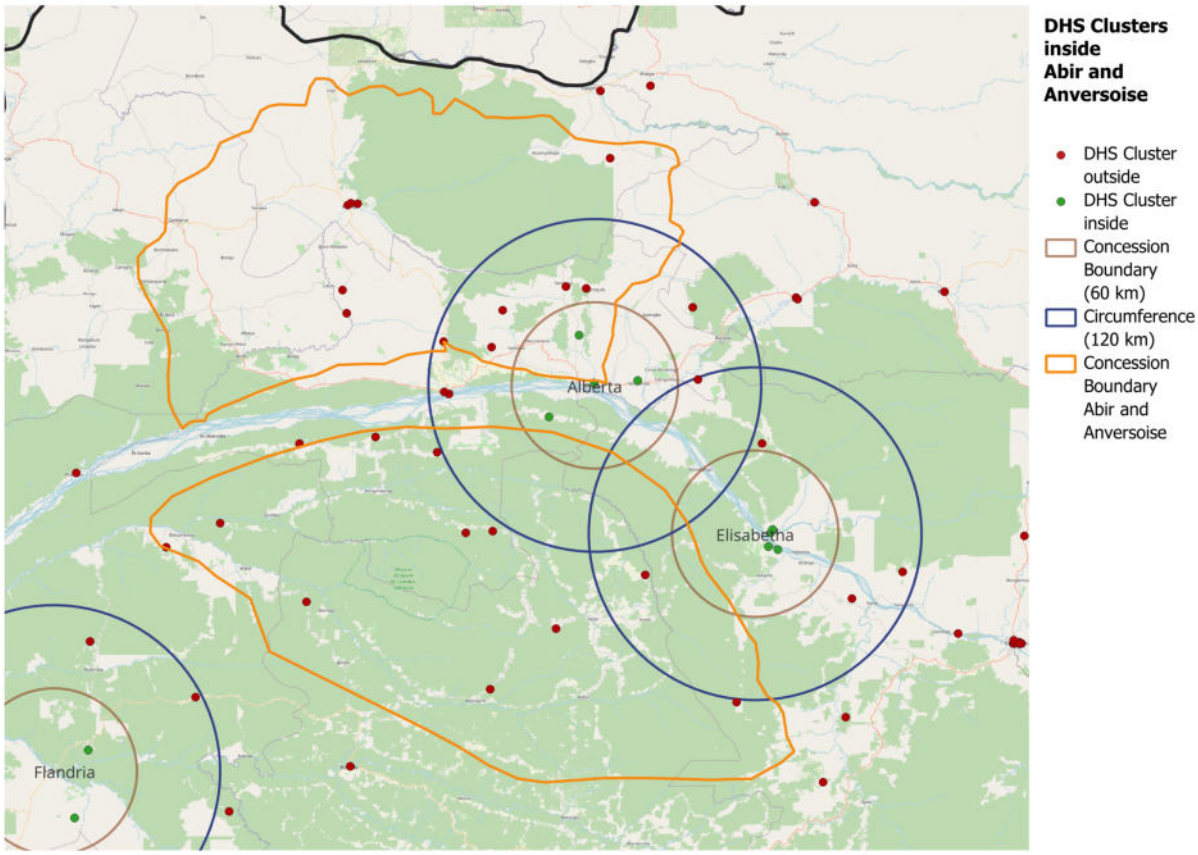


Figure B5: DHS Clusters inside Abir and Anversoise.

Notes: This figure displays the locations of DHS clusters around and within the Abir and Anversoise concessions as well as around and within HCB's palm oil concessions. The borders of the Abir and Anversoise concessions, which were examined by Lowes and Montero (2021), are marked in yellow. The circles outlined in brown represent the boundaries of the palm oil concessions, while the circles outlined in blue indicate auxiliary circles/circumferences with a radius of 120 km. Own figure created with the software QGIS.

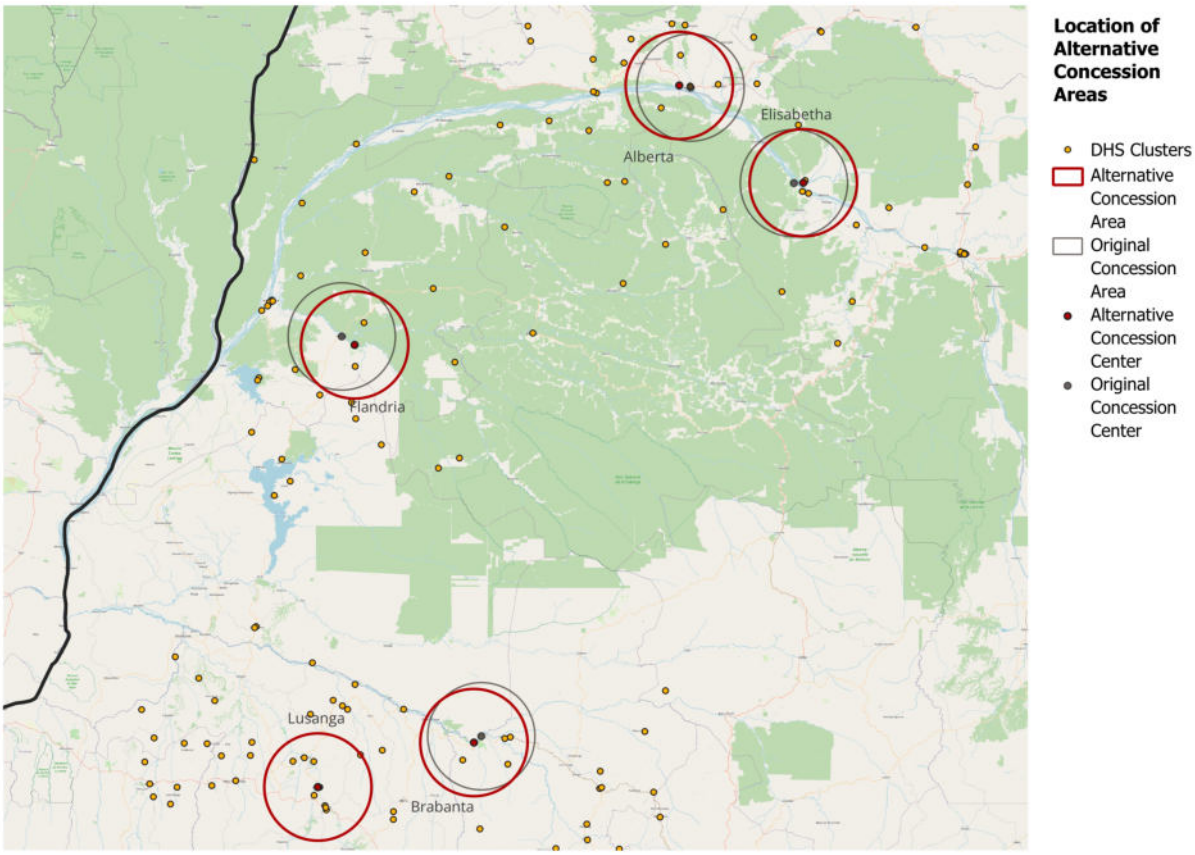


Figure B6: Distances between DHS Clusters and Concession Centers.

Notes: As described in Section 6.4, this figure shows alternative locations for the palm oil concessions outlined in red. The yellow points represent the DHS clusters. The black circles indicate the borders of the original concession areas. Own figure created with the software QGIS.

Source: Henriet, Benoît (2021). Colonial Impotence.

Figure B7: Example of adjacent Concessions.

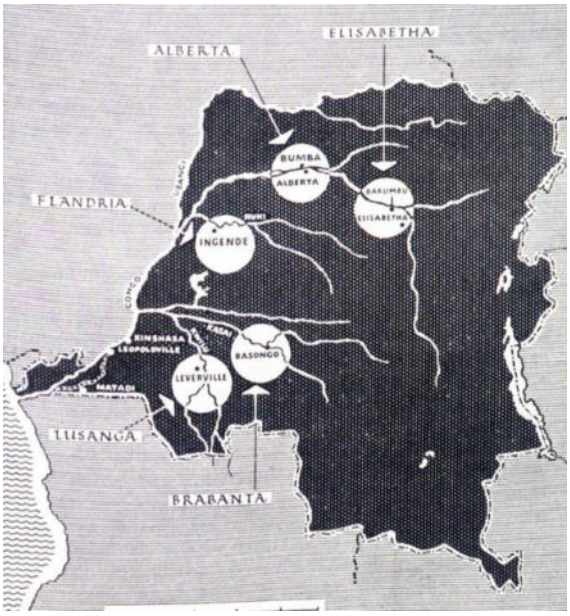


Figure B8: Assumed Location of the Concession Areas in the Book "Unilever Overseas" by Fieldhouse (1978).

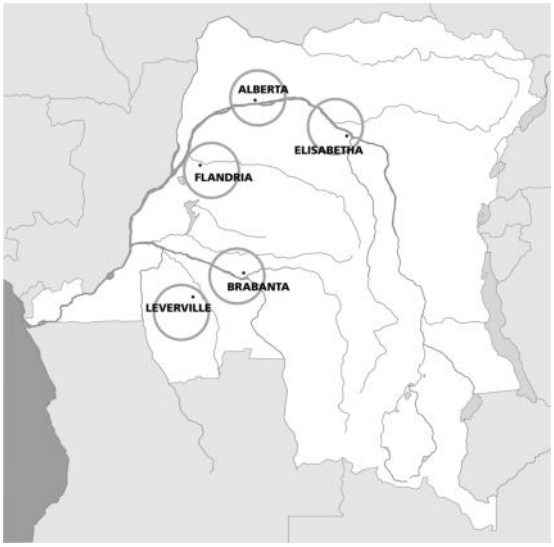


Figure B9: Assumed Location of the Concession Areas in the Book "Colonial Impotence" by Henriet (2021).

Appendix C Additional Theoretical Background

C.1 Extractive Institutions (Detailed Description)

The measures implemented by Lever Brothers and HCB, such as forced labor or the enforcement of monopolies, can be explained in economics with the help of the concept of extractive institutions. However, to understand extractive institutions, it is essential to first clarify what institutions are. Institutions primarily refer to the set of laws, rules, and social norms that shape and constrain human behaviour, guiding it toward a specific direction. These rules play a crucial role in an economy, as they establish the framework for human behaviour and interactions in a free market economy and help to prevent human actions that can be detrimental to the community. For instance, in many free market economies, the formation of cartels that manipulate prices for higher profits is prohibited, as it undermines the efficiency principles of a free market. Another example is the enforcement of property rights, which allows owners to plan for the long term with their property, making them more inclined to invest in it. When the probability of expropriation is high, owners are less willing to invest in their property since the likelihood of a profitable business emerging from it is low. The rule of law is also a crucial aspect of this, as it enables the punishment of rule violations. Additionally, the rule of law can prevent powerful elites from exploiting their fellow human beings at their expense. In general, institutions play a vital role in shaping human behaviour and interactions, encouraging desired behaviour and preventing undesired behaviour as much as possible. Institutions therefore also provide security for market participants, which in turn promotes transactions and reduces transaction costs (see Todaro and Smith, 2020, p.62f; Erlei, 2016, p.20; Roland, 2016, p.176).

The extractive institutions, which stand in opposition to the positive institutions, have often been used to exploit resource-rich areas for the benefit of the elites. For instance, under extractive institutions, property rights and the rule of law have not been developed, which, if established, could have prevented powerful elites from exploiting the local population through forced labour. The general population is also unable to benefit from the available resources and opportunities due to these institutions. This often results in extreme inequality in society, which may persist to this day, as local elites retain the system of extractive institutions even after their countries gain independence. Some countries were also unable to establish democracy and the rule of law, which in turn explains why there are fewer stable inclusive institutions to limit the power of the elites. Acemoglu and

Robinson (2013) refer to this as a "vicious cycle", where institutions are designed in a way that perpetuates the power of small groups, hindering the development of inclusive institutions. With their own advantage in mind, these elites thus often prioritize extracting surplus rather than acting in favour of their people, such as by investing in infrastructure and education, since doing so could also threaten their power (see Roland, 2016, p.106ff; Acemoglu and Robinson, 2017; Leschke, 2015).

Acemoglu et al. (2001, 2002) have demonstrated in their work that early and contemporary extractive and inclusive institutions play a vital role in explaining the varying levels of prosperity across different countries. Dell (2010) and Lowes and Montero (2021) have also illustrated that the long-lasting impact of extractive institutions is closely tied to underdeveloped land ownership rights, the perpetuation of power by the ruling elite, and low levels of public goods provision. In the case of Lever Brothers and HCB, it is evident that indigenous people were adversely affected by extractive institutions. Not only were they subjected to forced labour, but the monopoly held by HCB in the palm oil trade also restricted their economic opportunities. Hence, the use of extractive institutions by HCB in its concession areas may have prevented an improvement in socio-economic outcomes by discouraging investments in private and public goods on one hand, but also by undermining the development of inclusive institutions that are beneficial for everybody on the other hand.

C.2 Effect of Health, Education and Infrastructure on Development (Detailed Description)

Considering that Lever Brothers had to invest in education, health, and infrastructure, the basic assumption would be that these investments could have led to economic growth and better conditions regarding health and education in the area. In economics, all these factors are broadly regarded as contributors to lower production costs and more productive work, which increase economic welfare (see Ray, 1998, p.101ff; Dissou and Didic, 2013; Holmgren and Merkel, 2017). Many textbooks on economic development highlight the importance of human capital in the form of education and health on economic development.[5] Education normally helps individuals to obtain better-paid jobs since the skills they acquire, e.g. in advanced technology, are in higher demand in more industrialized economies (Perkins et al., 2013, p.267). More educated labourers theoretically provide

[5](see Todaro and Smith, 2020; Roland, 2016; Günther et al., 2021; Perkins et al., 2013)

more effective labour per person than uneducated workers and thus increase the GDP per capita. Moreover, experts, who may be necessary for technological progress and economic growth, cannot be replaced by an infinite number of unskilled workers. This again underlines the importance of education in the context of economic development (Meier and Rauch, 2005, p.217). Apart from individual higher earnings and the effects on economic growth, education also offers benefits to society. For instance, more educated people are more likely to adopt new technologies and are more receptive to innovations. They are also more likely to support beneficial institutions such as property rights, the rule of law, and care more about their own health. Educated parents also tend to place more value on the education of their children, thus providing them with better opportunities to get higher-paying jobs (see Janvry and Sadoulet, 2016, p.594; Meier and Rauch, 2005, p.217; Perkins et al., 2013, p.271).

When it comes to health, the most apparent assumption is that people can work and earn more when they are less affected by diseases, as they miss fewer working days and are more active and mentally alert. In low-income countries, many diseases like tuberculosis, HIV, or parasitic infections hinder people from being productive at hard work, which they mostly rely on. Poor health can also reduce cognitive ability and lead to increased absenteeism from school. Children's bad health conditions and malnutrition can later negatively impact their productivity as adults (see Perkins et al., 2013, p.320; Todaro and Smith, 2020, p.416ff; Janvry and Sadoulet, 2016, p.439). Additionally, just like education, better overall health also has positive externalities for society. For example, vaccination reduces the spread of certain diseases, from which also unvaccinated individuals benefit. A study by Miguel and Kremer (2004) evaluated a deworming program for school children in Kenya, and their findings indicate that, aside from improving health conditions, treated children had higher primary school attendance rates, and pupils who were not treated but attended the same school showed lower infection rates.

While it can be challenging to measure the causal effect of human capital on income and economic growth due to reverse causality and omitted variables (for example, better health conditions and education can lead to higher incomes, but the relationship can also be the other way around), there appears to be a tendency for a causal link between human capital and economic growth (see Todaro and Smith, 2020, p.429f; Janvry and Sadoulet, 2016, p.607ff). More recent literature reviews and meta-studies also emphasize the likelihood of a genuine effect of human capital in the form of education and health on

economic growth (see Ridhwan et al., 2022; Benos and Zotou, 2014; Valero, 2021).

Apart from the effect of health and education on development, establishing an infrastructure also plays a crucial role in the context of HCB virtuous intentions to improve living conditions in its circles. In economics, investments in infrastructure or the presence of physical infrastructure are believed to positively contribute to economic development (see Roland, 2016, p.489). Numerous studies have largely confirmed this linkage. However, the type of infrastructure also plays an important role, implying that not every infrastructure project leads to increased economic growth (see Foster et al., 2023; Timilsina et al., 2023, 2020; Calderón et al., 2015). A basic theoretical assumption posits that infrastructure lowers production expenses and enhances production efficiency, thus increasing overall welfare (Holmgren and Merkel, 2017; Kessides, 1996). For exmample, in the case of Lever Brothers in the Congo, this could mean that the construction of roads and telegraphs enabled HCB to sell their palm oil at a lower price on the world market.

It does not seem to be entirely clear through which channels infrastructure can be beneficial for households and also reduce poverty (see Dissou and Didic, 2013, p.19; Sawada, 2015). Apart from lowering production prices, the establishment of infrastructure can also have other effects that stimulate economic growth. For example, new markets can be accessed, demand for additional goods or new investments can be created, people can go to work further away, school attendance can rise due to better transport options, or clean water and sanitation contribute to the health system (Agenor and Moreno-Dodson, 2006). The recent meta-study by Foster et al. (2023) showed that rural road networks reduce poverty, increase household income and consumption, create more job opportunities and promote food security. Railways and roads also increase general output, particularly agricultural production, and promote specialisation and trade.

The previous paragraphs show how education and health contribute to economic development. However, health and education themselves can also be considered primary indicators for illustrating development, as, for example, it is demonstrated by the composition of the Human Development Index. This is also why the empirical analysis of this paper will show results for total years of education, rate of literacy and the height/age percentile as indicators for education and health.

Appendix D Additional Tables for the Main Results

D.1 Summary Statistics for Dependent Variables

Table D1: Summary Statistics for Dependent Variables

	Summary Statistics 60 km				Summary Statistics 30 km			
	Mean Inside	Mean Outside	Standard Error	P-Value	Mean Inside	Mean Outside	Standard Error	P-Value
Wealth Index	2.798	2.400	0.242	0.107	2.226	2.394	0.293	0.574
Number of Obs.	1247	2301	-	-	522	1186	-	-
Log(Weath Factor Score)	11.170	10.978	0.13	0.146	10.896	10.986	0.142	0.537
Number of Obs.	1247	2301	-	-	522	1186	-	-
Years of Education	7.135	6.785	0.597	0.562	6.368	6.102	0.841	0.755
Number of Obs.	1247	2297	-	-	522	1182	-	-
Literacy	0.735	0.731	0.049	0.941	0.659	0.669	0.07	0.885
Number of Obs.	1243	2289	-	-	522	1179	-	-
Height/Age Percentile (Women)	0.276	0.215	0.031	0.057	0.267	0.216	0.05	0.323
Number of Obs.	412	778	-	-	178	400	-	-
Height/Age Percentile (Children)	0.333	0.280	0.029	0.075	0.323	0.272	0.047	0.299
Number of Obs.	351	691	-	-	179	348	-	-
Number of Clusters	23	46			10	24		

Notes: Columns 1 and 2 as well as 5 and 6 show the mean values of the corresponding dependent variable for inside and outside the concession areas within a bandwidth of 60 km and 30 km. The associated standard errors and p-values for the difference of those means are reported in columns 3 and 4 as well as in columns 7 and 8. Standard errors are clustered at the DHS cluster level and reported in parentheses. More information on the variable definitions and data sources can be found in Section 4.1 and in Appendix A.

D.2 Balance Check for Mean Values

Table D2: Balance Check for Mean Values

	Balance Check Mean Values 60 km				Balance Check Mean Values 30 km			
	Mean Inside	Mean Outside	Standard Error	P-Value	Mean Inside	Mean Outside	Standard Error	P-Value
DHS Covariables								
Gender	0.670	0.683	0.017	0.178	0.661	0.680	0.028	0.260
Age	29.239	30.144	0.456	0.066	30.042	30.052	0.922	0.974
Age squared	971.532	1017.751	29.771	0.151	1019.644	1013.265	55.070	0.807
Dummy Survey Year	0.638	0.651	0.133	0.598	0.678	0.631	0.204	0.657
Population Density 2015	561.861	62.599	395.124	0.209	39.563	84.395	36.131	0.594
Level of Urbanization	0.106	0.082	0.051	0.259	0.021	0.126	0.078	0.660
Spatial Economic Output	206.401	216.487	6.347	0.092	211.196	218.676	11.289	0.283
Travel Times 2015	192.652	196.810	41.723	0.636	220.836	197.889	54.091	0.703
Mean Temperature 2015	25.648	25.564	0.101	0.573	25.703	25.547	0.118	0.280
Enhanced Vegetation Index	42.697	40.151	1.470	0.336	45.387	42.627	1.796	0.228
Further Geographic Covariables								
Elevation	416.130	460.783	32.466	0.046	420.000	459.875	68.945	0.136
Ruggedness	25.389	27.508	19.032	0.754	13.938	20.591	20.207	0.407
Land Suitability	0.171	0.115	0.032	0.263	0.153	0.082	0.019	0.233
Average Rainfall	130.785	121.184	4.680	0.285	133.904	109.872	7.754	0.264
Standard Deviation Rainfall	2.046	1.966	0.230	0.961	2.068	2.089	0.489	0.904
Malaria Suitability	19.026	18.412	0.428	0.193	19.141	18.658	0.359	0.186
Tsetse Fly Suitability	1.324	1.335	0.007	0.393	1.316	1.344	0.011	0.227
Road Density	144.847	139.648	24.778	0.955	107.354	119.939	19.140	0.814
Distance to Kinshasa	1009.775	886.211	16.055	0.456	989.701	838.560	25.541	0.678
Distance to Coast	863.370	857.010	22.118	0.623	870.630	881.703	20.369	0.388
Colonial Covariables								
Inside AA	0.043	0.130	0.084	0.283	0.100	0.208	0.164	0.459
Missionary Stations 1924	0.391	0.152	0.142	0.081	0.200	0.208	0.186	0.865
Colonial Population Density	0.357	0.800	0.387	0.339	0.176	0.845	0.150	0.098
Number Enslaved in 1000	8.638	2.902	6.276	0.310	12.555	0.270	15.295	0.328
Centralization	0.200	0.125	0.278	0.401	0.143	0.154	0.614	0.615
Polygynous	0.737	0.581	0.115	0.428	0.700	0.571	0.248	0.323

Notes: Columns 1 and 2 show the mean values of the corresponding variable for inside and outside the concession areas, considering a 60 km bandwidth. The associated standard errors and p-values for the difference of those means are reported in columns 3 and 4. The same pattern is applied to a bandwidth of 30 km in columns 5 - 8. Standard errors are clustered at the DHS cluster level for sociological variables, at the territory level for geographic variables or at ethnic group level for colonial variables. The RD regressions include concession fixed effects. More information on the variable definitions and data sources can be found in Appendix A.

D.3 Additional Bandwidths

Table D3: Main Results with Additional Bandwidths

Panel A: Wealth

	Wealth Index						
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6	RDD7
Inside Concession	−0.260	−0.154	0.837+	0.468	0.237	0.145	0.162
	(0.519)	(0.537)	(0.477)	(0.481)	(0.496)	(0.464)	(0.414)
Num.Obs.	477	1076	2307	2541	2732	3151	3245
Bandwidth	20.00	25.00	35.00	40.00	45.00	50.00	55.00
Mean Dep. Var.	1.912	2.039	2.475	2.545	2.534	2.548	2.523
SD Dep. Var.	0.960	0.992	1.246	1.274	1.254	1.237	1.234
Total Number Clusters	10	22	45	49	53	61	63
Clusters within Con. Area	2	7	14	16	17	21	21

	Wealth Factor Score						
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6	RDD7
Inside Concession	−0.511+	−0.377	0.272	0.088	−0.030	−0.079	−0.057
	(0.265)	(0.294)	(0.249)	(0.243)	(0.247)	(0.227)	(0.201)
Num.Obs.	477	1076	2307	2541	2732	3151	3245
Bandwidth	20.00	25.00	35.00	40.00	45.00	50.00	55.00
Mean Dep. Var.	10.723	10.794	11.019	11.064	11.052	11.048	11.035
SD Dep. Var.	0.430	0.454	0.610	0.633	0.624	0.608	0.607
Total Number Clusters	10	22	45	49	53	61	63
Clusters within Con. Area	2	7	14	16	17	21	21

Panel B: Education

	Total Number of Years of Education						
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6	RDD7
Inside Concession	1.586	1.302	2.993*	2.240	1.763	1.423	1.390
	(1.245)	(1.996)	(1.329)	(1.390)	(1.437)	(1.416)	(1.316)
Num.Obs.	476	1074	2303	2537	2728	3147	3241
Bandwidth	20.00	25.00	35.00	40.00	45.00	50.00	55.00
Mean Dep. Var.	4.901	5.389	6.788	7.059	6.981	6.878	6.842
SD Dep. Var.	3.825	3.888	4.115	4.106	4.082	4.015	4.010
Total Number Clusters	10	22	45	49	53	61	63
Clusters within Con. Area	2	7	14	16	17	21	21

Notes: Standard errors are clustered at the DHS cluster level and reported in parentheses. The regressions include concession fixed effects as well as age, age squared, gender and survey years as covariables. A triangular kernel is used. Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	Literacy						
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6	RDD7
Inside Concession	−0.037	0.073	0.205+	0.137	0.101	0.081	0.090
	(0.188)	(0.215)	(0.121)	(0.128)	(0.130)	(0.128)	(0.119)
Num.Obs.	475	1074	2298	2531	2722	3138	3231
Bandwidth	20.00	25.00	35.00	40.00	45.00	50.00	55.00
Mean Dep. Var.	0.568	0.598	0.714	0.734	0.728	0.727	0.728
SD Dep. Var.	0.496	0.491	0.452	0.442	0.445	0.445	0.445
Total Number Clusters	10	22	45	49	53	61	63
Clusters within Con. Area	2	7	14	16	17	21	21

Panel C: Health

	Height/Age Percentile for Women						
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6	RDD7
Inside Concession	−0.113*	0.197+	0.144*	0.098	0.068	0.055	0.049
	(0.047)	(0.103)	(0.063)	(0.064)	(0.069)	(0.072)	(0.072)
Num.Obs.	148	350	783	866	928	1055	1086
Bandwidth	20.00	25.00	35.00	40.00	45.00	50.00	55.00
Mean Dep. Var.	0.152	0.215	0.221	0.229	0.227	0.236	0.236
SD Dep. Var.	0.203	0.256	0.252	0.258	0.255	0.258	0.258
Total Number Clusters	10	22	45	49	53	61	63
Clusters within Con. Area	2	7	14	16	17	21	21

	Height/Age Percentile for Children						
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6	RDD7
Inside Concession	−0.135*	0.002	0.039	0.038	0.039	0.052	0.059
	(0.057)	(0.066)	(0.056)	(0.061)	(0.068)	(0.078)	(0.087)
Num.Obs.	162	350	684	741	794	909	945
Bandwidth	20.00	25.00	35.00	40.00	45.00	50.00	55.00
Mean Dep. Var.	0.272	0.291	0.283	0.288	0.288	0.296	0.298
SD Dep. Var.	0.353	0.357	0.346	0.349	0.350	0.356	0.355
Total Number Clusters	10	22	45	49	53	61	63
Clusters within Con. Area	2	7	14	16	17	21	21

Notes: Standard errors are clustered at the DHS cluster level and reported in parentheses. The regressions include concession fixed effects as well as age, age squared, gender and survey years as covariables. A triangular kernel is used. Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

D.4 RD Results for Alberta

Table D4: Main Results for Alberta

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.1845	−0.1485	0.7433+	0.1697**	−0.2144	−0.1221
	(0.2628)	(0.4632)	(0.3891)	(0.0605)	(0.1587)	(0.2656)
Num.Obs.	295	686	467	234	686	467
Bandwidth	25.00	60.00	30.00	21.52	60.00	30.00
Bandwidth Choice	Narrow	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	1.888	2.222	1.895	10.792	10.920	10.807
SD Dep. Var.	0.932	1.115	0.871	0.349	0.483	0.322
Total Number Clusters	6	13	9	5	13	9
Clusters within Con. Area	2	4	3	1	4	3

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.0280	0.0080	−1.1480***	−0.0675	−0.0601	−0.1606+
	(0.4391)	(0.4547)	(0.3185)	(0.0725)	(0.0732)	(0.0932)
Num.Obs.	728	684	465	782	685	467
Bandwidth	66.92	60.00	30.00	71.73	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	5.043	4.959	4.172	0.606	0.604	0.499
SD Dep. Var.	3.512	3.504	3.289	0.489	0.489	0.501
Total Number Clusters	14	13	9	15	13	9
Clusters within Con. Area	4	4	3	4	4	3

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.1410	0.1377	0.2564	0.0671	0.1228	0.3182***
	(0.1678)	(0.0938)	(0.2692)	(0.0563)	(0.0842)	(0.0527)
Num.Obs.	214	234	163	256	238	168
Bandwidth	54.12	60.00	30.00	64.17	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.262	0.260	0.219	0.325	0.321	0.290
SD Dep. Var.	0.259	0.251	0.230	0.364	0.362	0.352
Total Number Clusters	12	13	9	14	13	9
Clusters within Con. Area	3	4	3	4	4	3

Notes: Standard errors are clustered at the DHS cluster level and reported in parentheses. The regressions include age, age squared, gender and survey years as covariables. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). For the Wealth Index, the optimal bandwidth could not be calculated which is why an arbitrary bandwidth of 25 km is chosen. Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

D.5 Number of Clusters Inside Outside for each Circle

Table D5: Number of DHS Clusters Inside and Outside the Concession Areas

	Number of Clusters (60 km)		Number of Clusters (30 km)	
Concession Area	Inside	Outside	Inside	Outside
Alberta	4	9	3	6
Brabanta	4	2	3	0
Elisabetha	4	4	0	3
Flandria	2	15	1	9
Lusanga	9	16	3	6

Appendix E Robustness Checks: Estimation Specifications

E.1 Main Results with a Rectangular Kernel

Table E1: Main Results with a Rectangular Kernel

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	1.2892**	0.0702	1.2603*	−0.6039	0.0102	0.4939+
	(0.4993)	(0.4433)	(0.5177)	(0.4662)	(0.2337)	(0.2820)
Num.Obs.	1754	3548	1708	606	3548	1708
Bandwidth	31.07	60.00	30.00	21.11	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.360	2.540	2.343	10.741	11.045	10.958
SD Dep. Var.	1.198	1.238	1.204	0.418	0.602	0.569
Total Number Clusters	35	69	34	13	69	34
Clusters within Con. Area	11	23	10	4	23	10

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	1.0584	0.4210	3.5749*	−0.0165	0.0092	0.2640+
	(1.3555)	(1.2473)	(1.5742)	(0.1116)	(0.1124)	(0.1410)
Num.Obs.	3277	3544	1704	3655	3532	1701
Bandwidth	55.79	60.00	30.00	67.50	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.864	6.908	6.183	0.731	0.732	0.666
SD Dep. Var.	4.000	3.981	4.112	0.443	0.443	0.472
Total Number Clusters	64	69	34	72	69	34
Clusters within Con. Area	21	23	10	23	23	10

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.1615+	0.1053	0.1139	0.0575	0.0640	0.0579
	(0.0875)	(0.0753)	(0.0758)	(0.0792)	(0.0860)	(0.0820)
Num.Obs.	588	1190	578	1550	1042	527
Bandwidth	30.98	60.00	30.00	100.92	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.229	0.236	0.231	0.284	0.298	0.289
SD Dep. Var.	0.256	0.255	0.257	0.349	0.352	0.351
Total Number Clusters	35	69	34	100	69	34
Clusters within Con. Area	11	23	10	23	23	10

Notes: Standard errors are clustered at the DHS cluster level and reported in parentheses. The RD regressions include concession fixed effects as well as age, age squared, gender and survey years as covariables. A rectangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

E.2 Main Results with an Epanechnikov Kernel

Table E2: Main Results with an Epanechnikov Kernel

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.6971	0.0433	0.7978	−0.3031	−0.0809	0.2246
	(0.5107)	(0.4010)	(0.5562)	(0.2832)	(0.2029)	(0.2967)
Num.Obs.	2413	3548	1708	1076	3548	1708
Bandwidth	36.80	60.00	30.00	25.00	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.505	2.540	2.343	10.794	11.045	10.958
SD Dep. Var.	1.273	1.238	1.204	0.454	0.602	0.569
Total Number Clusters	47	69	34	22	69	34
Clusters within Con. Area	15	23	10	7	23	10

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.6598	0.9295	3.1132*	−0.0489	0.0640	0.2316+
	(1.2757)	(1.3344)	(1.5022)	(0.1095)	(0.1193)	(0.1345)
Num.Obs.	3667	3544	1704	4169	3532	1701
Bandwidth	66.07	60.00	30.00	79.94	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.893	6.908	6.183	0.711	0.732	0.666
SD Dep. Var.	3.982	3.981	4.112	0.453	0.443	0.472
Total Number Clusters	72	69	34	82	69	34
Clusters within Con. Area	23	23	10	23	23	10

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.1218+	0.0503	0.1759**	0.0547	0.0719	0.0451
	(0.0680)	(0.0760)	(0.0663)	(0.0810)	(0.0954)	(0.0610)
Num.Obs.	801	1190	578	1704	1042	527
Bandwidth	36.69	60.00	30.00	119.52	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.222	0.236	0.231	0.283	0.298	0.289
SD Dep. Var.	0.251	0.255	0.257	0.349	0.352	0.351
Total Number Clusters	46	69	34	109	69	34
Clusters within Con. Area	15	23	10	23	23	10

Notes: Standard errors are clustered at the DHS cluster level and reported in parentheses. The RD regressions include concession fixed effects as well as age, age squared, gender and survey years as covariables. A epanechnikov kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

E.3 Main Results for Additional Bandwidths with a Rectangular Kernel

Table E3: Main Results for Additional Bandwidths using a Rectangular Kernel

Panel A: Wealth

	Wealth Index						
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6	RDD7
Inside Concession	0.160	−0.070	0.265	−0.422	−0.550	0.262	0.080
	(0.774)	(0.608)	(0.680)	(0.679)	(0.640)	(0.413)	(0.424)
Num.Obs.	477	1076	2307	2541	2732	3151	3245
Bandwidth	20.00	25.00	35.00	40.00	45.00	50.00	55.00
Mean Dep. Var.	1.912	2.039	2.475	2.545	2.534	2.548	2.523
SD Dep. Var.	0.960	0.992	1.246	1.274	1.254	1.237	1.234
Total Number Clusters	10	22	45	49	53	61	63
Clusters within Con. Area	2	7	14	16	17	21	21

	Wealth Factor Score						
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6	RDD7
Inside Concession	−0.292	−0.282	0.028	−0.368	−0.437	0.042	−0.037
	(0.394)	(0.327)	(0.329)	(0.344)	(0.316)	(0.231)	(0.233)
Num.Obs.	477	1076	2307	2541	2732	3151	3245
Bandwidth	20.00	25.00	35.00	40.00	45.00	50.00	55.00
Mean Dep. Var.	10.723	10.794	11.019	11.064	11.052	11.048	11.035
SD Dep. Var.	0.430	0.454	0.610	0.633	0.624	0.608	0.607
Total Number Clusters	10	22	45	49	53	61	63
Clusters within Con. Area	2	7	14	16	17	21	21

Panel B: Education

	Total Number of Years of Education						
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6	RDD7
Inside Concession	1.586	1.302	2.993*	2.240	1.763	1.423	1.390
	(1.245)	(1.996)	(1.329)	(1.390)	(1.437)	(1.416)	(1.316)
Num.Obs.	476	1074	2303	2537	2728	3147	3241
Bandwidth	20.00	25.00	35.00	40.00	45.00	50.00	55.00
Mean Dep. Var.	4.901	5.389	6.788	7.059	6.981	6.878	6.842
SD Dep. Var.	3.825	3.888	4.115	4.106	4.082	4.015	4.010
Total Number Clusters	10	22	45	49	53	61	63
Clusters within Con. Area	2	7	14	16	17	21	21

Notes: Standard errors are clustered at the DHS cluster level and reported in parentheses. The regressions include concession fixed effects as well as age, age squared, gender and survey years as covariables. A rectangular kernel is used. Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	Literacy						
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6	RDD7
Inside Concession	−0.037	0.073	0.205+	0.137	0.101	0.081	0.090
	(0.188)	(0.215)	(0.121)	(0.128)	(0.130)	(0.128)	(0.119)
Num.Obs.	475	1074	2298	2531	2722	3138	3231
Bandwidth	20.00	25.00	35.00	40.00	45.00	50.00	55.00
Mean Dep. Var.	0.568	0.598	0.714	0.734	0.728	0.727	0.728
SD Dep. Var.	0.496	0.491	0.452	0.442	0.445	0.445	0.445
Total Number Clusters	10	22	45	49	53	61	63
Clusters within Con. Area	2	7	14	16	17	21	21

Panel C: Health

	Height/Age Percentile for Women						
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6	RDD7
Inside Concession	−0.130**	0.189*	0.080	−0.012	−0.016	0.039	0.033
	(0.050)	(0.089)	(0.083)	(0.084)	(0.090)	(0.079)	(0.077)
Num.Obs.	148	350	783	866	928	1055	1086
Bandwidth	20.00	25.00	35.00	40.00	45.00	50.00	55.00
Mean Dep. Var.	0.152	0.215	0.221	0.229	0.227	0.236	0.236
SD Dep. Var.	0.203	0.256	0.252	0.258	0.255	0.258	0.258
Total Number Clusters	10	22	45	49	53	61	63
Clusters within Con. Area	2	7	14	16	17	21	21

	Height/Age Percentile for Children						
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6	RDD7
Inside Concession	−0.135*	0.002	0.039	0.038	0.039	0.052	0.059
	(0.057)	(0.066)	(0.056)	(0.061)	(0.068)	(0.078)	(0.087)
Num.Obs.	162	350	684	741	794	909	945
Bandwidth	20.00	25.00	35.00	40.00	45.00	50.00	55.00
Mean Dep. Var.	0.272	0.291	0.283	0.288	0.288	0.296	0.298
SD Dep. Var.	0.353	0.357	0.346	0.349	0.350	0.356	0.355
Total Number Clusters	10	22	45	49	53	61	63
Clusters within Con. Area	2	7	14	16	17	21	21

Notes: Standard errors are clustered at the DHS cluster level and reported in parentheses. The regressions include concession fixed effects as well as age, age squared, gender and survey years as covariables. A rectangular kernel is used. Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

E.4 Main Results using a Quadratic Polynomial

Table E4: Main Results using a Quadratic Polynomial

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.7060	0.4258	−0.6070	−0.6747**	0.0237	−0.5731*
	(0.5464)	(0.5224)	(0.4168)	(0.2482)	(0.2754)	(0.2393)
Num.Obs.	2541	3548	1708	1254	3548	1708
Bandwidth	39.54	60.00	30.00	26.85	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.545	2.540	2.343	10.845	11.045	10.958
SD Dep. Var.	1.274	1.238	1.204	0.513	0.602	0.569
Total Number Clusters	49	69	34	25	69	34
Clusters within Con. Area	16	23	10	8	23	10

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	2.5525+	2.5047+	0.0833	0.2005	0.1347	−0.0791
	(1.4091)	(1.3453)	(1.4052)	(0.1349)	(0.1234)	(0.1548)
Num.Obs.	3819	3544	1704	4312	3532	1701
Bandwidth	70.97	60.00	30.00	85.88	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.831	6.908	6.183	0.712	0.732	0.666
SD Dep. Var.	3.984	3.981	4.112	0.453	0.443	0.472
Total Number Clusters	75	69	34	85	69	34
Clusters within Con. Area	23	23	10	23	23	10

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.2292**	0.0622	0.2283*	0.0094	−0.0358	0.0403
	(0.0774)	(0.0791)	(0.1089)	(0.0629)	(0.0628)	(0.1335)
Num.Obs.	866	1190	578	1821	1042	527
Bandwidth	39.41	60.00	30.00	128.40	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.229	0.236	0.231	0.282	0.298	0.289
SD Dep. Var.	0.258	0.255	0.257	0.348	0.352	0.351
Total Number Clusters	49	69	34	116	69	34
Clusters within Con. Area	16	23	10	23	23	10

Notes: Instead of a local linear regression, a local quadratic regression is used. Standard errors are clustered at the DHS cluster level and reported in parentheses. The RD regressions include concession fixed effects as well as age, age squared, gender and survey years as covariables. A triangular kernel is utilized and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Section A.
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

E.5 Main Results without Covariates

Table E5: RDD Wealth without Covariates

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.4878	0.1305	0.7026	−0.2738	−0.1000	0.0884
	(0.4656)	(0.3806)	(0.4884)	(0.2409)	(0.2016)	(0.2545)
Num.Obs.	2541	3548	1708	1254	3548	1708
Bandwidth	39.54	60.00	30.00	26.85	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.545	2.540	2.343	10.845	11.045	10.958
SD Dep. Var.	1.274	1.238	1.204	0.513	0.602	0.569
Total Number Clusters	49	69	34	25	69	34
Clusters within Con. Area	16	23	10	8	23	10

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.6247	1.0738	2.6191*	−0.0417	0.0827	0.2051+
	(1.0825)	(1.0963)	(1.3068)	(0.1001)	(0.1043)	(0.1232)
Num.Obs.	3819	3544	1704	4312	3532	1701
Bandwidth	70.97	60.00	30.00	85.88	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.831	6.908	6.183	0.712	0.732	0.666
SD Dep. Var.	3.984	3.981	4.112	0.453	0.443	0.472
Total Number Clusters	75	69	34	85	69	34
Clusters within Con. Area	23	23	10	23	23	10

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.1207*	0.0632	0.1957**	0.0475	0.0976	0.1728**
	(0.0604)	(0.0708)	(0.0641)	(0.0801)	(0.0805)	(0.0629)
Num.Obs.	866	1190	578	1821	1042	527
Bandwidth	39.41	60.00	30.00	128.40	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.229	0.236	0.231	0.282	0.298	0.289
SD Dep. Var.	0.258	0.255	0.257	0.348	0.352	0.351
Total Number Clusters	49	69	34	116	69	34
Clusters within Con. Area	16	23	10	23	23	10

Notes: In comparison to the main results, the estimates of this table are calculated without covariates. Standard errors are clustered at the DHS cluster level and reported in parentheses. The RD regressions include concession fixed effects. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Section A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

E.6 Main Results without Fixed Effects

Table E6: Main Results without Fixed Effects

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.4010	0.2361	0.5367	0.0132	0.0018	0.1195
	(0.4746)	(0.3588)	(0.4420)	(0.1925)	(0.1633)	(0.2110)
Num.Obs.	2541	3548	1708	1254	3548	1708
Bandwidth	39.54	60.00	30.00	26.85	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.545	2.540	2.343	10.845	11.045	10.958
SD Dep. Var.	1.274	1.238	1.204	0.513	0.602	0.569
Total Number Clusters	49	69	34	25	69	34
Clusters within Con. Area	16	23	10	8	23	10

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	2.2993+	2.8969*	3.9385**	0.0564	0.1873	0.2340+
	(1.1981)	(1.3057)	(1.2237)	(0.1009)	(0.1158)	(0.1265)
Num.Obs.	3819	3544	1704	4312	3532	1701
Bandwidth	70.97	60.00	30.00	85.88	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.831	6.908	6.183	0.712	0.732	0.666
SD Dep. Var.	3.984	3.981	4.112	0.453	0.443	0.472
Total Number Clusters	75	69	34	85	69	34
Clusters within Con. Area	23	23	10	23	23	10

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	−0.0227	−0.0066	0.0021	0.0190	0.0047	−0.0753
	(0.0550)	(0.0697)	(0.0496)	(0.0670)	(0.0729)	(0.0789)
Num.Obs.	866	1190	578	1821	1042	527
Bandwidth	39.41	60.00	30.00	128.40	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.229	0.236	0.231	0.282	0.298	0.289
SD Dep. Var.	0.258	0.255	0.257	0.348	0.352	0.351
Total Number Clusters	49	69	34	116	69	34
Clusters within Con. Area	16	23	10	23	23	10

Notes: In comparison to the main results, the estimates of this table are calculated without fixed effects. Standard errors are clustered at the DHS cluster level and reported in parentheses. The RD regressions include covariates. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Section A. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

E.7 Main Results without Fixed Effects and Covariates

Table E7: Main Results without Fixed Effects and Covariates

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.5798	0.2946	0.7289+	−0.0908	−0.0240	0.0304
	(0.4911)	(0.4126)	(0.3918)	(0.0737)	(0.1486)	(0.1052)
Num.Obs.	2541	3548	1708	1254	3548	1708
Bandwidth	39.54	60.00	30.00	26.85	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.545	2.540	2.343	10.845	11.045	10.958
SD Dep. Var.	1.274	1.238	1.204	0.513	0.602	0.569
Total Number Clusters	49	69	34	25	69	34
Clusters within Con. Area	16	23	10	8	23	10

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	2.2135*	2.7884*	3.8814***	0.0567	0.1861+	0.2463**
	(1.0779)	(1.1503)	(0.7875)	(0.0991)	(0.1124)	(0.0898)
Num.Obs.	3819	3544	1704	4312	3532	1701
Bandwidth	70.97	60.00	30.00	85.88	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.831	6.908	6.183	0.712	0.732	0.666
SD Dep. Var.	3.984	3.981	4.112	0.453	0.443	0.472
Total Number Clusters	75	69	34	85	69	34
Clusters within Con. Area	23	23	10	23	23	10

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	−0.0107	0.0029	0.0035	0.0200	0.0427	0.0411
	(0.0454)	(0.0652)	(0.0421)	(0.0623)	(0.0597)	(0.0499)
Num.Obs.	866	1190	578	1821	1042	527
Bandwidth	39.41	60.00	30.00	128.40	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.229	0.236	0.231	0.282	0.298	0.289
SD Dep. Var.	0.258	0.255	0.257	0.348	0.352	0.351
Total Number Clusters	49	69	34	116	69	34
Clusters within Con. Area	16	23	10	23	23	10

Notes: In comparison to the main results, the estimates of this table are calculated without including fixed effects and covariates. Standard errors are clustered at the DHS cluster level and reported in parentheses. A triangular kernel is utilized and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Section A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

E.8 Main Results without Clustered Errors

Table E8: Main Results without Clustered Errors

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.4979**	0.1212	0.6765***	−0.2374*	−0.0603	0.1531+
	(0.1516)	(0.1187)	(0.1778)	(0.0961)	(0.0559)	(0.0889)
Num.Obs.	2541	3548	1708	1254	3548	1708
Bandwidth	39.54	60.00	30.00	26.85	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.545	2.540	2.343	10.845	11.045	10.958
SD Dep. Var.	1.274	1.238	1.204	0.513	0.602	0.569
Clustered Std.Errors	-	-	-	-	-	-
Total Number Clusters	49	69	34	25	69	34
Clusters within Con. Area	16	23	10	8	23	10

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.7369+	1.2214**	2.8910***	−0.0462	0.0820	0.2096**
	(0.3829)	(0.4295)	(0.6554)	(0.0413)	(0.0501)	(0.0759)
Num.Obs.	3819	3544	1704	4312	3532	1701
Bandwidth	70.97	60.00	30.00	85.88	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.831	6.908	6.183	0.712	0.732	0.666
SD Dep. Var.	3.984	3.981	4.112	0.453	0.443	0.472
Clustered Std.Errors	-	-	-	-	-	-
Total Number Clusters	75	69	34	85	69	34
Clusters within Con. Area	23	23	10	23	23	10

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.1036*	0.0523	0.1801***	0.0503	0.0620	0.0414
	(0.0449)	(0.0398)	(0.0534)	(0.0616)	(0.0899)	(0.1622)
Num.Obs.	866	1190	578	1821	1042	527
Bandwidth	39.41	60.00	30.00	128.40	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.229	0.236	0.231	0.282	0.298	0.289
SD Dep. Var.	0.258	0.255	0.257	0.348	0.352	0.351
Clustered Std.Errors	-	-	-	-	-	-
Total Number Clusters	49	69	34	116	69	34
Clusters within Con. Area	16	23	10	23	23	10

Notes: In comparison to the main results, the estimates of this table are calculated without using clustered errors at the DHS cluster level. The RD regressions include concession fixed effects as well as age, age squared, gender and survey years as covariables. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Section A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Appendix F Robustness Checks: Context-Related Caveats

F.1 Main Results including Dummy for Abir and Aversoise

Table F1: RDD Main Results accounting for Abir and Anversoise

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.4970	0.1132	0.6879	−0.2378	−0.0594	0.1554
	(0.4796)	(0.3788)	(0.5339)	(0.2580)	(0.1876)	(0.2842)
Num.Obs.	2541	3548	1708	1254	3548	1708
Bandwidth	39.54	60.00	30.00	26.85	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.545	2.540	2.343	10.845	11.045	10.958
SD Dep. Var.	1.274	1.238	1.204	0.513	0.602	0.569
Total Number Clusters	49	69	34	25	69	34
Clusters within Con. Area	16	23	10	8	23	10

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.7026	1.1582	2.9444*	−0.0449	0.0741	0.2201+
	(1.1772)	(1.2493)	(1.4546)	(0.1020)	(0.1117)	(0.1335)
Num.Obs.	3819	3544	1704	4312	3532	1701
Bandwidth	70.97	60.00	30.00	85.88	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.831	6.908	6.183	0.712	0.732	0.666
SD Dep. Var.	3.984	3.981	4.112	0.453	0.443	0.472
Total Number Clusters	75	69	34	85	69	34
Clusters within Con. Area	23	23	10	23	23	10

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.1043	0.0543	0.1814**	0.0498	0.0669	0.0440
	(0.0643)	(0.0727)	(0.0667)	(0.0792)	(0.0898)	(0.0606)
Num.Obs.	866	1190	578	1821	1042	527
Bandwidth	39.41	60.00	30.00	128.40	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.229	0.236	0.231	0.282	0.298	0.289
SD Dep. Var.	0.258	0.255	0.257	0.348	0.352	0.351
Total Number Clusters	49	69	34	116	69	34
Clusters within Con. Area	16	23	10	23	23	10

Notes: Standard errors are clustered at the DHS cluster level and reported in parentheses. The RD regressions include concession fixed effects and the following covariables: age, age squared, gender, survey years and a dummy variable indicating whether a DHS cluster was earlier located in the Abir and Anversoise concession areas. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

F.2 Main Results taking into account DHS Random Displacement

Table F2: RDD Main Results accounting for DHS Random Displacement

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.4089	−0.5168	1.0303	−45.5579***	−0.2271	0.6597
	(1.2468)	(0.7183)	(1.2469)	(13.5305)	(0.4062)	(0.7462)
Num.Obs.	485	3281	1441	386	3281	1441
Bandwidth	22.73	60.00	30.00	21.72	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	1.961	2.582	2.401	10.746	11.068	10.995
SD Dep. Var.	0.921	1.246	1.230	0.415	0.609	0.587
Total Number Clusters	11	64	29	9	64	29
Clusters within Con. Area	4	22	9	3	22	9

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	1.3616	−1.0961	1.1386	−0.1294	−0.0959	−0.0488
	(3.0642)	(1.9741)	(3.2304)	(0.2597)	(0.1811)	(0.3432)
Num.Obs.	1437	3277	1437	2088	3267	1436
Bandwidth	30.54	60.00	30.00	36.59	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.327	7.031	6.327	0.734	0.742	0.677
SD Dep. Var.	4.132	3.961	4.132	0.442	0.437	0.468
Total Number Clusters	29	64	29	41	64	29
Clusters within Con. Area	9	22	9	14	22	9

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	−0.0058	0.0255	−0.0244	0.1202	0.1415	0.2026
	(0.1602)	(0.1084)	(0.1757)	(0.1094)	(0.1284)	(0.1645)
Num.Obs.	785	1109	497	1177	953	438
Bandwidth	38.23	60.00	30.00	82.62	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.241	0.245	0.251	0.298	0.301	0.294
SD Dep. Var.	0.263	0.258	0.265	0.354	0.352	0.350
Total Number Clusters	44	64	29	78	64	29
Clusters within Con. Area	15	22	9	22	22	9

Notes: Due to the random displaced of DHS clusters, the clusters that are located within a range of 10 km around the border are excluded. Standard errors are clustered at the DHS cluster level and reported in parentheses. The RD regressions include concession fixed effects and age, age squared, gender and survey years as covariables. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A. + p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

F.3 Main Results with Placebo Borders

Table F3: RDD Main Results with Placebo Border at 50 km

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	5.5800***	0.2834	−0.3879	3.5649***	0.2506	0.0062
	(0.1833)	(0.4717)	(0.8974)	(0.0566)	(0.2562)	(0.5609)
Num.Obs.	635	3266	1252	571	3266	1252
Bandwidth	19.40	60.00	30.00	18.74	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.189	2.569	2.540	10.873	11.059	11.071
SD Dep. Var.	1.046	1.241	1.292	0.458	0.607	0.642
Total Number Clusters	12	63	24	11	63	24
Clusters within Con. Area	9	22	15	8	22	15

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	−4.7364*	0.0943	−5.3284*	−0.4309*	−0.0398	−0.4667*
	(2.1780)	(1.4284)	(2.1564)	(0.2107)	(0.1261)	(0.2270)
Num.Obs.	1118	3262	1251	1377	3252	1249
Bandwidth	26.15	60.00	30.00	32.27	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.419	6.911	6.556	0.687	0.731	0.689
SD Dep. Var.	4.128	3.998	4.167	0.464	0.444	0.463
Total Number Clusters	22	63	24	27	63	24
Clusters within Con. Area	14	22	15	16	22	15

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.2408+	0.2072*	0.2258+	0.1370	0.1526	0.4063
	(0.1284)	(0.0841)	(0.1357)	(0.0960)	(0.1054)	(0.2519)
Num.Obs.	472	1092	416	1088	944	388
Bandwidth	32.78	60.00	30.00	77.58	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.246	0.235	0.243	0.300	0.297	0.303
SD Dep. Var.	0.260	0.256	0.259	0.353	0.355	0.363
Total Number Clusters	28	63	24	72	63	24
Clusters within Con. Area	17	22	15	23	22	15

Notes: The border of the concession areas was hypothetically shifted from 60 km to 50 km. Standard errors are clustered at the DHS cluster level and reported in parentheses. The RD regressions include concession fixed effects and age, age squared, gender and survey years as covariables. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table F4: RDD Main Results with Placebo Border at 70 km

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.0607	−0.4480	0.0716	0.0953	−0.2515	0.1575
	(0.4340)	(0.2835)	(0.4182)	(0.2477)	(0.1597)	(0.2406)
Num.Obs.	2178	3708	1766	2178	3708	1766
Bandwidth	34.76	60.00	30.00	35.23	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.384	2.471	2.378	10.965	11.012	10.967
SD Dep. Var.	1.195	1.228	1.217	0.581	0.597	0.596
Total Number Clusters	43	73	35	43	73	35
Clusters within Con. Area	11	25	6	11	25	6

Panel B: Education

	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.4690	−0.7191	0.7007	−0.0379	−0.0466	0.0266
	(1.1144)	(0.8332)	(1.1502)	(0.0817)	(0.0796)	(0.1324)
Num.Obs.	2537	3704	1762	3473	3692	1757
Bandwidth	40.48	60.00	30.00	58.02	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.689	6.800	6.833	0.726	0.720	0.722
SD Dep. Var.	3.984	3.997	4.114	0.446	0.449	0.448
Total Number Clusters	50	73	35	69	73	35
Clusters within Con. Area	14	25	6	23	25	6

Panel C: Health

	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	−0.1548*	−0.1271*	−0.1417	−0.0020	−0.0079	0.0112
	(0.0754)	(0.0516)	(0.0872)	(0.0366)	(0.0329)	(0.0466)
Num.Obs.	844	1247	598	1401	1108	515
Bandwidth	38.95	60.00	30.00	81.37	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.224	0.239	0.199	0.291	0.300	0.271
SD Dep. Var.	0.256	0.256	0.246	0.351	0.356	0.338
Total Number Clusters	49	73	35	90	73	35
Clusters within Con. Area	14	25	6	27	25	6

Notes: The border of the concession areas was hypothetically shifted from 60 km to 70 km. Standard errors are clustered at the DHS cluster level and reported in parentheses. The regressions include concession fixed effects and age, age squared, gender and survey years as covariables. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

F.4 Main Results with Alternative Concession Centers

Table F5: RDD Main Results with Alternative Concession Centers

Panel A: Wealth

	Wealth Index			Log(Wealth Factor Score)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.6304	0.3813	0.5720	0.6014*	0.1254	0.2440
	(0.5759)	(0.3178)	(0.5542)	(0.2756)	(0.1578)	(0.2922)
Num.Obs.	1245	3691	1335	724	3691	1335
Bandwidth	28.43	60.00	30.00	21.90	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	2.129	2.515	2.167	10.741	11.036	10.873
SD Dep. Var.	1.061	1.233	1.074	0.393	0.596	0.490
Total Number Clusters	25	72	27	15	72	27
Clusters within Con. Area	9	23	9	5	23	9

Panel B: Education

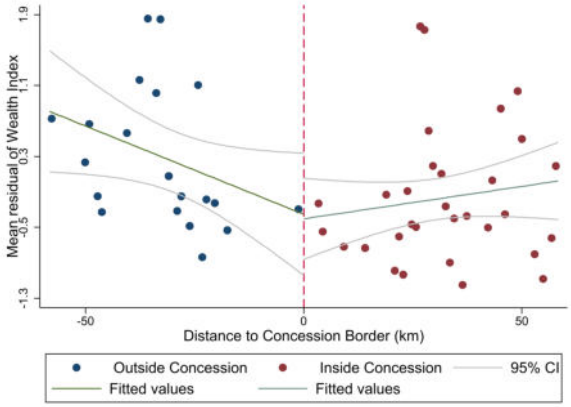
	Years of Education			Literacy		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	0.9341	1.8094+	4.3071**	0.0943	0.1628+	0.3578**
	(0.9189)	(0.9356)	(1.3310)	(0.0848)	(0.0850)	(0.1252)
Num.Obs.	4271	3687	1332	3941	3674	1332
Bandwidth	86.68	60.00	30.00	77.02	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	6.727	6.867	5.964	0.720	0.728	0.631
SD Dep. Var.	4.005	3.974	3.999	0.449	0.445	0.483
Total Number Clusters	84	72	27	78	72	27
Clusters within Con. Area	23	23	9	23	23	9

Panel C: Health

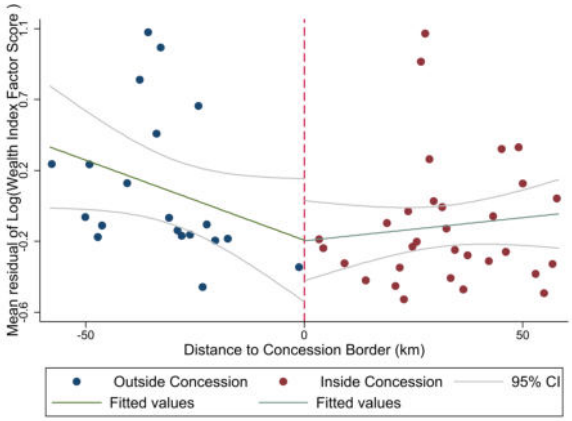
	Height/Age Percentile (Women)			Height/Age Percentile (Children)		
	RDD1	RDD2	RDD3	RDD4	RDD5	RDD6
Inside Concession	−0.0073	0.0494	−0.0238	−0.0361	−0.0490	−0.0614
	(0.0326)	(0.0395)	(0.0376)	(0.0420)	(0.0431)	(0.0545)
Num.Obs.	778	1242	447	1143	1088	441
Bandwidth	39.05	60.00	30.00	65.01	60.00	30.00
Bandwidth Choice	Optimal	Wide	Narrow	Optimal	Wide	Narrow
Mean Dep. Var.	0.226	0.239	0.171	0.301	0.299	0.263
SD Dep. Var.	0.259	0.254	0.213	0.355	0.354	0.338
Total Number Clusters	44	72	27	75	72	27
Clusters within Con. Area	16	23	9	23	23	9

Notes: Standard errors are clustered at the DHS cluster level and reported in parentheses. The RD regressions include concession fixed effects as well as age, age squared, gender and survey years as covariables. A triangular kernel is used and the optimal bandwidth is chosen by the MSE-minimizing method suggested by Imbens and Kalyanaraman (2012). Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

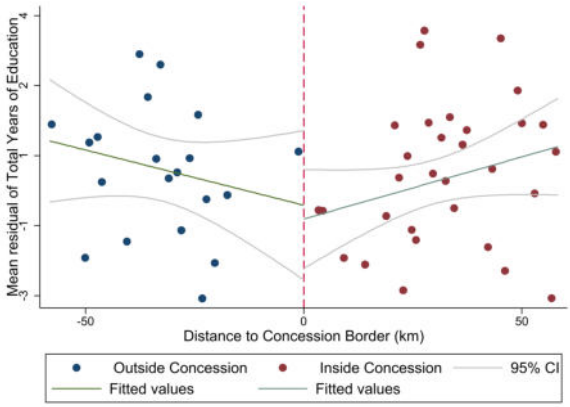
Appendix G Main RD Plots



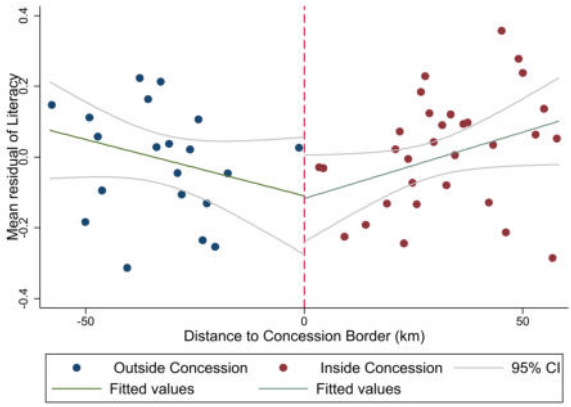
(a) Wealth Index



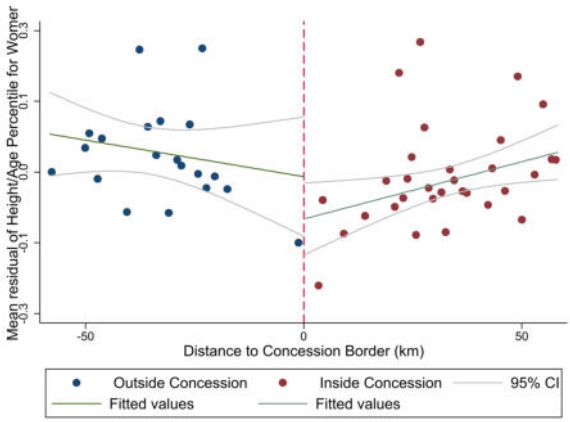
(b) Log(Wealth Factor Score)



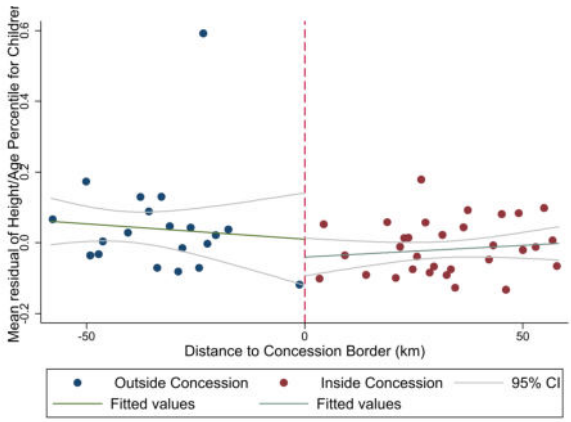
(c) Years of Education



(d) Literacy



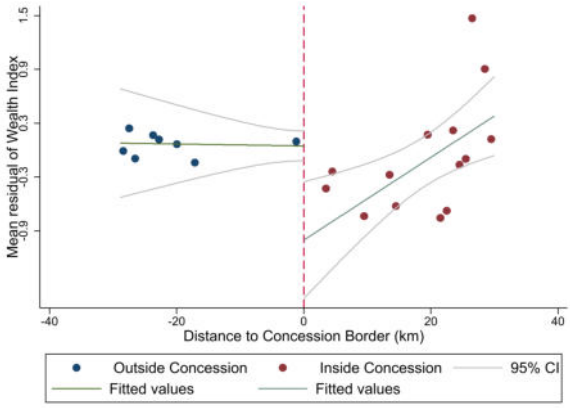
(e) Height/Age Percentile (Women)



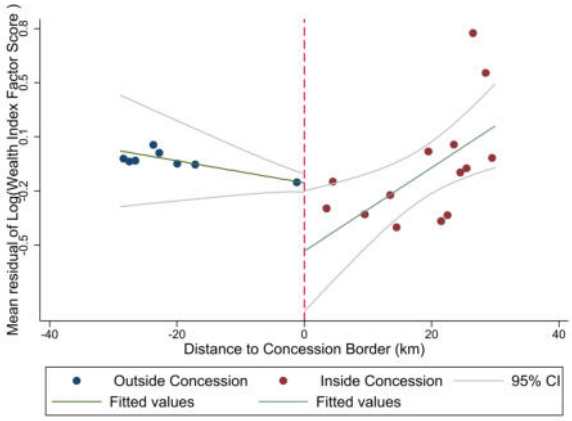
(f) Height/Age Percentile (Children)

Figure G1: RD Plots for a Bandwidth of 60 km

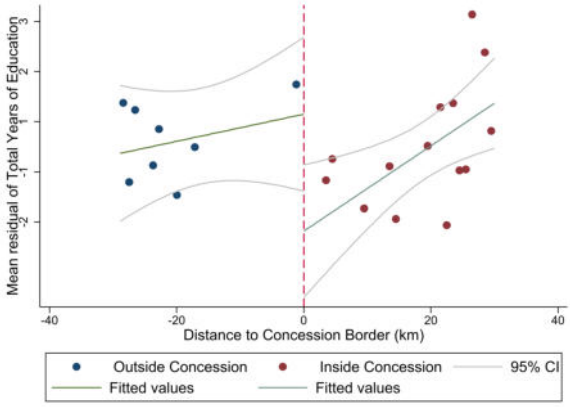
Notes: Each of these figures illustrates a typical RD plot for the dependent variables, using a bandwidth of 60 km. Since these plots include concession fixed effects as well as age, age squared, gender and survey years as covariables, the mean of the residual values for each bin is shown. Standard errors are clustered at the DHS cluster level. Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.



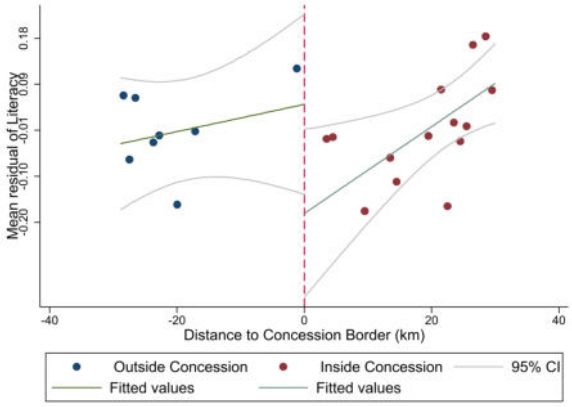
(a) Wealth Index



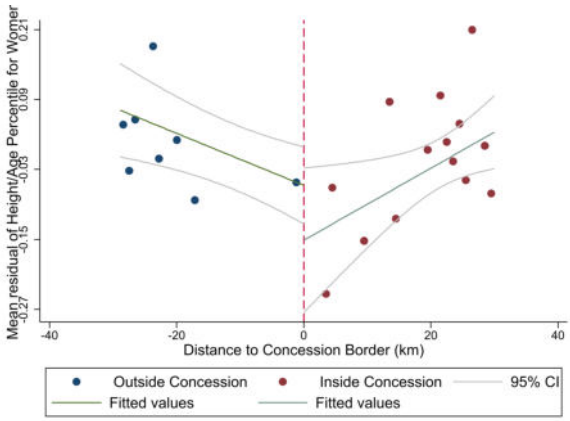
(b) Log(Wealth Factor Score)



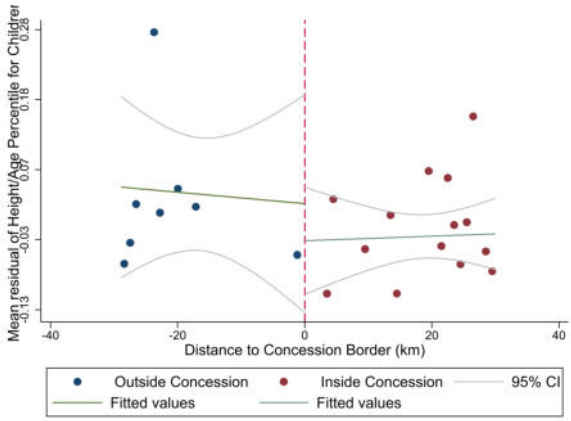
(c) Years of Education



(d) Literacy



(e) Height/Age Percentile (Women)



(f) Height/Age Percentile (Children)

Figure G2: RD Plots for a Bandwidth of 30 km

Notes: Each of these figures illustrates a typical RD plot for the dependent variables, using a bandwidth of 30 km. Since these plots include concession fixed effects as well as age, age squared, gender and survey years as covariables, the mean of the residual values for each bin is shown. Standard errors are clustered at the DHS cluster level. Exact definitions of the outcome variables can be found in Section 4.1 and in Appendix A.

Appendix H OLS Regressions

Table H1: OLS Regression for Wealth Index

	OLS1	OLS2	OLS3	OLS4	OLS5
Intercept	74.4208	27.3173			
	(73.0901)	(49.5796)			
Inside Concession	0.4719	0.4066	0.4270	0.4679	0.0824
	(0.4746)	(0.3642)	(0.5008)	(0.3538)	(0.1871)
Distance Concession Center	0.0013	0.0024	−0.0017	0.0002	−0.0005
	(0.0061)	(0.0054)	(0.0071)	(0.0053)	(0.0011)
Gender	−0.0237	−0.0707+	−0.0151	−0.0650	−0.0662*
	(0.0462)	(0.0411)	(0.0450)	(0.0393)	(0.0255)
Age	−0.0428***	−0.0356**	−0.0435***	−0.0336**	−0.0280***
	(0.0116)	(0.0107)	(0.0115)	(0.0110)	(0.0068)
Age squared	0.0007***	0.0006**	0.0007***	0.0005**	0.0004***
	(0.0002)	(0.0002)	(0.0002)	(0.0002)	(0.0001)
Survey Year	−0.0356	−0.0097	−0.0116	−0.0270	−0.0143
	(0.0364)	(0.0234)	(0.0353)	(0.0255)	(0.0152)
Population Density 2015		0.0003*		0.0004**	0.0002
		(0.0001)		(0.0001)	(0.0001)
Level of Urbanization		2.2420***		2.0747***	3.1460***
		(0.4466)		(0.4617)	(0.2610)
Land Suitability		−2.7496*		−4.5317**	−0.4598
		(1.0455)		(1.2743)	(0.5399)
Missionary Stations 1924		−0.0231		−0.3025	−0.0808
		(0.1929)		(0.2897)	(0.1513)
Number Enslaved (in 1000)		0.0082		0.0148+	0.0023
		(0.0063)		(0.0072)	(0.0039)
Tsetse Fly Suitability		2.3720		−5.0519	−3.2641+
		(2.1223)		(6.0870)	(1.8394)
Road Density		0.0009		0.0008	0.0005
		(0.0009)		(0.0012)	(0.0006)
Num.Obs.	3548	3548	3548	3548	8015
Bandwidth in km	60	60	60	60	full
Fixed Effects	-	-	X	X	X
Add. Covariats	-	X	-	X	X
Clusters	69	69	69	69	155
Mean Dep. Var.	2.540	2.540	2.540	2.540	2.427
SD Dep. Var.	1.238	1.238	1.238	1.238	1.251

Notes: This table shows different OLS specifications with*wealth index* as the dependent variable and *inside concession* as an explanatory variable that equals 1 if the corresponding DHS cluster is located inside the concession areas and 0 otherwise. Some of those regressions include additional covariates and fixed effects. Column 5 displays OLS results for the entire sample including all 155 DHS clusters. Standard errors are clustered at the DHS cluster level and reported in parentheses. Exact definitions of the variables can be found in Section 4.1 and in Appendix A.
$p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table H2: OLS Regressions for Log(Wealth Factor Score)

	OLS1	OLS2	OLS3	OLS4	OLS5
Intercept	−30.4256	−53.6181*			
	(40.2067)	(25.3002)			
Inside Concession	0.1904	0.1693	0.1697	0.2009	−0.0188
	(0.2482)	(0.1942)	(0.2647)	(0.1931)	(0.0979)
Distance Concession Center	0.0000	0.0013	−0.0015	0.0005	−0.0005
	(0.0031)	(0.0028)	(0.0036)	(0.0028)	(0.0007)
Gender	−0.0107	−0.0358*	−0.0065	−0.0332*	−0.0169
	(0.0203)	(0.0161)	(0.0200)	(0.0157)	(0.0140)
Age	−0.0178**	−0.0138**	−0.0181**	−0.0128**	−0.0116**
	(0.0055)	(0.0047)	(0.0055)	(0.0048)	(0.0036)
Age squared	0.0003**	0.0002**	0.0003**	0.0002**	0.0002**
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Survey Year	0.0207	0.0342**	0.0324	0.0256+	0.0345***
	(0.0200)	(0.0118)	(0.0194)	(0.0128)	(0.0082)
Population Density 2015		0.0002*		0.0002**	0.0001+
		(0.0001)		(0.0001)	(0.0001)
Level of Urbanization		1.2676***		1.1757***	1.7302***
		(0.2171)		(0.2152)	(0.1638)
Land Suitability		−1.1125+		−1.8302*	−0.1207
		(0.5376)		(0.7329)	(0.2636)
Missionary Stations 1924		−0.0093		−0.1187	−0.0261
		(0.0944)		(0.1360)	(0.0825)
Number Enslaved (in 1000)		0.0022		0.0054	0.0010
		(0.0026)		(0.0030)	(0.0018)
Tsetse Fly Suitability		0.6899		−3.4103	−1.5707
		(0.9645)		(3.2460)	(1.0965)
Road Density		0.0003		0.0003	0.0002
		(0.0005)		(0.0006)	(0.0004)
Num.Obs.	3548	3548	3548	3548	8015
Bandwidth in km	60	60	60	60	full
Fixed Effects	-	-	X	X	X
Add. Covariats	-	X	-	X	X
Clusters	69	69	69	69	155
Mean Dep. Var.	11.045	11.045	11.045	11.045	11.009
SD Dep. Var.	0.602	0.602	0.602	0.602	0.670

Notes: This table shows different OLS specifications with *log(wealth factor score* as the dependent variable and *inside concession* as an explanatory variable that equals 1 if the corresponding DHS cluster is located inside the concession areas and 0 otherwise. Some of those regressions include additional covariates and fixed effects. Column 5 displays OLS results for the entire sample, including all 155 DHS clusters. Standard errors are clustered at the DHS cluster level and reported in parentheses. Exact definitions of the variables can be found in Section 4.1 and in Appendix A.
+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table H3: OLS Regressions for Years of Education

	OLS1	OLS2	OLS3	OLS4	OLS5
Intercept	−209.2908	−327.6161*			
	(196.0180)	(149.3306)			
Inside Concession	2.4856+	1.2060	1.1477	0.9115	−0.6757
	(1.2865)	(1.0364)	(1.2499)	(1.1190)	(0.5309)
Distance Concession Center	0.0328*	0.0328*	0.0120	0.0287	−0.0025
	(0.0158)	(0.0151)	(0.0173)	(0.0171)	(0.0033)
Gender	−2.3877***	−2.4765***	−2.4051***	−2.4752***	−2.7649***
	(0.1724)	(0.1613)	(0.1746)	(0.1598)	(0.1113)
Age	0.0748	0.0834+	0.0733	0.0822+	0.1026**
	(0.0496)	(0.0456)	(0.0479)	(0.0463)	(0.0318)
Age squared	−0.0014+	−0.0015*	−0.0014+	−0.0015*	−0.0020***
	(0.0008)	(0.0007)	(0.0007)	(0.0007)	(0.0005)
Survey Year	0.1063	0.1764*	0.1549+	0.1633+	0.1921***
	(0.0975)	(0.0739)	(0.0864)	(0.0800)	(0.0490)
Population Density 2015		0.0004		0.0007	0.0004
		(0.0004)		(0.0004)	(0.0003)
Level of Urbanization		5.0368***		4.8535***	5.8727***
		(1.1293)		(1.0584)	(0.7443)
Land Suitability		2.9808		−1.4392	2.3720
		(4.2398)		(4.1505)	(1.6115)
Missionary Stations 1924		0.3741		−0.3255	−0.1654
		(0.7697)		(0.6592)	(0.4794)
Number Enslaved (in 1000)		0.0131		0.0247	0.0071
		(0.0129)		(0.0198)	(0.0094)
Tsetse Fly Suitability		−1.6144		3.6389	−8.9820
		(7.7282)		(19.3010)	(5.6137)
Road Density		0.0028		0.0079+	0.0024
		(0.0022)		(0.0042)	(0.0020)
Num.Obs.	3544	3544	3544	3544	8002
Bandwidth in km	60	60	60	60	full
Fixed Effects	-	-	X	X	X
Add. Covariats	-	X	-	X	X
Clusters	69	69	69	69	155
Mean Dep. Var.	6.908	6.908	6.908	6.908	6.570
SD Dep. Var.	3.981	3.981	3.981	3.981	3.971

Notes: This table shows different OLS specifications with *years of education* as the dependent variable and *inside concession* as an explanatory variable that equals 1 if the corresponding DHS cluster is located inside the concession areas and 0 otherwise. Some of those regressions include additional covariates and fixed effects. Column 5 displays OLS results for the entire sample including all 155 DHS clusters. Standard errors are clustered at the DHS cluster level and reported in parentheses. Exact definitions of the variables can be found in Section 4.1 and in Appendix A.
p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table H4: OLS Regressions for Literacy

	OLS1	OLS2	OLS3	OLS4	OLS5
Intercept	−4.6914	−7.9422			
	(15.9886)	(13.9187)			
Inside Concession	0.1789	0.0864	0.0955	0.0635	−0.1068+
	(0.1112)	(0.1056)	(0.1191)	(0.1164)	(0.0539)
Distance Concession Center	0.0028+	0.0029+	0.0015	0.0023	−0.0004
	(0.0014)	(0.0016)	(0.0017)	(0.0017)	(0.0004)
Gender	−0.2495***	−0.2554***	−0.2505***	−0.2549***	−0.2862***
	(0.0227)	(0.0225)	(0.0234)	(0.0226)	(0.0161)
Age	−0.0184***	−0.0177***	−0.0186***	−0.0177***	−0.0175***
	(0.0051)	(0.0047)	(0.0050)	(0.0048)	(0.0034)
Age squared	0.0002**	0.0002**	0.0002**	0.0002**	0.0002***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Survey Year	0.0028	0.0064	0.0054	0.0054	0.0091+
	(0.0079)	(0.0069)	(0.0079)	(0.0075)	(0.0053)
Population Density 2015		0.0000		0.0001	0.0000
		(0.0000)		(0.0000)	(0.0000)
Level of Urbanization		0.3335*		0.3205**	0.4454***
		(0.1136)		(0.0988)	(0.0673)
Land Suitability		0.3486		−0.2263	0.2648
		(0.3619)		(0.3232)	(0.1797)
Missionary Stations 1924		0.0286		−0.0756	−0.0318
		(0.0827)		(0.0664)	(0.0538)
Number Enslaved (in 1000)		0.0019		0.0031	0.0010
		(0.0013)		(0.0019)	(0.0009)
Tsetse Fly Suitability		0.0620		0.8984	−0.9702
		(0.7710)		(1.7469)	(0.6037)
Road Density		0.0001		0.0006	0.0001
		(0.0002)		(0.0004)	(0.0002)
Num.Obs.	3532	3532	3532	3532	7972
Bandwidth in km	60	60	60	60	full
Fixed Effects	-	-	X	X	X
Add. Covariats	-	X	-	X	X
Clusters	69	69	69	69	155
Mean Dep. Var.	0.732	0.732	0.732	0.732	0.693
SD Dep. Var.	0.443	0.443	0.443	0.443	0.461

Notes: This table shows different OLS specifications with *literacy* as the dependent variable and *inside concession* as an explanatory variable that equals 1 if the corresponding DHS cluster is located inside the concession areas and 0 otherwise. Some of those regressions include additional covariates and fixed effects. Column 5 displays OLS results for the entire sample including all 155 DHS clusters. Standard errors are clustered at the DHS cluster level and reported in parentheses. Exact definitions of the variables can be found in Section 4.1 and in Appendix A.
p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table H5: OLS Regressions for Height/Age Percentile (Women)

	OLS1	OLS2	OLS3	OLS4	OLS5
Intercept	20.6019+	22.2538*			
	(10.2663)	(9.7410)			
Inside Concession	0.1087	0.1278	0.1597+	0.1262	0.0552
	(0.0865)	(0.0906)	(0.0855)	(0.0873)	(0.0392)
Distance Concession Center	0.0007	0.0018	0.0013	0.0022+	0.0004+
	(0.0011)	(0.0012)	(0.0011)	(0.0011)	(0.0002)
Age	0.0022	0.0027	0.0013	0.0025	0.0081*
	(0.0055)	(0.0056)	(0.0055)	(0.0054)	(0.0033)
Age squared	0.0000	0.0000	0.0000	0.0000	−0.0001
	(0.0001)	(0.0001)	(0.0001)	(0.0001)	(0.0001)
Survey Year	−0.0102+	−0.0085+	−0.0105*	−0.0073	−0.0051
	(0.0051)	(0.0047)	(0.0051)	(0.0047)	(0.0033)
Population Density 2015		0.0000		0.0000	0.0000
		(0.0000)		(0.0000)	(0.0000)
Level of Urbanization		0.1513		0.1387	0.1233*
		(0.1160)		(0.1191)	(0.0596)
Land Suitability		0.4818+		0.6465+	0.0234
		(0.2267)		(0.2996)	(0.1004)
Missionary Stations 1924		0.0111		0.0240	−0.0309
		(0.0524)		(0.0588)	(0.0267)
Number Enslaved (in 1000)		0.0006		0.0006	0.0007
		(0.0009)		(0.0010)	(0.0008)
Tsetse Fly Suitability		0.2031		0.1545	0.1618
		(0.4192)		(1.1071)	(0.2766)
Road Density		−0.0002		0.0000	0.0000
		(0.0001)		(0.0003)	(0.0001)
Num.Obs.	1190	1190	1190	1190	2689
Bandwidth in km	60	60	60	60	full
Fixed Effects	-	-	X	X	X
Add. Covariats	-	X	-	X	X
Clusters	69	69	69	69	155
Mean Dep. Var.	0.236	0.236	0.236	0.236	0.241
SD Dep. Var.	0.255	0.255	0.255	0.255	0.259

Notes: This table shows different OLS specifications with *height/age percentile of women* as the dependent variable and *inside concession* as an explanatory variable that equals 1 if the corresponding DHS cluster is located inside the concession areas and 0 otherwise. Some of those regressions include additional covariates and fixed effects. Column 5 displays OLS results for the entire sample including all 155 DHS clusters. Standard errors are clustered at the DHS cluster level and reported in parentheses. Exact definitions of the variables can be found in Section 4.1 and in Appendix A.
p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001.

Table H6: OLS Regressions for Height/Age Percentile (Children)

	OLS1	OLS2	OLS3	OLS4	OLS5
Intercept	31.2683**	32.0180**			
	(10.1217)	(10.8933)			
Inside Concession	0.0561	0.0306	0.0835	0.0261	0.0435
	(0.0679)	(0.0611)	(0.0809)	(0.0620)	(0.0426)
Distance Concession Center	−0.0001	0.0003	0.0003	0.0002	0.0001
	(0.0008)	(0.0008)	(0.0009)	(0.0009)	(0.0002)
Age squared	0.0002***	0.0002***	0.0002***	0.0002***	0.0002***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Survey Year	−0.0153**	−0.0144**	−0.0157**	−0.0132*	−0.0141***
	(0.0050)	(0.0051)	(0.0048)	(0.0051)	(0.0039)
Population Density 2015		0.0000		0.0000	0.0000
		(0.0000)		(0.0000)	(0.0000)
Level of Urbanization		0.1160		0.1262	0.1924**
		(0.1086)		(0.1149)	(0.0684)
Land Suitability		0.3225		0.2577	0.2496*
		(0.2043)		(0.2679)	(0.1034)
Missionary Stations 1924		−0.0378		−0.0546	−0.0447
		(0.0420)		(0.0506)	(0.0298)
Number Enslaved (in 1000)		−0.0005		−0.0007	−0.0005
		(0.0008)		(0.0012)	(0.0006)
Tsetse Fly Suitability		−0.3097		0.4873	0.0762
		(0.2757)		(1.2053)	(0.2516)
Road Density		−0.0002		−0.0002	−0.0002*
		(0.0002)		(0.0002)	(0.0001)
Num.Obs.	1042	1042	1042	1042	2436
Bandwidth in km	60	60	60	60	full
Fixed Effects	-	-	X	X	X
Add. Covariats	-	X	-	X	X
Clusters	69	69	69	69	155
Mean Dep. Var.	0.298	0.298	0.298	0.298	0.282
SD Dep. Var.	0.352	0.352	0.352	0.352	0.348

Notes: This table shows different OLS specifications with *height/age percentile of children* as the dependent variable and *inside concession* as an explanatory variable that equals 1 if the corresponding DHS cluster is located inside the concession areas and 0 otherwise. Some of those regressions include additional covariates and fixed effects. Column 5 displays OLS results for the entire sample including all 155 DHS clusters. Standard errors are clustered at the DHS cluster level and reported in parentheses. Exact definitions of the variables can be found in Section 4.1 and in Appendix A.
$p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Declaration of Authorship

I hereby declare that I have written this paper without any help from others and without the use of documents and aids other than those stated in the references. I have mentioned all the sources used and I have cited them correctly according to established academic citation rules. Moreover, the topic or parts of it are not already the object of any work or examination of another course. The paper format and electronic version of the paper are the same.

In the work presented here, I used ChatGPT or another AI as follows:

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Dominik Bantel